

Astr 210 Intro to Cosmology

Numbered Transcript

ASTR 210 • Dr. Brian Keating

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Lecture 1: Cosmology and Its Instruments

At a glance — Keating introduces cosmology and its instruments.

- 1.1 Welcome, everybody. It is a great pleasure to be here. I've traveled literally trillions of micrometers and billions of seconds to be here, and we are going to explore this universe together. I hope it's going to be a fun trip. It is going to take us back in time and throughout the entire observable universe. I hope it's going to be quite interesting, and I hope there'll be a lot of good questions. That is the best part of my job is answering questions.
- 1.2 By way of background, I am a cosmologist, not a cosmetologist, despite the look, the hair and the makeup that they've done here. But there is a similarity between those two words, as you may have noticed. I often get asked for either tips on hair and makeup or to read the zodiac sign of my guests, neither of which I'll be doing. But cosmology and cosmetology do share the same prefix, "kosmos." And there's a reason for that. Because in Greek, kosmos means beautiful or appearance, and the universe is incredibly beautiful. And so it's not really such a surprise that they should have this prefix and this relationship together. But again, if you came expecting hair and makeup, you're going to be disappointed.
- 1.3 What we are going to talk about is what I call this cosmic circle. Cosmic circle involves the universe and ends with us, and may begin again with us as we strive to understand the universe using very simple tools. Actually, tools we're all born with. There are two telescopes in our head, these little refracting telescopes with lenses and detectors. Then we have this supercomputer on our shoulders. The combination of those tools can allow us to not only comprehend what we observe in the universe through more powerful computers, in a certain sense, and more powerful telescopes, certainly. But also make predictions about what the universe will be like perhaps billions of years from now.
- 1.4 I like to say I've been teaching cosmology for a billion seconds. It sounds really, really long. And when you were a kid, if you tried to go to a bar and say, "Well, I'm over a billion seconds old," the bartender may or may not serve you. But a billion seconds is a 30-year time period. So I've been doing this for quite some time. I love doing it. It's my favorite subject. I hope to communicate some of the infectious questions that we're all born with. And it really takes formal education sometimes to beat that curiosity out of us. I want to instill that curiosity back. We're going to be really involving very simple tools—telescopes,
- 1.5 computers, and brains. Simple in some sense, very complicated in others. We're going to ask how those tools help us gather evidence, and what that evidence can tell us about the past history of the universe and perhaps its deep future. And more mesmerizingly than almost anything, are there other universes? And can we contact

them? And if we contacted them, what would that mean? We're not going to do any alien autopsies or anything in this class, but we are going to cover a lot of fascinating questions.

- 1.6 I aim to do this with some equations, but not too many. I cannot teach a class in cosmology and not use the famous equation, $E = MC^2$. You'll be encountering that—and understanding, hopefully, how it was derived and what it means so that you may more effectively use it when you are educating other people about what you've learned in this class.
- 1.7 So the grand scope of cosmology is perhaps best depicted by where it fits in, at least in popular culture. The repository of all knowledge is, nowadays it's Wikipedia. It used to be Encyclopedia Britannica when I was a kid, but now it's Wikipedia. And if you go to the page that says "Science," you don't see biology, you don't see DNA strand, you don't see a T-rex skeleton, all of which we will encounter in this class in one form or another. I'll explain why. You see a picture of the evolution of the universe. It's pretty surprising. If this picture, if Wikipedia existed even 30 years ago when I began my career, this would have been completely unbelievable. Why would you include cosmology?
- 1.8 For you see, in that time, in the 1990s, the age of the universe wasn't known exactly. It was thought to be at least 10 billion years old, but it could be 20 billion years old. Now, if you're young and someone says, "Well, you're 10 years old, maybe it could be 20," that sounds really good. If you're something like my age or older, you don't want to be overestimating that person's age. So it was laughable that you would even include cosmology as what's called a precision science, a scientific process by which we come to understand and maybe even categorize and exemplify the scientific method. That was laughable.
- 1.9 Nowadays, we know the age of the universe to within the equivalent of being able to look out at any one of you in the audience and say, "I know within a few hours of when you were born, not just your age, but the day, the month, the day, the year, the hour." So it's incredible how much we've come. And I'll explain how that's happened by way of the technology and tools we will encounter in this class. So this is what appears in Wikipedia. And why is that? We're going to cover, by the end of this course, you'll understand why cosmology features so centrally in the description of modern science. And we'll dig into this diagram that's provided from NASA that shows our universe and its evolution throughout the cosmic history.
- 1.10 But it also makes predictions about what's going to happen in the future and what happened at the beginning of time that, surprisingly, Wikipedia may have gotten wrong. We want to understand what are the limitations that led to the, perhaps, the flaws in this diagram. So hopefully we can organize a letter-writing campaign and get that changed. But what is cosmology?

- 1.11** I already said kosmos means beautiful or appearance. But cosmology is the laws of physics when applied to the universe as a whole. The properties of our universe are determined by the laws of physics. We're going to encounter all four main laws of physics—the law of electromagnetism, which governs light; the law of gravity, which governs objects and how they move; and there are two nuclear forces that we'll encounter when we talk about what the matter is in the universe that we're made up of, that everything we've ever encountered is made up of, and what other types of mysterious matter that may exist in the universe that goes by the name of dark matter, and other forms of energy that are equally mysterious, called dark energy.
- 1.12** So by the end of this class, you will understand the observational techniques, the scientific data that have been collected, and how they've led to a model that's so precise and accurate, and the distinction between those will become clearer, that it is the very hallmark of what science is. And towards the end of the class, we will start to encounter the biggest picture topics in all of science. How do we know what we know? How can we make predictions that are based upon objects that we can't even see, encounter? We don't live long enough to experience the full scope and grandeur of the universe. And yet, we can make fantastical predictions, based on our observations of the past about what will happen in the future.
- 1.13** Then lastly, as I said a few moments ago, a huge question, perhaps the most interesting question, not just in cosmology, my field, but in all of science, is the question of other universes. This will involve not just the question of physics, not just even the laws that we're not going to spend as much time on, what's called quantum mechanics, but it may involve bigger picture questions than we even think of being associated in a scientific course such as this—namely, questions of origins, of fine-tuning, of a creator, of philosophy, of theology even. Now, this is not going to involve discussions of that, but it's impossible to ignore them. We'll talk about that in the context of other possible universes and other life forms.
- 1.14** I always say, if you could only choose one class, and I'm a pretty biased person in this regard, but if you could only choose one class to study in all of your higher education, it would be cosmology, hopefully this class. And that's because it encompasses everything, from the microscopic laws of physics, to the laws of how we understand the laws of nature, namely philosophy, epistemology, but also questions of other universes, of purpose, of meaning. So these are incredible questions, and they're all subsumed within this one course.
- 1.15** Now, we'll get a taste of each one of these and then, to give you the appropriate background to then pursue and study these in more depth, perhaps in future courses that we'll offer. I mentioned it's not going to cover much in religion, but it's impossible to ignore the following. When I ask most people—I'm going to ask you to raise your hands in just a second. Raise your hand if your favorite day on the calendar is either your birthday, Christmas, or some anniversary of something. So raise your hand if that's

true. Your favorite day every year on the calendar. OK. Raise your hand if it's not one of those things, something else. OK. And raise your hand if you're scared to raise your hand in public. OK, fine. Most people will say that. They'll say their birthday, their child's birthday. I always have to say my anniversary, or my wife gets mad.

1.16 What are those? Those are beginnings. Those are origins. We're all fascinated with origin stories. And again, we're not going to delve into religion here, nor politics. But we can't avoid the ancient cosmological models that form the bedrock of Western civilization. One of these begins with the beginning of the universe, and that's the Old Testament's description called Genesis. We'll actually be talking and using that word "genesis" throughout this course, not in a religious sense, but merely to say, the generation of something, the origin of something, the origin story.

1.17 Why is your birthday so interesting to you? Well, you get presents and whatnot, but it celebrated the moment in time before which you could possibly have no memories. There was no yesterday. And the question is, is there a cosmic birthday? You may have heard of the Big Bang theory. You may have watched the show, but it's actually not clear if there was a Big Bang, namely one origin, one genesis, one unique birthday of the cosmos, or more than one. Some speculate there might be an infinite number of them ongoing right now somewhere in what's called the multiverse. So origin stories are fascinating for natural reasons.

1.18 Now, I always point out that this story, this story begins, of course, it's the origin of Judeo-Christian theology. But most of Judeo-Christian theology is not concerned with the Big Bang or the origin of the universe. So why does God, if you believe God authored it, why does he start with this description? He could have described, "You're not allowed to eat that delicious animal called a pig," for the Jews, or he could have said something about Jesus Christ. No, he said specifically, or this starts with the beginning of the universe. "The Earth was without form and void. Then God created the heavens and the Earth." Is that true?

1.19 Now, obviously, we don't rely on the Bible as a science book. But soon thereafter, there became other cosmological models, other thoughts about what happened before I got here—I, the collective I, or we. What happened before human beings were really sentient and were able to contemplate, for the first time, how did the universe get there, not just how did they get there? Here's an Egyptian creation myth, about 700 years after the Biblical narrative. It depicts a cyclical universe, a universe coming into and out of existence.

1.20 What if I told you that's still in vogue? That type of model, not with these hieroglyphic characters and so forth, but there are notions practiced by my eminent colleagues at the world's top institutions that posit that the universe has emerged from an unending cycle of Big Bangs, if you like, collapsing universes, crunches, expansions, crunches, collapses, unyielding throughout time. We'll explore that as we go through the rest of this course.

- 1.21** Fast forward a couple hundred more years, you come to Aristotle. Aristotle and the Greeks believed in a geocentric universe. They were the ultimate ego-maniacs. They thought everything revolved around them. But they started to have inklings of scientific thoughts, that there were physics principles that applied on Earth that could also apply throughout the cosmos and could also make predictions about the past history of that cosmos. Namely, they knew about gravity. They knew about motion. They started to wonder, well, what put motion in motion? Was there a first cause? That seemed ludicrous. So they supposed that the universe was static, eternal, had existed for all time, unchanging, unyielding.
- 1.22** And furthermore, because things had a tendency to fall down, that they were pulled by this force we call gravity, that the universe was somehow centered. No matter where they went, it was always falling down. They knew the Earth was spherical. We'll get to that in just a bit. Therefore, the only thing that could be at the center of everything was the Earth. And of course, they built many, many myths around that hypothesis. We now know that's wrong. But how do we know it's wrong? How do we know?
- 1.23** You're educated, you wouldn't be here if you weren't at least technologically savvy, if you weren't punctual, if you didn't have means of transportation, some sort of income, etc. I submit that you're in the upper echelon, not just of human beings today, but of all human beings that have ever lived. You have the benefit of all the knowledge that's come before you. And soon you'll have the benefit of all my knowledge and even more. Now, I ask you, if I told you, I'm going to take away your favorite golf club or whatever, I'm going to deny you a latte for the rest of your life, could you prove that the Earth is not the center of the universe? Could you even prove that the Earth isn't flat?
- 1.24** If I say to you, "This looks flat. I'll give you a million dollars if you can prove to me in the next 10 minutes, or explain how to prove to me that the Earth isn't flat and the Earth isn't—" Now, I'm not going to do that. I don't have Peterson Academy's full bank account at my disposal. But inside, ask yourself that question. As some of the most educated human beings in the world, in world history, because of the benefits that all we enjoy—I'm not going to call in anybody to prove it, but you know if you could or not. Or are we just merely like this ancient woodcut, which was colorized in the era of Photoshop? It depicts the quest that we're on to go from a comfortable vantage point here on Earth and use our minds to contemplate things that are far away, old, cold, distant. This is the journey that we're on.
- 1.25** Hopefully, by the end of the course, you'll have a sense to be able to determine whether or not this makes sense and what exciting things await us in the future. Because we're all playing a part in this. Since the very ancient times, we're all participants in this story, not the least of which because we pay taxes and we live in a society that funds the most incredible instruments ever built. We'll spend some time talking about the James Webb Space Telescope, the Hubble Space Telescope. These are all constructed by NASA, by the European Space Agency, the version of NASA that they

have. It's an incredible contribution that wouldn't be possible without the help from all of you, who are participating in this course. So why not get the full benefit of all your hard-earned money that you paid to understand things?

- 1.26** Fast forward about 1,000 years or more to this guy with the great hairstyle, Isaac Newton. You'll see that cosmologists have excellent hair, when we encounter some of these characters that we're going to be dealing with in this course throughout the next couple of hours of lectures. So Newton realized that actually there are things things that move in the universe. They're called planets. Actually, planet is a Greek name, and planet means something that wanders, just like a plane. A plane moves, it wanders. It goes from place to place. It doesn't mean a flat plane. It means something that wanders and moves. So he had to come up with the reason why these things move and explain why, in fact, the universe might not be centered on the Earth, but only appear that way. He came up with a set of universal laws, which is amazing. What does that mean? He said his laws apply not just in England, not just in Florida, California, where I'm from. They apply throughout the entire universe. And guess what? Some of them still hold to this very day, 400 years nearly after he first posited them.
- 1.27** Then came this guy. You'll see different pictures of his hairstyle in just a minute. Again, cosmetology. Never forget the link between cosmos and cosmetology. So there'll be a lot of great hairstyles. He realized that the universe couldn't be as Newton said, certainly not as Aristotle had said. And he came up with a theory of gravity that would allow for a universe, not just the planets to move, but the universe to move as a whole. And that was called general relativity. We will start and we will encounter this throughout this course. It is the underlying bedrock, the scientific underpinning of everything we discuss in this class. And it was only discovered 100 years ago. It's only 100 years old.
- 1.28** And we'll see throughout this course, what happens frequently in science, is that a person, a group of people, have an idea, and that idea seems to apply and hold true. It's based on the encounter of that idea with evidence. If I drop something, it falls down. Seems like a simple proposition. But how do you explain it without recourse to things like the Greeks had or the pagans had? "There's a god that controls these phenomena. There's a god that makes thunder. There's a god that makes rain." No, it's better to have a science of meteorology because that explains more things with fewer assumptions, more elegantly, more beautifully, and more compactly. And those are all virtues of a good theory.
- 1.29** But even Einstein makes mistakes. And we'll see in what way his mistake has come, in some sense, to haunt us because his theory contained within it the seeds of the eventual destruction of the universe. But don't worry, don't worry. That won't take place for at least another 100 billion years. So we're safe. We keep paying our taxes and we will keep enjoying this wonderful cosmos. We'll understand that in just a bit.

- 1.30 In the cosmos, I said, I've been teaching for a billion seconds. Sounds like a lot. It's only about 30 years. And in fact, we don't use seconds or feet or meters or things like that in the cosmos. They're simply inadequate. So just as you wouldn't describe yourself as somehow seconds old, you use a bigger unit like years, which is about 31 million seconds per year. And that allows me to be teaching this for 30 years, about a billion seconds. Similarly, we're not going to use miles or feet. We're going to use astronomical terms. In fact, one of them is called the astronomical unit, for the most minimal length scale that we can possibly encounter. And that is the distance between the Earth and the sun. That's one astronomical unit. You may have heard it, 93 million miles.
- 1.31 Because the universe is so capacious, we also will use sometimes the speed of light as a measuring yard stick, which is uncomfortable, perhaps, because you're used to thinking of a speed as so many being miles per hour or meters per second, but we can actually multiply that by time and get a very large length scale—many, many billions or trillions of miles. And so we'll combine the two together, years instead of seconds, and light speed instead of more conventional human accessible speeds, and we'll get what's called a light year. And that light year, we will then expand because that will be terribly, pathetically inadequate to describe things in the cosmos.
- 1.32 A light year from here, if you go one light year from the sun, you don't even get to the nearest star. Not even close. The nearest star is about 4 light years away, and that's traveling at the speed of light for one year. Don't even leave our solar neighborhood. And we'll explain that. In fact, we're going to need not millions of light years, but billions of light years. And we'll explain that as we go along. We'll talk about energies and masses. And we know, even before we came in, even though we don't know why, maybe, that energy and mass are related via Einstein's most famous equation.
- 1.33 We'll see that he didn't actually create that equation in the form that everybody has on T-shirts and whatnot, but he created it in another form to depict how to think about mass, because we'll see that mass will change in the universe, that there are different types of mass, and different types of matter, and different matter and energy. And one of my sons has a T-shirt that says, "You matter unless you multiply yourself by the speed of light squared, then you energy." And that will be come into play because we will not speak about energy in terms of calories or things that we're used to. I doubt anyone here really is familiar with joules, capital J, J-O-U-L-E-S, not jewels like the gem. But we'll be dealing with things millions of times more energetic than anything we could possibly contemplate or have experience with.
- 1.34 And then when you take energy scales and you convert them into how much energy an object is producing over time, that's called its luminosity. We'll talk about things in terms of the sun, its luminosity. And the sun, as spectacular as it is, as responsible for all of our lives as it is, every form of energy we use, except for nuclear energy, the sun itself is kind of a middle-ranked star. It's not that big, not that small, not that hot, not

that dim. And we'll describe things yet in terms of it, as a standard. Naturally so, because it's our nearest neighbor in the universe, at least in terms of stars.

1.35 So I said earlier, cosmology is a science. A science has to take observations, confront those observations with predictions, refine future predictions based on the observations and the confrontation with the theory, and refine and reprocess. The goal of science is not to prove things right. I'm not coming here to prove that there was a Big Bang, or prove that there wasn't one, or prove that there is a multiverse. I'm here to lay out possibilities, present evidence, present a reasonable chain of logic, and equip you to examine upcoming pieces of evidence that will be coming from the Webb telescope, from projects my colleagues and I are building, perhaps, and to assess and appreciate how much human beings have been able to achieve, and perhaps, perhaps to participate. Because never before in history have so many citizens, lay people, been involved in actual scientific discoveries. We'll talk about that in this course.

1.36 So as I said before, the Greeks thought the universe was centered on them. I have some colleagues that think the universe is centered on them, too. We're not going to be talking about psychology in this class, though. But they wanted it to be there. They had what's called a theoretical prejudice, that they expected the universe to be organized according to certain laws. And we, too, expect the universe behaves in certain ways, and we adapt and adjust our models to fit them. Over time, we've come to realize that the Earth is not the center of the universe, that planets, the only things that seem to move in the sky besides the sun, and of course, we know that's because the Earth is in motion today, that wasn't really discovered until the 1700s.

1.37 So we're talking 2,000 years after Aristotle, to prove him wrong. That's what scientists do, physical scientists. We don't prove people right. We prove other people wrong. Whatever is left is right, or is provisionally right. And then we try to prove that person wrong, too. So Aristotle was wrong. Newton came along, showed Aristotle was wrong. Einstein came along, showed that Newton was wrong. People have come along and showed many times that Einstein is wrong, believe it or not. And so on and so on. That's the goal, is to refine, reproduce, and improve upon what's come before you. As Newton said, he saw farther than other people because he "stood on the shoulders of giants." We attempt to stand on their shoulders and learn more about what we can see, seeing farther off into the cosmic horizon.

1.38 It was only until the 20th century, less than 100 years ago—in fact, we just celebrated the 100th anniversary of the discovery that the Milky Way is not the universe. A hundred years ago, people thought the entire universe was one galaxy. Einstein thought it was. And that theoretical prejudice—which came from the limited number of astronomical observations at the time, that the universe was the Milky Way galaxy—that led him to make what he called his biggest blunder, which was to provide a fudge factor, literally, in his famous equations, that made the universe stationary, static, eternal. And that we know is wrong. And he admitted it was wrong. And it's good

because it shows that even he makes mistakes. And I like to think about what a career he could have had, if he didn't make that mistake. He could have been very successful. Now, he is, of course, one of the few people that is celebrated by lay people, but also by professional scientists such as myself.

- 1.39** So we'll talk a lot about the so-called Hot Big Bang and what that means. Is the Big Bang the origin of time? Are there other Big Bangs going on throughout a vast cosmic landscape called the multiverse? These are the questions we're going to address, maybe not answer, because we have to rely on upcoming data sets, including, as I say, data sets my colleagues and I are working on at this very moment at observatories all over the planet and in space. So it's perhaps the most exciting time to be doing what we'll be doing in this class and to be learning about it—if nothing else, then to have a deep abiding appreciation for how far science has come.
- 1.40** OK, so, as I said, we're going to talk about things not in terms of inches, feet, meters, even light years. We're going to be talking about thousands or tens of thousands of light years. In fact, we're going to go out billions of light years into the cosmos. We're going to ask a question that you probably thought about on one of those late-night college dorm room dish sessions. Is there an edge of the universe? What happens if I go beyond the edge of the universe? As I said, are there other universes? How far away are they? How would we know if they contacted us? Can we know what will happen in the deep future?
- 1.41** And this is all in the context of laying out the question that I like to think about. What happens if I teleported you back in time, back in space, perhaps? What would happen to this kind of alternative history? If I were to teleport you back and talk to Aristotle and say to him, "Everything that you learn in this course, how would that change this alternative history?" How would it affect the future of our understanding of the laws of nature and being able to understand and predict new laws that we can't even contemplate at this time?
- 1.42** Because the physics that we do now and that you'll encounter here would blow the mind of everybody, perhaps with the recent exception of some of my brilliant students and colleagues. But even just a few decades ago, the contents of this course would have been almost unimaginable. And so, as I say, it's the most exciting time to be studying this. And yet I don't want you to think it's all going to be easy because it's not. Astronomy, of which cosmology is a subset, is one of the hardest sciences to do. It's harder than doing biology. No offense to biologists, but it's harder for a very specific reason.
- 1.43** We are not able to do experiments. I cannot do an experiment the way I can with, say, a fruit fly, as we'll encounter in just a bit. I can't say, let me expose one fruit fly to some condition, and I have to be careful because people at PETA or somewhere else may be watching this, but I won't say what I used to do in biology class with a frog or

something like that. But the frog usually lived after my dissection. I'm such a terrible biologist. So biology is hard, believe me. But we can't even do an experiment.

1.44 What is an experiment? You change some variable and you ask what happens to the thing that you change and to the control that you didn't change. That's really hard because I can't change the temperature of the sun, right? And say, well, what would happen to the properties of our galaxy, or the rotation speed of the Earth, or what have you, if I change the temperature of the star. But astronomy compared to cosmology is a piece of cake because there are literally 100 billion 100 billion stars in the observable universe. So 100 billion squared stars. So you can start to do statistics. You can ask questions. When you see a yellow star, do all yellow stars have the same mass? You can actually do that

1.45 and calculate that and plot that. Do they all put out the same heat? Even though one looks really far away, could it be really close to us? You can do those types of experiments. What do you do with a universe? The "uni" in universe means one. There's one universe. So how do you do an experiment on the universe? So in cosmology, we only have one universe. That's what the word uni means, after all, one. And "verse," by the way, means rolled up. It's weird. It means everything rolled into one, I guess. I wasn't around when they created the etymology that word.

1.46 But cosmology, there's only one universe, or is there? That's a topic we're going to ask and try to answer scientifically. For the first time in history, it's possible to do so. I like to think of what we're doing as paleontology with corrupt archeologists abounding. In other words, we're looking at fossils. Those fossils have traveled through time and space, and they've gotten corrupted. They've gotten afflicted. People have interfered with it. It's like raiders, raiding the Lost Ark. I can't show that for copyright reasons, but here's what my Artificial Assistant TA came up with. So there's all sorts of artifacts that we'll see. Forms of matter, forms of light, other exotic things you may have never heard of, like waves of gravity. What does that mean? And they get contaminated as they come to us. And how do we deal with that?

1.47 We'll be talking a lot about what's called a space-time diagram. Sounds complicated, but you do it all the time. Because, as I said, you have two telescopes in your eyes, and your eyes are little refracting telescopes. They have lenses. Those lenses focus light, visible light, onto an exquisite set of sensors, rods and cones in your retina, and they help you detect the light. And then due to the finite speed of light, fast but finite, when you look back in space, you look back in time. Light travels one foot in a billionth of a second. So if there are twins here—my wife and I are blessed, we have twins—and I like to move one behind the other by one foot, and that one is exactly, appears to be, one billionth of a second older. So she loves that, because she's the younger sister.

1.48 So when you move something back in space, it's equivalent to moving it back in time, or you're seeing things older. We've all seen pictures of archeologists. Here's some archeologist seeing new, fresh green grass here. There's a layer here. As he looks out

farther and farther away, gets older, different-looking, strange-looking. And that will happen in the cosmos, too. That's a space-time diagram.

- 1.49** Let's do a more distant example than those examples of the Raiders of the Lost Ark or that archeologist looking at a dig site. As I said, the astronomical unit is defined to be the distance between the sun and the Earth. It's the first cosmic distance scale of real interest. The moon to the Earth is much shorter, and the moon is only a quarter million miles. And the distance between the sun and the Earth, being what it is, takes light about 8.3 minutes to get from the sun to the Earth, which is interesting, because when you consider this disgusting-looking creature called a mayfly, some mayflies have an average lifespan of about 8 minutes.
- 1.50** You could see that, again, I wasn't consulted when they designed this object, this creature, but there are some of them that have an adult reproductive system that also is accompanied by its jaws or mandibles, which are vestigial. Has anyone taken a biology class? Anyone tell me what vestigial means? I'm just a simple cosmologist. Vestigial, I think, means it's superficial. It doesn't really, it's superfluous. It isn't really necessary. Supposedly human beings have a vestigial tail, tailbone. I know some of my colleagues have a visible tailbone, but I'm not going to get into that.
- 1.51** Vestigial means it's not necessary. So if you have mouth parts that don't work, they're only there for historical reasons, you're not going to live that long. That's the extent of my biological knowledge. And in fact, some of the males only live 8 minutes. In other words, their whole job is to reproduce, and then they reproduce primarily to make other females, I suppose. And so a lot of them die in 8 minutes. Now, imagine two of them, twin mayflies. One is born on the moon, which is a quarter of a light, or about one light second away, rather. And one is born near the sun, same distance away as the sun. They're separated at birth.
- 1.52** You look at this one, all of a sudden, you hear the tapped music playing in the background. I should have had that. This one is dead. What do you know about that one? Is this one alive or dead? You see this one is dead 8 minutes after it was born. What do you know about that one? Is it alive or dead? Who thinks it's alive? Raise your hand if you think that one on the sun is alive when you see this one next to you is dead. Good. Because this one's dead, too. You just don't know about it yet. They were born at the same time. You already see this one's dead. You know that this one is dead, too, but its news of its departure hasn't reached you, like they said about Mark Twain. Or he said, "News of my demise is slightly premature."
- 1.53** So this one is also dead. And that's because you're looking back in space, means you're looking back in time. You're looking back at the sun, and that sun may fly is dead. And it's also terrifying to think, if you were, right now, if we're here enjoying a sunny day in Miami, and all of a sudden, right now, the sun disappeared, no longer exists. We wouldn't know about it for 8 minutes. We wouldn't see it. We wouldn't feel it. No effects would happen. Nothing would change for 8 minutes. In fact, if it was converted

to the same mass as our sun is now, we'll see later, we wouldn't even change anything about the Earth and its rotation or its orbit or anything. You would just see it would be dark. "Oh, it must be an eclipse."

- 1.54** We look out farther into space. As I said, things look different. They look very puzzling. We don't see stars. This is the Hubble Deep Field. Every single speck of light you see here is a galaxy. Some of the galaxies are identical to the other ones. They have different colors and so forth. We'll talk about that. But they're on average, average. Each galaxy is a cluster of perhaps 100 billion stars, just like our Milky Way is. And yet some of these things look very different. Some are bigger, some are small, some are redder, some are bluer. What does that mean? We're going to go and look out into the universe as far as we can look. What prevents me from seeing somebody in the back row? Well, someone's in the front row. You look younger than someone's behind you. She looks younger than you, and so on and so on.
- 1.55** And we can look out into space as long as there's nothing in our way. We can keep looking as far back as there was stuff to look at. If we look back to the very origin of our particular universe, we won't see anything. This is depicted here in another form of space-time diagram. Close things are new. They are recent, farther things away. Going back in time, we'll encounter something called the cosmic microwave background. And eventually, we'll come back to something called inflation, which is the very first entry in that Wikipedia article on science. And we'll see, is it really true that our universe began with what's called inflation? And that is one of the hotter topics in my field of cosmology today.
- 1.56** Before we do that, I want to do a quick tour, starting again close, new, recent, modern, going back farther and farther into space. So it's hard to see in the glaring, beautiful white light studio here, but there are a little stars here. And you can see stars, you can see constellation. That's in Manhattan. This is me near midtown Manhattan, looking out. I like to think, if you turned your head upside-down, it's almost like there are constellations of human-made light down here, and then there'd be this vast expanse of darkness. But this goes to show, and even in the middle of a city, wherever you are watching this or participating in this, you can see stars. Those stars may or may not even be there.
- 1.57** Their light may be traveling through space, but the source of them may no longer exist, in that the stars that produce them, or the galaxy in the case of the Milky Way's nearest neighbor Andromeda, may not even exist, and we won't know about it. But our life should go on, and we can't get too morose about that. But the point is, we can still see these things. We can see a mysterious Milky Way nightscape shown here. This is taken by a friend of mine, Brent, in Arizona. What I love about this is these silhouettes and shadows, and it's linking together, as we'll see, the universe of light and a universe that you can't see—of shadows, of darkness. The application and the implication of that to

things such as dark matter and dark energy—shadow forces we cannot see, but we see their effects, just like the effects of these figures that instill something on us.

- 1.58** We connect and we see the beautiful cosmic landscape. It does something to us at a primordial level because we're all born with this appreciation for these grand questions, and it only takes formal education to beat that out of us. Here's a picture of the whole Earth. Now you could say, "Well, now I could prove that it's a sphere, but not really. It's not so simple." We'll discuss that later. Our nearest cosmic neighbor, if you like, is the moon. The moon tells a certain tale of its origin, its formation, and the formation of the solar system. We'll talk briefly later on, and I'll actually pass out samples to you in the audience today, of the very few objects that astronomers get to actually do experiments on. Some of them are meteorites, and I brought some here for you guys.
- 1.59** Those meteorites are part of a very, very select set of pieces of astronomical data that we actually can experiment on. Meteorites came from a series of chain of events that I'll describe in some detail later on, that involved the formation of a star and its cataclysmic explosion. Not recently, not even in recorded history, not even in the history of the universe since the Earth was formed because it is made of the same material that formed the Earth. In fact, it provided this raw materials for the sun and the Earth to exist. Then that was called a supernova. We'll talk about that.
- 1.60** Some of the fragments of them became meteoroids, traveling throughout what would become our solar system. And after the moon was formed, some of them impacted, landed on the moon, and some of them impacted on Earth. Some of them made their way to various online scientific supply sources, and some of them eventually made their way to me, and I will provide them to you to study because we can learn about the formation of the moon, of the solar system, and of the universe from these things, just as we learn about the universe from the light we gather in our telescopes.
- 1.61** But that's it. We can't do experiments. We can't change the temperature of that supernova. We can't change the moon-Earth distance, the sun's composition. We can't do it. So we have to use our brains and our models and our observations —that much harder. We look at the sun. We see not this boring yellow disc. We see it's extremely active. It affects the weather, the climate on the Earth. It can affect geo-communications, satellite navigation systems. Its activity produces solar storms, solar winds, and it's an active star. It's not, as I said, a boring yellow disc.
- 1.62** We go farther out, we see stars that we can't even identify. They look like pinpricks of light. How do we know how big they are? How do we know how far away they are? We're about to encounter that. Since mankind only invented Netflix in the early 2000s, astronomers of the past used to occupy their evenings by chilling out and looking up and drawing connections between the patterns of starlight that they saw. And we still do these types of things today. But there's nothing intrinsically meaningful about the pattern of constellations, say, depicted here. But nevertheless, it helps us organize, to

categorize, to do sort of a taxonomy, which is what astronomy does have in common with other sciences. We have name things, classify things, provide our best guesses at the laws that govern them, in order to make predictions about what the universe used to be like and what it's going to be like in the deep, deep future.

- 1.63** The first scientific method application was really done when an observation by Galileo was used to test and not prove a theory, but to falsify a theory. The theory at the time in 1609 and 1610 was that the organization of the universe was thus, this Earth-centered universe where everything was moving away from the Earth. We knew about the 5 planets. There were distant stars that didn't seem to move. The planets in Greek means mover. And Ptolemy, up through Galileo, up through Copernicus, really, hypothesized that the Earth was the center of the universe. Then came Copernicus and said, "No, this is wrong. Actually, the way the universe is truly organized is that the sun is the center." Heliocentric, not geocentric.
- 1.64** A heliocentric universe is very different. In fact, it's simpler. It's more organized. It has more unification to it. The laws that were required to explain these observations in this model, the geocentric model, it got most of the predictions right. In other words, you could predict Jupiter will be there at this time, but it required tremendous amount of work called epicycles, knobs, dials, wheels, to tune a model that was inherently fatally flawed. Whereas Copernicus came up with this model that no, it's actually the sun that's the center of the universe. Then you need fewer epicycles, fewer knobs and wheels and dials to turn.
- 1.65** Actually, we know that that's right, but it wasn't proven. That's not our job. We don't prove things. Mathematicians prove things. Philosophers prove things. Physicists, of which I am one, and physical scientists don't prove anything. We falsify things. By falsifying this model, it didn't prove this model. Does that make sense? Just because you show somebody's wrong, it doesn't mean you're right. There's an infinite number of ways to be wrong. There's only one way to be right. It turned out this is right, but it too has flaws that need to be corrected by a later scientist, in this case, Isaac Newton. And then Newton's laws, applied more globally, didn't require the wheels that even the Copernican model required, but still had some flaws, or lacune, to use a fancy term, that were corrected by Einstein. And as I said, guess what? Good old Albert is not the final word either.
- 1.66** So how did this actually occur? Well, in the 1609 and 1610 time frame, Galileo didn't invent a refracting telescope. That was actually, interestingly enough, connected to the invention of the printing press. And all the way back to the very first slides I showed, the Gutenberg Bible, the Bible itself, was responsible for the overthrowing, some say, of the Biblical worldview that the sun went around the Earth. And it did so in a very interesting fashion. What happened was, Galileo was living in, as I said, in the mid, well, the early part of the 1600s. He heard about this new device that had been invented in

the lowlands, in the Netherlands. It was invented by a spectacle maker, an optician, who took two lenses and took the two lenses and put them in front of each other.

1.67 What do you know? He got a spy glass. A spy glass, a telescope, tele-distant scope, vision. It allowed him to see things, as Galileo said, "as if they were right next to them, even though they were far away." Now, how does this connect to the Gutenberg Bible, and to the Bible generally? Well, if you think back, if you've ever been to your optometrist, they take out an eye chart. The eye chart has all these weird letters, and there's always an E at the top. That's the only one I always remember. There'll be an E at the top, and then there'll be T, O, E, whatever, all these different letters. And they'll ask you, "What do you see and how far you can go away from it?" And what it means now to have 20/20 vision —so 20/20 vision means you see something at 20 feet away that the average person sees at 20 feet away. That's one way of classifying, "Are you better than average or are you worse than average?"

1.68 If you need to be at 5 feet away to see something that an average person can see at 20 feet means you need some eyeglass, some corrective lens. Well, they didn't have eyeglass charts back then. They only had one printed work that was available around the world, and it was the Gutenberg Bible. So if you could read the Gutenberg Bible without putting a lens in front of your eye, you had perfect vision. At that time. So it was the Bible that allowed an eyeglass standard, which then allowed the mass production of lenses, which then allowed this guy, Hans Lippershey, to put two lenses together and see, "Oh, it makes things far away look like they're close." But it took Galileo, oddly enough—Galileo was the first person that took those two lenses at night and looked up at things he could see in the sky.

1.69 But it wasn't enough to have these eyeglasses. This is me at his villa, which would become his prison outside of Florence, Italy, using a replica of it. They wouldn't let me touch the original. There's only one surviving telescope of Galileo's left. And in the cold evenings of northern Italy in 1609 and 1610, in January 1610—this is over 400 years ago—he took those two lenses up, connected them to his eye, which is conveniently connected to his brain, and realized when he looked at the planet Jupiter, Jupiter was always accompanied by these tiny little dots. And I would have said, because I'm a lesser scientist for sure than Galileo, "Oh, that's cool. It's near some stars. Let me see what the neighbor is doing next door."

1.70 But Galileo said, "No," methodically, night after night, mapping out these positions, seeing that there were always four of them, they were doing a periodic motion around them. He realized he was looking at, as he called it, a tiny solar system. A solar system seen edge-on, looking this way, and he was seeing little dots of light moving around not the Earth, not even the sun, but around Jupiter. Did that prove the Copernican theory right? Absolutely not. It just proved that the Earth wasn't the center of the universe. These objects, the Galilean moons, now called the Galilean moons—back then, he was politically savvy, and he named them after the patron who had supported

his research. Like, I would name it after NASA or something today. Or he named them after the Medici family who sponsored his research. He called them the Medici moons. Now, we called them the Galilean moons.

- 1.71** He realized they were like a miniature solar system, orbiting not around the sun, not around the Earth for sure, torpedoing forever the Ptolemaic model. So as I said, he didn't prove this was right, and he proved this was wrong. That's what we're here to do, prove a lot of different models right, wrong is our goal. That's the actual sketch of the telescope. That's his signature. It's only about as long as your arm. And I call it a Biblical bludgeon because some say this was the start of the Enlightenment. And the Enlightenment led to a sort of overturning of the Biblical worldview that some say features an Earth-centered universe. These are some more of his sketches. He sketched the planet Jupiter many, many nights. It's kind of boring, actually. And he wrote this in this book called *Sidereus Nuncius*, which means "the starry messenger."
- 1.72** He also sketched the phases of the moon and what the moon surface looked like through a telescope, which proved the moon was not a perfect sphere. It wasn't crystalline and glowing and luminescent. It was actually just reflected light from the sun. He measured on the moon. The moon had mountains, craters, valleys. He measured their heights. 1610. He proved that it was not perfect, and that overturned a lot of the Galilean, a lot of the Aristotelian model of the universe. He even observed Saturn. He thought Saturn was a triple planet because he could not see that it had rings, as we now know it does. And he sketched the phases of Venus as well, which could only be thus if Venus orbits closer to the sun than the Earth. Again, a blow against the Aristotelian Ptolemaic cosmology that was geocentric. By no means a proof that the Earth was the center, the sun was the center, but it was a step in the right direction.
- 1.73** Nowadays, we use massive telescopes. I'm blessed in San Diego to be very close to what was the biggest telescope in the world up until about 50 years ago, 45 years ago. That's a 5-meter, 200-inch diameter telescope in Mount Palomar. That's me and my friends there, visiting it. This is a telescope. You wouldn't think it. It doesn't look much like a telescope. This is a telescope that observes not light, not meteorites, not sun. It observes waves of gravity produced by collisions of massive black holes orbiting around one another and eventually crashing together in a cataclysmic explosion located billions of light years away. This is called a gravitational wave observatory. There's actually two of them. That one is in Louisiana.
- 1.74** And what this does is, when a black hole crashes with another black hole, waves of space-time ripples, called gravitational waves, travel at the speed of light, never die, and eventually some reach the Earth. Actually, if you're here when one of these—actually, millions of these are happening right now, but their effects are so weak because they're so far away, we don't feel them. Actually, what they do is they squish and squash the Earth, the entire Earth, but not as much as I've illustrated here. This is an image from my first book. It's called *Losing the Nobel Prize*. I'll talk a little bit about it

because I'm proud of it and because it has a lot of illustrations of things that I wish had existed when I started to learn about cosmology, as I say, 30 or so years ago.

- 1.75** Here's another view. These are currently the biggest telescopes on Earth, and there are two of them. They're located in Hawaii. If you've ever been to Hawaii on Mauna Kea, you can see these white little balls, and these are twin telescopes. Each one is 400 inches in diameter. They're amazing. But they're not going to be the biggest for very much longer. We're building actually two much, much, much, much bigger telescopes. And the telescope that we're making now in the next generation are going to show us an unrivaled view of the universe. The next ones that are coming out are not 10 meters in diameter. They're going to be 24 meters in diameter.
- 1.76** Now, what's curious about that is that, when you have a telescope, the amount of light that is let in—if you have a telescope, it's characterized by the diameter of its aperture. It's so many inches, feet, what have you. Galileo's was 1 inch in diameter. And yet he could see all the craters on the moon, the phases of Venus, et cetera. And you can actually see the same things from any city on Earth. You can not only see what Galileo saw, but you'll feel, I guarantee you, you will feel what Galileo felt. You'll feel the visceral connection that what you're seeing is another world, when you look at the craters on the moon. And again, you can see that from the middle of Manhattan as well as you can see it from a dark desert in San Diego.
- 1.77** So the diameter of the telescope is important. That's how wide it is. What really matters is the ratio of the diameter of the telescope, squared, divided by the wavelength of the light that you're looking at. This has units, as you'll see. This is in a meter, say, and this is in meters as well. So this whole thing is meters squared over meters. So this has dimensions of meters. So this is so many meters. This is basically, how finely resolved the telescope can see depends on the ratio of the diameter to the wavelength, that quantity squared. And so a telescope that is twice as big, the Mount Palomar telescope, as I said, is 5 meters. This is on Mount Palomar. Then the Keck telescope is 10 meters. So this has D^2 , 2 times that. So it actually collects four times the amount of light that the Keck telescope—is collected by the Keck telescope than the Mount Palomar telescope.
- 1.78** These ones, the giant Magellan telescope, and the next one, which is going to be called the 30-metre telescope—I'll give you one guess how big the diameter of that telescope is. It's 30 meters—that has 9 times as many photons, particles of light, can get detected. You can see much, much finer things by that same ratio. So it's an incredible age of astronomy to be able to live as we do right now and study the universe with these tools. Now, as I said, these aren't even anywhere close. They don't exist yet. They will in the next 10 years, those optical telescopes. But they're dwarfed by even an average-sized radio telescope. These are radio telescopes in New Mexico. If you've ever seen the movie *Contact* with Jody Foster—so you should know, she's based on a real-

life character who's a friend of mine. His name is Jill Tarter, who is an astronomer who does search for alien life in the universe, a fascinating scientist.

1.79 But these telescopes are tiny and puny compared to what used to exist, or this one was unfortunately collapsed about 7 or 8 years ago in Hawaii. This was in the movie *Golden Eye*, if you've ever seen that, James Bond movie. And this one, they haven't made any movies about, but this is the biggest single dish telescope. These are radio telescopes. These don't look at light waves. They look for radio waves, much, much longer waves. And so even though they're much, much bigger, because it depends on the wavelength, the wavelength is so much longer, they actually don't have—they have much better resolution. Finer, detailed objects can be seen, but they're not actually—they don't give you as quite a big resolution gain, as you might naively think.

1.80 And of course, you've heard of the Hubble Space Telescope, which has been in orbit since 1990, and this sees visible light to optical light. This is only, I say, only half the diameter of the biggest telescope 50 years ago. It's 2.5 meters, about 100 inches in diameter. But to launch something in space costs at least 100 times more than the same size telescope on Earth. So this was a billions of dollar project back then, dwarfed by its successor, which is about 3 times bigger, called the James Webb Space Telescope. That was launched on Christmas Day in 2021, so it's almost 2 years in orbit, and it's already revolutionized how we understand the formation of stars, the composition of the atmospheres of exoplanets.

1.81 We're seeing the atmosphere of another planet for the first time with telescopes like the Webb Telescope. We see them in different wavelengths. We'll be talking about that. We'll be doing some demonstrations with different types of light-detecting sensors and objects that go into what are called spectrometers. Everybody has seen prisms before. That's what they do. They disperse the light into different colors. We'll look at this in the next lecture, how you can use the color pattern to determine what something is made of. When you do that, you can not only use these as a tool to do chemical composition analysis of distant objects, you can use it to measure the expansion of the universe. And that'll be a key topic in this.

1.82 In a few years, the Nancy Roman Space Telescope will be launched. It'll observe spectrum of billions of galaxies. And that will tell us, again, not just what they're made about, not just that the universe is expanding, not just how fast the universe is expanding, but will the universe's expansion eventually end because the universe as a whole is ripped apart? Something to look forward to. And we'll finish up with this animation from NASA of flying through these different survey fields of these galaxies and what we might expect to find with these vast telescopes in the very near future, zooming through the universe.

1.83 This is a picture of a nebula region, a star-forming region. Nebula is Latin for cloud. These clouds of gas and dust are the byproducts of the formation of new solar systems. The previous slide shows what the Hubble telescope saw. The Webb telescope sees an

even greater detail in different wavelengths. We'll look at these stellar nurseries and what they inhabit between them. This is the Pleiades. Does anyone know why I'm showing a very sensible-looking station wagon next to the Pleiades, or Seven Sisters? One of you perhaps might even drive one of these because, in Japanese, the Pleiades are known as Subaru. I love Subarus because it's the only car named after a constellation or an asterism. Allegedly, they had—Chevy had a car in the '70s or '80s called the Nova, which means a bright, glowing star. But it didn't sell well in Mexico because, in Spanish, it meant doesn't go, "no va." Be careful when you name your car after an astronomical object.

1.84 Lastly, going out from the Earth, we see different stars. We're going to do some experiments. Yes, we'll do some experiments here to depict the ways that astronomers know the distances to these objects. We'll encounter other objects, beautiful clusters of galaxies and stars, supernova remnants, exploding stars. We'll understand the structure of our Milky Way Galaxy for the first time, known with great precision. Other galaxies, this galaxy is the most distant object that human beings can see without the aid of a telescope. This object's light was released to travel to your eye. You can see it only from a dark site, unlike the other objects I talked about, but you can see it with your naked eye.

1.85 The light that it is emitting that we can see on a dark site tonight, for example, was emitted 2.5 million years ago. There were primordial hominids ranging around the plains of Africa when that light was emitted. It may not exist, as I said. We have every reason to expect that it does, but nevertheless, it's a possibility. We'll zoom through other galaxies at greater and greater distances, clusters of galaxies. Just as there are clusters of asteroids, there are clusters of stars, there are clusters of galaxies. Question I have for you, are there clusters of universes? That's something we're going to encounter in this class.

1.86 More clusters of galaxies. And the matter that we'll encounter is not all that you can see. This is a zoom through the famous Hubble Deep Field. We'll conclude with some images from the Hubble telescope traveling, zooming through space and time, as if we're traveling through a spaceship, covering not stars in this image, but galaxies—each one perhaps as vast as our own Milky Way, 100 billion suns glowing bright. And this is a comparison of how much has changed between the Hubble Space Telescope image of that very same field on the sky with the James Webb Space Telescope. So James Webb reveals everything that Hubble saw, but it sees things that Hubble couldn't see. And we'll explain why in the next lecture.

1.87 And finally, going back to what I study, it's called the Cosmic Microwave Background, which is a relic. The oldest light in the universe is not a galaxy. It's not a star. It's a microwave field of energy that comes to us in all directions. And lastly, we'll zoom out to the edges of our observable universe and contemplate, is there a physical edge? We'll compare with our greatest brains and supercomputers what they reveal—again,

not to prove a cosmologist right, but to prove every other model wrong. This is a zoom-through of an equivalent slice of the universe, but this one is in a box, is in a computer, one of our most advanced computers simulating all the ingredients of the universe. It's matter, it's light, it's energy, it's gravitational waves, it's age, it's future destiny, incorporated as scanning over billions of years. This is one of the most powerful tools that we have to assess the accuracy of our models.

1.88 Lastly, we'll conclude this lecture with the image that Wikipedia cited at the very beginning of this lecture, as the paradigm representative of all of science—cosmology. I love it. The question is, are all these different epochs of the universe, from the age of the development of galaxies and planets and universities and academies and the cosmic dark ages and the afterglow light pattern of the cosmic background radiation — did it begin with this epoch that Wikipedia said? So not everything Wikipedia has to say is true. Don't look at my Wikipedia page if you need more examples of that. And the question of where our universe came from, and is our universe alone? That's where we're going. Thanks for joining me on this tour. See you in the next lecture.

Lecture 2: Measuring the Universe

At a glance — Measuring the universe.

- 2.1** OK, so last lecture, I asked you to envision what the most favorite day on the calendar is for each and every one of you in person or watching at home. And for most of you, it was your birthday or some anniversary, some origin point. Now, I want to ask you to imagine you just woke up. You're on a new planet. Perhaps it's your first day ever existing. And I don't know how you got to have the comprehension to understand that today's your first day in the universe. But imagine what you would do that very first day, that very first hour, trying to survey where are you? Who are you? And most importantly, perhaps, what are you made of?
- 2.2** And so that's the cosmic ouroboros here, as I showed in the first day of lectures, that we perhaps are, at the first time in the universe, when we understand in the history of the universe, when we understand, using our conscious minds, that the universe might have come from nothing. And there may be other universes as well. Those universes produce energy and matter, matter that becomes living, matter that then, once living, becomes a conscious brain or mind. That brain or mind can comprehend the universe.
- 2.3** The universe can make predictions about the past and the future, and so on and so on, in an evolutionary process that continues to this very day. So imagine you woke up. Depending on when you woke up in the universe, where you were and perhaps who you are, you might have come upon a time several thousand years ago, the so-called classical period, after the Greeks had sort of really firmly established a universe with just a few moving things —the planets, the wanderers, as they called them, and the sun and the moon. And that was it. That was their universe. And they wanted to survey the universe to understand it.
- 2.4** Well, they didn't have Hubble Space Telescopes. They didn't have laser measuring micrometers. They had nothing of the sort. They had very simple tools. They had their eyes and their brains. And with that, they accomplished a remarkable amount of surveying and laid the groundwork for what will come next into the Renaissance and beyond. So without these Persian astronomers, these Persian surveyors, to understand, using the laws of—that they had already come upon the laws of trigonometry, of geometry, the laws of logic, formulated by Aristotle and others, they really, as I say, laid the groundwork or the star work for what would come later. They had simple tools. They had models. The astrolabe is basically a model. And they had surveying things like the sextant and other things, and they would make very accurate geometrical measurements of angles.
- 2.5** But let me remind you, this is about 300 years before the invention of the first accurate clock, so they weren't able to time things. Even Galileo failed to have access to any kind of timepiece that could measure accurate amounts of elapsed time. So it was very

difficult to do accurate physics, since most of physics is taking place within what's called space-time, if you have no way to measure half of what space-time is made of, namely the time component. So it was very challenging, and yet they did a tremendous amount.

- 2.6 Moving just before the Renaissance, the Renaissance is sort of an important period because it was when Galileo ushered in sort of the age of science and scientific reason, and that was accomplished using the telescope. These were the last astronomers to really only use their eyes, their brains, and very simple non-telescope tools. And they were very colorful characters. There are people like Kepler, Johannes Kepler, Tycho Brahe, and they built these great observatories. But unlike the Mount Palomar observatory, the Keck observatory, the LIGO gravitational-wave observatory, these are basically like resplendent castles. And so you see down here a picture of an astronomer using
- 2.7 this sort of really incredibly architected segment of a circle, a quadrant, and that quadrant, like this sloping wall here, was ruled with markings and indentations to allow the astronomer, as the building, sometimes itself, the whole building would rotate throughout the day, and they would make accurate measurements of the height above the horizon of various objects, including the sun, the moon, the stars, and the planets. And then the tools of the trade were then coupled to the most powerful tool in the known universe, our brain. And they were used to make models of the universe. And sometimes they got a little bit carried away in their models and their love affair with the mathematical precision of what mathematicians can do.
- 2.8 As I said in lecture one, physicists can't prove a darn thing. Mathematicians prove things all the time. In other words, you can prove that one plus one equals two everywhere and throughout the universe. Prove it beyond a reasonable doubt. But you can't prove that there are no purple unicorns that experience anti-gravity on the surface of Saturn's moon Enceladus. You cannot prove those kinds of things in science. You can say that's very unlikely, but you can't really prove it. So there's a time, and there's an impulse that, even to this day, astronomers, physicists, have, which is to try to find some way to make your model so beautiful that it becomes math. And Kepler did that by asserting that the planets moved in orbits that were fixed according to the regular geometric solids.
- 2.9 So geometric objects amenable to proofs and mathematics were very appealing to physicists, and they still are. We won't cover things like string theory. You may have heard of it. These are things that are purely mathematical abstractions with no connection to physical reality as far as we know right now. And yet physicists get overcome with emotion when you talk about them because they're pure. They seem to be like an object of mathematics that fell into the physical world. Well, so too did Kepler think that there were sort of shells of pyramids nested within cubes, nested

within spheres, nested within icosahedrons, et cetera, et cetera. And that was the reason that the planets had the orbital dynamics that he observed.

- 2.10** It actually came pretty close. But it's utter nonsense. There are no great crystalline cubes that hold the planet Mercury in its orbit or have some ratio between them. Nevertheless, Kepler's investigations, using the optical data acquired from the eye of Tycho Brahe and his compatriots, he was able to formulate a set of laws, of mathematical laws, that we use to this very day, that determine the properties of gravitationally bound planets. Just two planets, or a planet and a star. The laws of Kepler still hold to this very day. And in fact, a satellite named after him—Kepler—has discovered more than 5,000 exosolar planets, planets outside of our solar system.
- 2.11** A great deal of them have properties just like the Earth. The same size, the same shape, the same orbital distance from the sun. Some of them may have atmospheres. Some of them may have life forms on them. And so what Kepler did is rightfully heroic and is still used to this very day, although the reason that he was right is wrong. These planets are not held in this model, according to Kepler.
- 2.12** I showed you last time my picture of me gazing through a telescope at Galileo's final prison in Arcetri, Italy. There's an ancient woodcut of it, and there's me observing from a replica. In the Renaissance Enlightenment, the age of reason, the scientific age was ushered in by Galileo—I mentioned that last time—who was the first scientist to use hypotheses confronted with evidence collected by tools. He's really the first modern practitioner of the scientific method. Newton came along and invented a different type of telescope called a reflecting telescope. That's the kind that the Webb telescope is made of. So that uses lenses.
- 2.13** As I said earlier, we didn't really know, as we survey out from the Earth beyond the moon, that the Earth is in motion. And again, I laid that down as a homework problem, a challenge to you last time, and here's the answer. I issued a challenge for you to prove to me that the Earth isn't flat. If I was very stubborn and offered you a million of the Peterson Academy's money, which I'm not authorized to do, that could you prove it? Some could, some couldn't. A great deal fewer probably could prove that the Earth is in motion or explain how the Earth was proved to be in motion. And in fact, humanity as a whole didn't really prove that until the 1700s. So think of tens of thousands of years of human evolution, and only in the last 300 years or so have we proven that the Earth is in motion beyond a reasonable doubt. And that's done through what's called aberration, stellar aberration.
- 2.14** And we've done this throughout history. We've looked at laws that seem to apply on Earth and ask the question, "Do they extend throughout the entirety of the universe?" Fast forward another 300 years. We come to the, basically the modern era, where the question of whether or not and where we are within an expanding universe was first conjectured. And the characters there are the famous Einstein and Hubble, and we'll

encounter some others along the way as well. I call this period the birth of the Big Bang.

- 2.15** And now, if I had to claim what epoch are we in now? It's perhaps the epoch of the multiverse, of multiple universes. For just as Copernicus posited that the Earth is nothing but a planet orbiting around a star, and Bradley and others discovered that the sun is but one of many, perhaps hundreds of billions, of stars in the Milky Way Galaxy. That the Milky Way Galaxy, Hubble and others showed, as we'll see, is just one of a hundred billion galaxies in the universe —why not? Why not, indeed, posit our universe is just one of 100 billion or more universes? Can we confront that assertion with astronomical data? That's our job.
- 2.16** So the first thing that you would do when you wake up in the universe, you stretch out, you're all alone, and you want to survey where you are. Are you on a deserted island, or as they used to call galaxies, a deserted island universe? Astronomy is hard. Here are two—well, there's a lot of astronomical objects in this picture. Most of these dots are stars. This is a comet. This looks like a comet. But is it? How could you tell what it is? The problem with astronomy is there's no sense of depth. Things that look bright may be close to you, like the sun, but if you move the sun to a distance of where one of these stars is, it would be much, much dimmer, in fact, because it's a slightly dimmer-than-average star. So we need other pieces of information to determine that this smudge is a comet, and this smudge is the Andromeda galaxy, 2.5 million light years away from Earth.
- 2.17** How do you do that? So we need what's called a cosmic distance ladder, a ladder that extends firmly from where we are on Earth, using just the laws of geometry, extending that from the Earth to the moon, from the moon to the sun, from the sun to the nearest star, from those stars to clusters of stars in the galaxy as a whole, from our galaxy to other galaxies, and then to the observable edge of our universe, which may or may not be something like a true boundary in what's called the multiverse. We call this the cosmic distance ladder, not to scale, OK? We want to start with a rung that's planted firmly on the ground, extends all the way out to the universe as a whole.
- 2.18** There's a problem. We cannot use the same cosmic distance ladders, rungs at each step of our climb out into the cosmos. You need different techniques at different stages along the way, and each one is applicable over a finite region of the ladder. In other words, the ladder's rung that's at the very farthest away from us, to gauge the distance to the nearest, farthest source of light in the observable universe, which is called the cosmic microwave background radiation —that rung is totally useless down on Earth or in our solar system or even in our galaxy, because that is only capable of measuring things on the millions or billions of light year distance scales. Remember, the speed of light times the unit of time is a measure of distance.
- 2.19** So we can't even, and we certainly can't use a rung that's appropriate on Earth to measure things that are billions of light years away. So we have to see if there are

overlapping rungs where we trust the bedrock foundational base of the ladder, and we improve or increase our credulity in each step of the ladder as we go out. The very first step is with something called parallax. Parallax is a purely geometrical phenomenon. Parallax relies on the fact that if you make a measurement from two different vantage points of something that's at some distance away, it will appear to move with respect to things that are farther away than it.

- 2.20** So the very first rung on the cosmic distance ladder is utilizing the very fact that the speed of light is constant. It's very fast, but it's constant. So we can actually measure things in our solar system by bouncing or reflecting light off of them and then measuring that elapsed time between emission of the light pulse, radar pulse, energy from our source to the object, and then dividing by two, the time that it takes to get from point A to point B and back. So you can actually do that using radar, using a large telescope in reverse. Telescopes normally collect beams of light, or radio waves, but in this case, you can use it to transmit.
- 2.21** So you transmit a pulse of energy in the radar, in the radio wave. That's what the R in radar stands for. You transmit this black series of lines as a pulse of energy reflecting off an object, like a planet, planet Mercury, for example. Then you measure, how long did it take to shoot out the radar pulse and to receive it back? My colleagues have actually done this throughout the entire solar system. But in particular, they've measured the distance to the moon extremely accurately, because when the first Apollo astronauts went to the moon, they dropped little mirrors near the lander site. These mirrors have a peculiar property that, almost no matter where you look at them, if you shoot them with a laser beam, it'll exactly bounce the laser beam back and send it back to Earth for whoever shot it at it. It's like the reflector on a bicycle, the back of a bicycle. It reflects light almost perfectly back to where it was emitted from.
- 2.22** So my colleagues have done this with lasers, and they measured the position of the moon to 1 mm precision, so they can measure the average distance of the moon to within 1 mm using this technique. It's exquisite. So it's incredibly accurate. Again, we can use one rung of the ladder. We can't use 1 mm precision to measure the distance to a nearest galaxy, though. It's impractical. So that helps us define, if you were to, say, bounce energy off of the sun, which we could do, or we could measure all the planets and bounce energy off of them. We measured exquisitely the distance between the Earth and the sun. And there's a couple different time periods and dates on the calendar that are marked here.
- 2.23** But the orbit of the Earth is found in this fashion, and we can construct from it the base unit of the astronomical distance ladder, called the astronomical unit, which you may have thought is 93 million miles. That's a very close approximation, but again, everything in science is provisional. Copernicus thought the orbits of the planets were not around the Earth. He was right. He thought that they were circular. He was wrong. They're actually elliptical. That took Isaac Newton to determine that. Actually, Kepler

and Newton together—Kepler discovered the laws that the planets were in elliptical orbits, but not the reason for them. Newton provided the reason via what's called the law of universal gravitation.

2.24 So it's not exactly a perfect circle. So, sometimes in the calendar, we're closer to the sun than at other times. So here, perihelion is the closest distance that the Earth gets to the sun in its elliptical orbit. It occurs in January. Why is it weird that the closest the Earth gets to the sun is in the middle of winter? Is that weird? Wouldn't you think it would be in the middle of summer? Well, of course, in winter in the northern hemisphere, it's summer in the southern hemisphere, so you'd be half right. But of course, it doesn't have anything to do with that. The reason we feel summer and winter is not because of our distance to the sun, but because of the tilt of the Earth's axis and the amount of solar radiation that we get. We'll talk about that maybe in another course.

2.25 So this is the astronomical unit, and it's the first rung on the cosmic distance ladder. For the second rung, we're going to do a little demonstration. So I am going to be over here in front of this whiteboard, and I'm going to put some exquisitely drawn astronomical objects on here called stars. I'll make even different colors. Some will be black. These are going to represent stars in the Milky Way Galaxy, far beyond the Earth-sun distance, many, many astronomical units away. How far away they are, we don't know yet. These will be fixed stars in contradistinction to planets, which wander against the background of fixed stars. Now, those of you over there, because you paid extra, you are able to do this experiment. Unfortunately, you guys cannot because the line of sight doesn't allow you to.

2.26 Everybody here is equipped with two telescopes. As I said, your eyes, your eyes are at some known distance apart from each other. We could measure them if we so desired. What I want you to do is take your right thumb or your left thumb, it doesn't matter, and cover up my face beyond these fixed stars. Hold your hand out and cover me up and close your eye. Close one eye, whichever eye you like. Close that eye. Put the thumb of your right hand to cover my face. Now open your other eye. Do you see that I shifted a little bit back and forth? That determines two things. One, how far away we are from each other, but also which is your dominant eye.

2.27 So one of your eyes—for me, it's my right eye. If I open it up and I close, alternate between my left and right eye, I see the same thing if both eyes are open, as I see when I close my left eye. So my right eye is my dominant eye. When I'm shifting back and forth between them, that's because you're using one vantage point and another vantage point to triangulate where I am. You're making a tiny triangle. The two eyes in your head are two different telescopes. So one eye is like this, one eye is like this. And when you're looking at something in the foreground, you see a different angle. So here's my beautiful drawing of me as a stick figure, and you're going to see me at

different angles with respect to these fixed stars in the background. That's called parallax.

- 2.28** I'm going to erase this because this embarrasses even me, and I don't want this preserved for all posterity. That's called parallax, illustrated here. We don't do it with people. We don't do it with our eyes. What's the equivalent to the two different eyes in your eye sockets, in your skull? The different positions of the Earth in its orbit around the sun. By looking at distant stars, some are closer than others. Say there's one over here. This is not to scale. The nearest star is 4 light years away. That star makes some angle with respect to the background stars when you look at it in June, and it makes a different angle when you look at it in December. That's an exact one diameter of the Earth's orbit. That's the equivalent to the distance between your two eyes in the analogy of using your thumb.
- 2.29** If you measured your eye distance separation accurately, and you could measure the angle accurately, and your hand wasn't moving, and your thumb wasn't doing, jittering around, you could measure the distance to me. That's one way to measure the distance away. It wouldn't be as accurate as using a measuring tape, but it would give you some level of accuracy. This level of accuracy, even for the tools of the Renaissance, was enough to show that the stars are incredibly distant from us, that these angles get smaller and smaller the farther away the objects are. Something close will make a huge change when you close your eyes back and forth.
- 2.30** So when you do that, this is the second rung. Again, this diagram is not to scale. So one parsec—you may have heard of parsecs—one parsec is the distance at which a star has a parallax angle of one arc second. What is an arc second? An arc second is one-3600th of a degree. The symbol is, looks like a quote, equals one degree divided by 3600 seconds, which is the number of seconds in an hour. And this is because the Babylonians were fascinated with the number 60, and 60 came from their fascination with the number 12. And many people don't know this. When you count, we count in base 10. Why do we count in base 10? It's just a historical fact. We could count in base anything. Why do we use base 10? I'll give you 10 guesses. It's because we have 10 fingers.
- 2.31** The Babylonians, the Persians, they used base 12. Why? Why would you ever use base 12? Well, they're actually pretty smart. The bigger the base, the larger numbers you can represent. If you take out your fingers, you have 14 digits, 14 individual digits. Two are on your thumb. That's why it's not considered a finger. Again, this is not a biology class. My biology skills are incredibly limited. But you could count a different number for each one of your digits. So if they wanted to represent the number 4, they'd go like this. When they were counting, they didn't go 1, 2, 3, 4, like my kids do and I do, to be honest. They'll go like this. Oh, that's 4, because it's 3 digits here and 1 digit there. So they can count all the way up to 12, 4 times 3.

- 2.32** And they could do 24 with their second hand. And they were really good at this, so we're not so good at it. Anyway, if you multiply 4 by 5, I guess if you took all your hands and arms, now you're getting up there. So they were—they're the reason that we have 60 minutes in an hour, 60 seconds in a minute. Therefore, 3600 seconds in an hour. So an angle of one arc second, if you see the Earth and it's separated from the sun—this is a thought experiment—and it's separated by one arc second, you are one parsec away. That corresponds to an enormous distance, 3.26 light years. And guess what? If you're at 3.26 light years away, you're not even at the closest star to the Earth or the sun. That's Proxima Centauri. That's about 4 light years away.
- 2.33** So with this technique, we've been able to measure things like the distance to objects, clusters of stars. What was the name for the Pleiades in Japanese? Anybody drive here in that? Super. So why is it important to measure clusters of stars? Well, if you measure clusters of stars, you know one thing, just by looking at them. They're all at the same distance. You should get the same average distance to each one of the stars here. It allows you to do a cross-check, to validate. Your suspicion in physics should always be that you're wrong, not that you're right, that you're wrong. And you should go to great lengths to prove yourself wrong before somebody else does.
- 2.34** So this is how they measure distances. And this is the second rung of the distance ladder. Wait, there's more we'll encounter. The very first star to measure out with a ruler using the parallax rung of the distance ladder had a parallax angle of less than one arc second. Therefore, it was at a slightly greater distance to the Earth. This one is something like 10 light years away, or something like 12 light years away from the Earth. And it was measured in 1838, so barely 100 and what, 50 years ago? Now, we've built another set of satellites can do this much better than the human eye and a telescope on Earth can do. And in fact, this satellite is called Gaia, named after the goddess of the Earth. It's measured 1 billion stars in the Milky Way Galaxy and measured their distance to an accuracy of 1,000,000th of an arc second. Not one arcsecond, 1,000,000th of an arcsecond. It's incredibly precise.
- 2.35** And these data from this telescope are available for the public. And actually, scientists can't actually keep up with how much data is being produced. So people can actually get involved at this site here and actually volunteer to do what's called citizen science, do cataloging, do prospecting the data to examine it, and maybe make a scientific discovery. Unlike almost anything. It's very hard to do that in particle physics unless you happen to have a large hadron collider in your basement, which you definitely don't. You're not able to really measure the mass of a particle or detect a neutrino, but you can detect these—you can help scientists measure and constrain how far away objects are on the cosmic distance ladder.
- 2.36** Next thing we want to do is go farther out. We come to our galaxy, the sky satellite has measured exquisitely. One billion, which is about 1% of all the stars in our galaxy. Next, go out another rung, other galaxies. And here things get interesting. Here is depicted

the Milky Way Galaxy. There's a bunch of galaxies nearby to the Milky Way. Some of these are what are called satellite galaxies. These are galaxies kind of like celebrities have the paparazzi that follow them around. These are tiny little galaxies that are attracted to the Milky Way Galaxy, which it, in turn, is attracted to the Andromeda Galaxy, destined for a collision in about 10 billion years or so. Again, keep paying your taxes and your tuition at Peterson Academy.

- 2.37** These satellite galaxies are all tied together through the force of gravity, which we'll discuss in just a bit. But how do you measure the distance to these objects? Gaia, the most advanced parallax measuring instrument, cannot measure light from beyond the Milky Way satellite galaxies, called the Magellanic Clouds, which are not visible in the northern hemisphere, but they're exquisitely beautiful objects in the southern hemisphere. These objects are at a distance greater than the diameter of the Milky Way. So they're not a part of the Milky Way itself. As I said, they're stragglers, hangers-on, paparazzi around the Milky Way Galaxy.
- 2.38** Fortunately, Gaia can measure the distance to these objects, and then these objects have other objects in them that allow us to have a stepping up onto another rung of the cosmic distance ladder, and that's called the luminosity distance. Luminosity distance is obtained by measuring the amount of light called the wattage, or the luminosity, coming from an astronomical object and using that, and the decay of that light, with distance. So if we can get the lights turned down a little bit, will make this demonstration a little bit more vivid.
- 2.39** Each one of these candles here—it's not a real candle or else I wouldn't touch it like that—is a little light source with a known amount of luminosity. Luminosity is the amount of energy released per unit time. So these are equipped with some LED light that emits based on the battery that's inside of here. There's only so much energy, and it'll be released over the finite period of the battery's lifetime. And every second, it puts out so many, in this case, trillions of photons, and they're all in the visible light regime. Now, an object like the sun is producing a vast amount more. In fact, it's not a 100-watt light bulb. It's not a 1,000-watt light bulb. It's a 4 times a 40, 10 trillion, 10 trillion-watt light bulb. Don't get too close to it.
- 2.40** And it is not even the most luminous object in the universe. Now, the object puts out luminosity. Written on here is so many watts, one watt for each one of these candles. They're all the same intrinsic luminosity. You detect with your telescope and with your eyes, which are telescopes, a certain amount of light per second as well. That's called the flux. The flux is the amount of light detected or received per square meter of your telescope. Now, your iris and your pupil is only a few square millimeters, and yet it can allow in trillions of photons just the same. Now, here I've depicted it because the last thing an astronomer wants to do is tell someone to go outside and look with a telescope at the sun. Do not do that. You can look at a candle, but don't look at the sun ever.

- 2.41** The telescope has a fixed area of its lens and a finite distance from the object you're looking at. Now, over here, we have these three identical candles. Because of the diminution of light with distance, we can construct a second, oops, a second candle. We can construct a separate rung of the distance scale, called the luminosity distance. By knowing the intrinsic luminosity and measuring the received flux, we have a new ruler, a new tape measure, and that's provided by pure geometry. The candle, or star, emits a certain amount of light. That light is distributed over a given area. That area increases with distance.
- 2.42** As the radius of a sphere increases, its surface area increases as the square of the distance. That means, at a given distance, if you have a distance one unit away from the candle or from the star, the object's light is not changing. It's still emitting the same amount as its luminosity permits. But the amount of detected flux is going down. It's getting diluted over a given amount of area. One meter out here of square meter of area gets much, much less flux than 1 meter over here. Do we agree with that? And it actually decreases as the square of the distance. Hence, it's called the inverse square law. The flux received is equal to the luminosity produced by the candle or star, diminished by the distance squared to the object, divided by the distance squared.
- 2.43** So if you're looking from the back, the candle in the front is identical luminosity of the candle in the back. But it should appear something like 10 times dimmer, 3 squared, 9 times dimmer, than the one that's in the very front. And that law allows us to construct a new distance scale in the cosmos called the luminosity distance. Pure math. But you have to know one thing very accurately—the intrinsic luminosity. Stars do not come with their wattages labeled on them. Galaxies do not come with their wattages rated on them. There could be huge variations. Even if things look identical, the math is easy to do. And without error, we know pi to trillions of digits. Where else can an error be made? In the flux. You cannot measure with intrinsically perfect accuracy the flux that you detect. There will always be some uncertainty, and there are other effects that come in as well. But this, effectively, schematically, is how we measure the distances beyond the parallax regime on the third rung.
- 2.44** You do that by comparing to a known candle. Those known candles have a name. They're called standard candles—standard meaning, we understand them just like we understand this. Now, for some things in the universe, we have done this. We know if we put a laser on the moon, we know exactly how bright it was when we put it there, and we measure its distance to the Earth. Of course, we already knew that from bouncing radar pulses or laser pulses off of the moon. But what you do is, you do a combined two-rung measurement, first with parallax, which is pure geometry. There's some errors there in measuring the angle, but it's well-known. And you compare that. So the rung that's called parallax is much, much more firm than the one that's called the standard candle method, or the luminosity distance measurement, simply because stars don't come with their wattages listed on the bottom.

- 2.45 So now we're out two rungs. We still have to determine what makes for a good standard candle. So astronomers have been measuring these for over 100 years. There's a source, a type of star which is almost as good as a light bulb with its wattage written on it. It's called a Cepheid variable star. Cepheid because it's first discovered in the constellation Cepheus. Cepheus, everybody knows, is the water carrier in the ancient Greek mythology. I'm just kidding. I just, I mean, I have to look that up every time I mention it.
- 2.46 These are stars that have a peculiar property that makes them extremely interesting to astronomers. They don't actually have their luminosity written on them, but they pulse. They increase and decrease and blink almost, not completely on and off. They do so with a consistent periodicity, a consistent time period between maximum brightness and minimum brightness. Physically, what's happening is, the star is swelling and contracting. When it swells, it gets more or less bright, and when it contracts, the opposite occurs. This doesn't tell us its intrinsic luminosity, but it allows us to do something very, very important. It's very easy to count. Astronomers are very good at counting things. All you have to do is count how many days it'll take between two successive peaks or two successive troughs.
- 2.47 Then you can look up in a table how bright that star is, at its maximum and its minimum. It's as close to a standard candle that's not an actual candle that we have. And it's been measured for over 100 years, first measured by one of the greatest unsung heroes of astronomer, of astronomy. And her name was Henrietta Swan Leavitt. And she died, unfortunately, very, very young. But she was the first person to notice this correlation. What she plotted here is her original data. She's plotting effectively the period of a cycle of these stars and their intrinsic luminosity, how bright these stars are.
- 2.48 And she's done what all good scientists do. She's plotted different lines through the data to guide the eye, but really, that's constructing a model. She found what's called a correlation, 100% relationship between the period that the Cepheid blinks on and off at and how bright it is intrinsically. And this was done in over 100 years ago. This allowed astronomers for the first time to measure the distance to this first type of star, called the Cepheid variable. First one measured in the constellation Cepheus, which is in the Milky Way Galaxy, which means you can do what? What could you do with it? You could also measure it using what? The very most base level of the ladder is? Parallax.
- 2.49 The parallax method is purely mathematical, purely geometrical. She did that. She measured the distance, knowing parallax, which is just the separation between December and January, December and June, and the Earth's orbit. And she measured it and compared it with these plots, and she got the first accurate measurements. And this is how you do it. You measure the period. This is from May 4 to May 31, so good part of a month. You look up in the table that she constructed, and her colleagues over the last 100 years have constructed, and guess what? This one that you're seeing here

is not in the constellation Cepheus. It's not even in the Milky Way. It's nowhere near the Milky Way. It's in a galaxy called M100. That's millions of light years away.

2.50 So her technique allows us to rule the entire cosmos, to make a measurement out millions of light years away. As long as the galaxy has one of these Cepheid variables in it, you don't need parallax. You can't bounce radar off of it. But you measure its distance, exquisitely accurately. Some fun facts about supergiant variable stars known as Cepheids. So, effectively, all you need to know is that we have measured millions of these things, not just in the Milky Way, which we can use parallax to determine and calibrate the distance ladder, but in other galaxies. And as we'll see, these variable stars allow us to understand that the universe is not only expanding, it's accelerating and may indeed one day rip apart. But stay tuned. That's what they call a cliff-hanger. That'll be coming in upcoming lectures.

2.51 So she first did this in the small Magellanic Cloud, did Henrietta Leavitt, and this is a galaxy, a satellite galaxy, the kind of hangers-on galaxies and the entourage of the Milky Way. So there's a subtlety that we don't actually know the true intrinsic luminosity of even the Cepheids, but we know the relative luminosity of these Cepheids, such that if you measure one in one galaxy—so say you measure one, a Cepheid variable, in the large Magellanic Cloud, which is another satellite galaxy, and then you measure another one in the Andromeda Galaxy, you don't know either one's wattage written on it, but you do know that they will have the same period, say, that means intrinsically they're the same brightness. You don't know what that brightness is. But you know each one of these galaxies has a Cepheid variable that's equally bright.

2.52 Then you know they have the same luminosity. So they have the same wattage. You don't know what the wattage is. Will you measure the same flux? If this one is at the same distance as the large Magellanic Cloud, you'll measure the same flux, but they're not. I said that the Milky Way Galaxy is about 2.5 million light years away from the galaxy we call Andromeda. But it's only a couple hundred thousand light years away. A couple hundred thousand. Not a big number. It's a huge number. But relative to Andromeda, it's puny. Therefore, you can make the flux ratio. You can measure what flux do I detect in my eye or my telescope of the same wattage light bulb, if it's at the distance of M31, which is the Andromeda Galaxy, or the large Magellanic Cloud?

2.53 You find that they're different. You actually find that the flux ratio, it's 230 times dimmer in the Andromeda Galaxy than the large Magellanic Cloud. So which is farther away, Andromeda or large Magellanic Cloud? Andromeda is farther away. By how much farther is it away? The square root of this number, which is about 15. So 15 times farther away is Andromeda than the large Magellanic Cloud. This is enormous. These are enormous distances. In fact, we realized for measurements like this that the Milky Way Galaxy was a separate galaxy from the Andromeda Galaxy. Nowadays, we take that for granted. But they used to think it was inside of the Milky Way. In fact, they called Andromeda a spiral nebula, meaning a cloud of gas.

- 2.54** And it is reminiscent of the very first slide I showed you. The very first slide I showed you was of a comet. And I said, is this comet the same distance away as that smudge, Andromeda? In the 1600s, the analogous debate over whether or not these objects were in our galaxy or external to our galaxy was whether or not comets were in our atmosphere or in space. And, of course, now we take it for granted. Comets are dirty snowballs that orbit around, perhaps for decades, around the sun, come from something called the Oort cloud originally. And as they get close to the sun, they heat up and they emit this tail of light that we can see. And sometimes, as a harbinger, that's actually where the name disaster comes from, "dis" meaning bad, "aster" meaning star.
- 2.55** And so people see a star or a supernova or doing something like that, a comet, they would think bad things would happen. That was their scientific hypothesis. We proved that wrong throughout the course of the scientific age. But the analogous debate was whether or not comets were in the atmosphere or not, in the 1600s. Then it became our galaxies in the Milky Way. And now, what will the next question be? Is our universe inside of something else called the multiverse? Again, cliff-hanger for future lectures.
- 2.56** So how far out can you measure? What do we need to measure something in the cosmos? You need to have something you can measure, which can only be there if the universe has objects that can be measured. So that means that you can go as far out as you could possibly go, as long as there's something to measure. And then you might ask, well, what happens if you go beyond that? Can you measure something that's not inside of our universe? The answer, theoretically, is yes, as again we will come to later on in this course. So the structure of the universe. Is our universe finite? Is it infinite? Is it eternal? Or is it finite in age or in space? That's what we're coming to next.
- 2.57** So there's another experiment that everyone on Earth can do, any time of year, no matter what city, what place they are. And it's to think about and actually observe data relative and relevant to the following statement : "The night sky is dark." You might not believe it under these blinding lights here. But at night tonight, no matter where you are, except if you're at the South Pole, where I've been twice in my life, at this time of year, the night sky will not be dark because it only has one night and one day per year. But ignoring my friends at the South Pole right now, tonight, wherever you are watching this, it's going to be dark. Why is that surprising? Why is that even a paradox? Isn't that the definition of night?
- 2.58** This can only happen under very special circumstances. We take it for granted, but we'll see, in just a moment, we should be absolutely flabbergasted by this fact. So, in the 1500s, a philosopher, natural scientist, Thomas Digges, made a woodcut and made a scientific hypothesis that here we are centered on the sun. This is after the Copernican revolution. And considering that the sun was the center of the universe, these are different orbital distances. Here's Saturn. Jupiter is here. Mars is here. Earth is here. The moon is this orbit around the Earth. And you keep going out. I can't read Latin, but over here, he does say, "This orb of stars, fixed infinitely up, extendeth itself in altitude,

spherically, and therefore innumerable." And these are the fixed stars beyond this region.

- 2.59** So his model, which was very much the commonly accepted model, the Big Bang of its day, the theory of evolution of its day, maintained that the universe had an infinitely extending set of infinite stars. Now, in such a universe, what would you see? You wouldn't see this. You wouldn't see a dark night sky with just a few pinpricks of light. You would see the entire night sky would be filled with stars. And I'll give you an analogy to that here. If you go north or west to the Everglades or you go to a forest, imagine you're in the center of a forest and you're just looking around. Remember, we just woke up in this universe, and we're looking around. What do we see? Well, we see a tree. We see a trunk of a tree. We're just scanning back and forth. All the way around us, we see trees. OK, interesting.
- 2.60** Now, what if we're in an infinite forest with an infinite amount of trees? No matter where you look, you're going to see the trunk of a tree. It's like if I lectured in front with an infinite number of people, no matter where I looked, there'd be a person. Even if some are behind others, there's still people there. So again, what does this have to do with the night sky? Well, imagine—this is a picture of me at the famous Hayden Planetarium, where I gave a lecture a long time ago. So this was my dream as a little kid. I got to do it. And in the Planetarium, like I said earlier, it just identifies certain patterns that human beings have come up with. But there's only a finite number of stars. Because the human eye can only see about 5,000 stars out of the 100 billion stars in our galaxy. We certainly can only see one other galaxy, and that's the Milk—and that's the Andromeda Galaxy.
- 2.61** But if they're just finite number of stars glowing on the dome of a Planetarium, the night sky should be dark, as I showed in the picture of me with my back to New York City, when I was caught between the moon and New York City, as the song goes. But imagine an infinite number of stars that have existed for all time, shining their light, raining it down. How bright would the universe be at night, or the night sky be at night, I should say? The answer is infinitely bright.
- 2.62** So to calculate this, you can make a certain model. Imagine every cubic parsec. Remember, a parsec is a unit of length approximately 3.26 light years. So light traveling as fast as it does for 3.2 years will trace out 1 parsec of distance. A parsec is the distance you have to go to to see the 93-million-mile Earth-sun distance subtend one-3600th of a degree. So these are enormous distances. So imagine that there's so many stars per cubic parsec, 1 parsec wide, high, and deep. Now imagine that each star has a radius, on average, of capital R for the sun. The sun's radius is almost a million kilometers. Sun is enormous. A million kilometers. And then convert 1 million km, or 700,000 km, to parsecs. It's very small. It's only 2 times 10 to the minus 8th. So two 100,000,000th of a parsec. So it's in order to divide things.

- 2.63 Now imagine there's so many stars. This is a cross-section of a sphere, a spherical shell. And now imagine you go out some distance to this shell, and it has so many stars in it. What we're going to do is count up, we're going to build up the entire universe, making it out of little shells or, if you like, little peels of an orange and just keep adding on peels of orange until we go out an infinite distance. Do you see where I'm going with this?
- 2.64 There's going to be some finite number of stars in each spherical shell. Each one has some brightness, each one has some width, and we're going to keep going out from the Earth on outwards. The flux, as I said before, when we measured the third rung in the cosmic distance ladder, the flux is what we observe. It's the luminosity of that star divided by the surface area of that spherical shell. The total flux is going to be how many stars per shell times the flux per shell. The total flux is going to be this semi-complicated-looking algebraic equation. And the total flux that we're going to get to is the thickness times the number density—how many stars per cubic megaparsec—times the luminosity of each star. Now we're going to multiply that by the number of stars.
- 2.65 So a single star has the luminosity, L . This is how many stars per cubic parsec and this is the flux, the thickness of the shell, and this is the distance to that shell. So T is this distance, R is the distance from the center to the radius. The total flux is going to be obtained by summing up all these shells, starting at your feet and going out to infinity. You see what's going to happen. We've assumed an infinite number of shells, if you have an infinite number of stars in an "infinitely extendeth universe," as Digges said, so any number multiplied by infinity is infinity. So therefore we've proven that the night sky should not be dark. Actually, the daytime sky should be as bright as the nighttime. There should never be a night, is a better way to phrase Olbers' paradox. It should never get dark at night because no matter where you look, there's a star.
- 2.66 And we assume those stars are just like the sun. And imagine if everywhere you look was as bright as the sun. You think it's bright in here. Imagine if the whole sky was glowing the same brightness as the sun at all times of day. There'd never be a night. We only have a night because there's one source of sunlight and it's pretty far away. And when the Earth rotates enough, we're on the occluded side, the shadow of the Earth between the sun and us. So this is crazy, right? This is why it's a paradox. It should be infinitely bright. We observe that it's not infinitely bright. Therefore, something we've assumed must be flawed.
- 2.67 So here's just a simple, well, it's not simple, but it's just a statement that if you had 1 quadrillion, which is 10^{15} shells, which is a far cry from infinity, right? Infinity is a very big number, especially near the end, that would cover, and each one only covered a quadrillionth of the sky—so a tiny, tiny bit amount of stars—you would still have the sky be completely covered with sunlight, so the entire night sky should be as bright as the sun. So something's going wrong here. That's what makes it a paradox.
- 2.68 You may have heard of Zeno's paradox. Has anyone heard of Zeno's paradox? Zeno's paradox is the statement that, for me to get from where I am to the blackboard or to

this whiteboard, first to get there, I have to cover half the distance. Do we agree? To get from here to there, I also have to cover half the remaining distance, and then I have to cover half the remaining distance. Half the remaining distance. So Zeno's paradox is that you'll never reach your destination because the sum will always take you infinitely close, perhaps, but never to converge, to actually get to your destination. That's obviously a paradox, too. We go throughout space all the time, so that is solved in a different way.

2.69 But paradoxes are based upon some flawed assumption. The key that we have to do as cosmologists is ascertain which assumption is wrong. Are there more that are wrong than just one? And how do we improve upon this to get an accurate estimate of how bright the night sky should actually be? And that will actually lead us into a notion of an expanding universe. That woodcut that I showed assumed a couple of things. There's an infinite number of stars. The universe is infinitely large. The universe is infinitely old. Maybe you can think of other assumptions. What are some other ways that you could wiggle your way out of Olbers' paradox? Which other ideas could do it?

2.70 One that I frequently get is, so what if there's an infinite number of stars? What if one's behind another one? You won't see it, right? Does that solve Olbers' paradox? Yes or no? Would the night sky be dark in a universe where some of the stars are obstructed or eclipsed by stars in front of them? No. You'd see the one in front, like these penguins here. Everywhere you look still ends in a star. So maybe there's something in the way. There's something to dim it. What could cause the diminution of the light to effectively block it out, as you said? So there could be something. In fact, you're right. There is something that dims the light between us and any star in the universe.

2.71 It's called interstellar dust. It's actually the villain of my first book, Losing the Nobel Prize, in a way I'll explain in the very last lecture. Dust does do exactly what you said. Dust absorbs light. It changes the light and endows the light with certain other properties that we'll explain later. But there's a problem, and that problem is known as the third law of thermodynamics, which is that, when you have something that absorbs light, it heats up. And if that material, like dust or some other magic fairy dust or whatever, as long as it absorbs light, by the laws of thermodynamics, it will start to warm up. How much will it warm up to? Exactly the temperature of the star behind it that's illuminating it. So, eventually, it will come to glow with the fiery intensity of the star itself, if there's an infinite number of stars.

2.72 So it's a good guess. It can explain some potential wiggles out of the Olbers' paradox, but it actually does not fully explain it because it would cause this heating effect. So it was a very good guess. But as I say here, it will come to what's called thermal equilibrium. The heat source will cause the dust to glow with the same intensity as the star. Now, another explanation. What if the universe isn't infinitely large? Who thinks that would solve Olbers' paradox? If the universe isn't infinitely large, would the night

sky be dark? Who thinks yes, raise your hand, please. Who thinks no, raise your hand. Who doesn't like to raise their hand in public again?

2.73 Well, could partially explain why we don't see it, but the surface brightness doesn't actually depend on the distance, as long as there's an infinite number of stars. Now, you might properly say, how could you have an infinite number of stars in a finite universe? And, in fact, you'd be right. You can't have an infinite number of stars in a finite—you can't have an infinite number of ice cubes in a finite universe, right? Because they occupy some space. But we actually showed a couple slides back with this 10, one-quadrillionth number, that you actually don't need a fully infinite number of stars. So actually, you can have a finite number of stars in a finite-sized universe, and you could still have this whole sky glowing with the intensity of the sun.

2.74 Next, we assumed that the universe was eternal. It existed forever, as Dobbs said. We also have assumed that the universe is static. It's not changing. It will turn out that those explanations may resolve Olbers' paradox, that the universe may not be infinitely old. In fact, spoiler alert, it isn't infinitely old. And, spoiler alert, it's expanding. Whether it'll expand forever, and whether it will live forever, exist forever, are going to be subjects of our upcoming lectures. And that is a cliff-hanger for lecture number three. Thank you very much.

Lecture 3: Expansion and Origin

At a glance — Expansion and the origin of the cosmos.

- 3.1 When last we left off, we were hanging by our fingernails on the edge of a cliff, trying to understand how or why we might think that the night sky should be dark. And we came up with some very startling conclusions that led us to think that maybe, just maybe, the assumptions that went into our understanding of how big the universe is, or how old the universe is, or how static or not the universe is, might be completely wrong. And that's the subject of today's lecture.
- 3.2 Lecture 3 concerns the evidence that we have that points us to an unavoidable conclusion, that the universe is not static, that it is, in fact, changing. And the way that it's changing is that it's getting bigger, it's getting larger with time. And the conclusion that we'll draw from that is not just about what will happen in the future, but what happened in the very, very, very distant past, namely at perhaps what may be the origin of time itself. So, the expanding universe, lecture 3.
- 3.3 Recall that Olbers suggested that it might be paradoxical that the universe is, in fact, ever dark, especially at night, when it appears as if the universe is punctuated by tiny little pinpricks of light and nothing more. And we said that might indeed strike us as paradoxical when we come to the conclusion that the universe cannot be filled with an infinite number of stars that have existed for an infinite number of time, amount of time.
- 3.4 But we need more data. As we mentioned, there were so few things in the universe that actually changed that the Greeks came up with a name for those things. They called them planets. The implication is, except for those 5 things that they could observe moving in the heavens, the rest of the universe was stationary, static, unchanging with time. Things would change with the seasons as the sun orbited around the Earth, or so they thought. But the fixed stars never changed their appearance or their properties. Now, we take it for granted that that's not true. But how did we come to know that, in fact, that was a grave mistake?
- 3.5 We have to go back to Copernicus himself. The Copernican principle became known as a way to think about our place in the universe. I call it the ultimate big brother complex. Any of us with big brothers—I'm also a big brother, but I have a big brother—knows that your big brother will always tell you that you're not that special. And indeed, that's a fairly accurate interpretation of the Copernican principle. There's nothing special about us—not just we're all individuals, and that's incredible. Although for my favorite movie, *The Life of Brian*, when they all chant that we're all individuals, I love that segment. We're all, of course, unique and infinitely precious.
- 3.6 But from the perspective of physics, we're all the same. Every planet is the same, every star is the same. No place can claim to be the center of the universe, no matter how big

our cosmic egos are. So it really was the pinprick that deflated our cosmic egos. And cosmologists later formalized this, to say that the universe not only didn't have a center, that that notion is outdated, that there couldn't also be perhaps a center in time. In other words, there might not be anything special about our current epoch in the universe. And we'll explain why that's important later.

- 3.7** We describe these as their mathematical properties. Their mathematical properties can be then treated as mathematical foundations on which to build a predictive set of equations that will allow us to make speculations about the extent of the universe and its properties. The two most important of these properties are so-called symmetry principles. They are the expressions that, just like the fact that we are not special in the grand cosmic tapestry, that the universe is invariant does not change, is symmetric in both time and space in the following sense. That if you do an experiment, if you drop an object, it will fall the same way on a Monday as it will on a Tuesday. It's independent of time.
- 3.8** It will fall in a certain way, no matter what that object is made of. It'll not depend on where you are in the universe. If it's done in Florida or California or anywhere else, the experimental properties, the speed of light, the force of gravity, these will not change. They may be local variations, but the overall laws of nature are symmetric to displacement, to change in position, and change in time. We call those properties symmetry properties. You are familiar with some of those. And the more symmetric something is, the easier it is to describe. A sphere looks the same no matter which angle you look at it. A cube does not. There are different ways, although a cube is more symmetric than some completely irregular object.
- 3.9** We talk about the universe as possessing two properties. One is called homogeneity, which means the properties of the universe don't vary from point to point. They are homogeneous, like homogenized milk is milk that's been thoroughly blended and mixed. And then there's another property that suggests that the universe should look the same, no matter which direction you look, albeit there'll be slight variations. That property is called isotropy. To my knowledge, there is no isotropic milk. Maybe that's a business idea that I can pursue.
- 3.10** Here's two examples. On the left, the pattern is sort of looking symmetric, in that if you shift in space, you get roughly the same pattern no matter where you are, as long as you shift by an integer multiple of the characteristic size scale that you have there. So this pattern is not, is not homogeneous on small scales, but you imagine if you went really far away, it would start to blend together, maybe be like a gray square. This, on the right, this bulls-eye pattern, it is isotropic at the center. No matter where you look, it will appear the same. So that's the property of isotropy. And on the left is inhomogeneity.
- 3.11** But both of these patterns, if you went really far away, they'd all, both look relatively the same, a grayish sort of square blend between the two. And so too with the cosmos,

although it may not be particularly obvious because we're here on Earth. And if you go a couple of dozen miles above the Earth's surface, there's no Earth, and you start to get in space above about 60 miles. And so things look very inhomogeneous, but we're talking about scales much, much bigger than the Earth, much bigger than the solar system, much bigger than our galaxy, even, and on sufficiently large scales of order, 100 million light years. How vast the universe is gives you a very slight inkling into the tiniest size scale fluctuations that we see really don't matter on cosmic scales.

- 3.12** And it's like the air in this room. This room is composed of air molecules, and there are space between the air molecules. And we wouldn't say that the air in this room is completely homogeneous, because if you zoom down small enough, it wouldn't be homogeneous at all, and you'd have to look at the individual molecular dynamics. But on a large enough scale, like a human scale, the properties on microscopic scales average out. We can talk about the average pressure or temperature or density of the gas. So the analogy is for air molecules, you'd have to be many, many millions of times bigger than the smallest possible size scale in the physical problem you're describing.
- 3.13** For the cosmos at large, we need to be on hundreds of millions of light years before we can average over the properties of the universe and say it looks homogeneous and isotropic. Here's a plot of galaxies, and it's a space-time diagram, in that at the very center of the apex, between these two wedges, these are slices through the sky, looking at galaxies and measuring their distance. Each blue speck is a galaxy and, written down proportional here, is the length of distance they are away. It goes from zero light years out to 5 billion light years away. And again, they're small variations, but on a big enough size scale, the properties, the density of these things, when averaged according to an accurate prescription, it is isotropic and homogeneous only on scales of about 1/10th or so of one of these ticks.
- 3.14** So on scales of about 100 million light years, or 300 million light years, 300 megaparsecs, the universe is isotropic and homogeneous. And just in this plot, there's over 100,000 galaxies. And this is a tiny slice through a tiny 2-degree-wide wedge of the universe. And you can imagine how many galaxies there must be if you observed over the entirety of the universe, of the entirety of the sphere of sky that surrounds us. There's more evidence in different wavelength regimes. In other words, not just looking at visible light, you can look at infrared light, count up the light from stars and galaxies.
- 3.15** You also see, again, variations on small size scales and then much, much smaller amplitude variations on larger size scales, such that if you put yourself in the middle, if you went billion light years away from Earth, and you looked around and you asked the question, "What's the average density? What does the pattern look like?", it won't look identical to what we see on Earth, but the properties, the statistical properties, will be identical. The average number of galaxies per 1000 megaparsecs will be the same, no matter where you are in the universe. That's a manifestation of isotropy and

homogeneity, together with the Copernican principle, that the laws of physics don't depend on who you are, where you are, what you are.

- 3.16** That comprises what's called the cosmological principle. The laws of physics are invariant throughout the entirety of the universe. If they weren't, if they only applied on Earth, that would make our jobs much, much harder. So the universality of the facts of physics lead us to believe that we can do quantitative measurements of both the past of the universe and its future. Going back as far as you can using light, we'll come to in the next lecture what's called the cosmic microwave background radiation. This is not an image of the sky as seen looking at stars or galaxies. This is the average amount of radiation, in the form of microwave energy, which is shortwave length radio waves, that we see looking around.
- 3.17** And actually, again, there's tiny variations here, but the average amount of red spots or blue spots that you see per square inch on the screen here, or per square degree on the sky, are exactly equal over the entirety of the sky. And this will come into play later on as we describe how we know about the very, very distant past of our universe's history. So we see no evidence for violations of this. We say the universe is isotropic and homogeneous. Putting those together, we come up with the cosmological principle. Any observer, no matter who you are, where you are, what you're made of, should see the exact same properties. In general, specifically, you'll see different aspects of it.
- 3.18** So the question is, what does this have to do with the expansion of the universe? And how can the expansion of the universe be compatible with a universe that is the same, no matter when or where you look at it from? And that will come into play as we discuss not only the mathematical and physical laws of the universe, but also some of the philosophical principles. How do we know what is true? What are the tools that we use to seek out what is true? This is very important in cosmology, as I say, because there is only one universe. We can't do proper experiments and ask, "What would the universe look like if this galaxy were moved over here, or if we changed all the stars in this galaxy to an anti-matter galaxy instead of a matter galaxy? That is not possible to do.
- 3.19** Luckily, we have statistical quantities, multiple quantities of stars, of galaxies, but there's only one universe. At least, we think that there's only one universe. We'll see evidence, perhaps, of how, not that we may have detected other universes, but how we could detect other universes. And that will be encountered when we discuss the philosophy of cosmology.
- 3.20** Now, we can only see stuff if there's stuff to be seen. We said that the objects that we looked at to develop the cosmic distance ladder were stars, and that's true. And that's really, basically, the only time we're going to really encounter stars in this class. So, unfortunately for those of you that wanted to perhaps learn more about stars, we're actually going to treat the fundamental objects in this course—they're galaxies, which

are giant compendiums, conglomerations, of stars. But the stars themselves are relatively insignificant to us.

3.21 Now, up until the 1920s, it wasn't known that these objects were actually outside of the Milky Way. The Milky Way was thought to be the entire universe, and these objects were called spiral nebulae—clouds, nebulae, of gas and dust, perhaps. And in truth, they are. There's gas and dust in these objects and they are spiral shapes. That's real, but they're not in the Milky Way. They're external to it. So this was a huge, huge, contentious battle, back 100 years ago exactly. Are these swirling nebulae basically the exact same composition, properties, and distance as these objects? This is the great Orion nebula, one of the most beautiful and striking examples of nebula.

3.22 And it even has sort of an appearance. There's some nebulosity here a little bit. Maybe you could convince yourself there's some swirling going on here. This is purely gas and dust, with a few stars. And the question is, is this thing—this is the Whirlpool Galaxy—not far away and easily seen through a modest telescope? Or is that basically the same, or is it an entire "universe," quote, unquote, just like the Milky Way? That was the debate. And in fact, it was so contentious that the National Academy of Sciences convened a great debate in 1920, the presidential election year. Imagine, it was very contentious, and they convened a debate that lasted for a long time, and it actually ended in a draw. They could not end up determining what was the answer. Were these objects in our galaxy or external to our galaxy?

3.23 The debate ended in a draw. And it wasn't until Henrietta Leavitt and her work with the work of Edwin Hubble, who we'll come to in just a bit, came about just a mere 3 years later, in 1923, that the debate was finally resolved, that those spiral nebulae were island universes, galaxies unto themselves. Now, why is that hard? As I said, we don't have any depth perception. A comet looks like it's in the atmosphere, like a meteorite is, a meteor is. So a meteor in the atmosphere has some of the same features. A shooting star looks vaguely similar to a comet. Of course, one is great distance farther away.

3.24 The problem is, it's very hard to do distance measurements, as we learned when we studied the cosmic distance ladder. To an object of a given luminosity, a given wattage, if you like, the Cepheid variables that we encountered in lecture number two, you cannot tell their intrinsic luminosity just by looking at them. It may be that a dim Cepheid variable is truly farther away than a brighter one, or it may be that it's intrinsically dimmer and it's just as close as the so-called brighter ones. So the work of Edwin Hubble was quite phenomenal. This was in 1923, and the Mount Wilson Telescope, 100-inch telescope on Mount Wilson in Los Angeles, which you can go and visit to this very day—and he measured a distance of about 3 million light years using the Cepheid variable technique pioneered by Henrietta Swan Leavitt.

3.25 This, then, was beyond the farthest extent of the Milky Way Galaxy as measured at that time. The Milky Way's diameter was measured to be about 100,000 light years from

edge to edge. So it has this bigly spiral shape to it. I'm not going to embarrass myself too much by drawing it. And it's about 10 to the 5th light year, 10 to the 5th light years, across. The Andromeda Galaxy, on the other hand, over here, is about 30 times farther away. So this distance here was measured to be 30 x the distance between the farthest stars in the Milky Way from the Earth, and its diameter, 30 times the diameter of the Milky Way Galaxy. So it could not be in our galaxy.

3.26 They both had Cepheid variables. There was one over there, and there's many, many in the Milky Way. And so you could use the standard candle technique, rung two, measuring parallax to a Cepheid variable inside the Milky Way, and then a Cepheid variable inside of the Andromeda Galaxy. And that told you the ratio, 30 times the distance of the diameter. It was 30 times the distance of the diameter of the Milky Way. So it's about 3 million light years away, not in the Milky Way, external to it. Settled the great debate. These objects are island universes, just like the Milky Way.

3.27 Now, they noticed something else at that time. I told you that cosmologists have fabulous hair, not surprising because of the prefix "cosmos," which makes beautiful, cosmetology. But they also have cool names, like Henrietta Swan Leavitt, and this gentleman here, Vesto Slipher. I don't know anybody whose name's Vesto, but it's a cool name. We considered it for one of my kid's names, but old Vesto did phenomenal work, very painstaking work. Between 1912 and the great debate in 1920, he measured about two dozen galaxies, which weren't known to be galaxies. They were just nebulae. And he noticed a very peculiar observation.

3.28 He noticed the colors of the galaxy, on average, were distributed in an interesting way, that most of the galaxies that he saw were redder than the Milky Way Galaxy, and only a handful were bluer. And that's weird. First of all, how do you measure the color of a galaxy? Well, the way that you measure the color of the galaxy is by using its spectrum. The spectrum is basically the rainbow of light of an object. Now, for this object, this is a prism. The prism disperses light and bends it to a different angle physically, which then can be projected some distance away on a wall, and that comprises what's called the spectrum, or the spectral graph of that object.

3.29 Here's a visible wavelength spectrum. White light comes in. White light is comprised of all the colors of the rainbow. That's how the rainbow gets its name. And it has an average length, wavelength between peaks of the waves, of about 550 billionths of a meter. Nanometer is a billionth of a meter, symbol NM. The shortest wavelength that the human eye can see is about 400 nm. Shorter wavelengths than that are called ultraviolet light, and it can be very damaging, as you know, to your skin, to your eyes. And wavelengths longer than 700 nm beyond the red spectrum are called infrared, infra meaning beyond the red.

3.30 And these are very important wavelengths in astronomy. One very cool fact about a prism is that you could take two prisms. If you were to take this prism with white light coming in, and then you were to take another prism and turn it upside down and put it

in this beam of colored rainbow light coming out, what do you think would come out on the other side? Any guesses? White light. That's pretty amazing. And Isaac Newton, who was no slouch —he did an interesting experiment. He took a prism and another prism, oriented just like I said. White light coming in. White light comes out.

- 3.31** But then he said, let me block off one of these colors. And he put an obstruction in that blocked off the yellow light, and the light that came out was missing yellow light. He invented and discovered color theory. And an incredible gentleman by the name of William Herschel, who worked with his sister Caroline, was the first optical astronomer to use telescopes in history. She was an amazing scientist herself. The Herschels took something and did an amazing experiment. They took a thermometer, and they moved the thermometer, and they put the thermometer not here. They put a thermometer here, in the infrared portion of the spectrum. And what do you think happened to the thermometer's mercury? It rose.
- 3.32** They discovered that heat is what we call the byproduct of wavelengths of light that the human eye can't see. And you're all familiar with that. If you go outside here in Miami, you can close your eyes, and you can find the sun just with your hand, which doesn't have any retina cells in it. It's a detector of heat. Your hand is a crude thermostat, and it's a crude telescope, right? It's not great, but you could use it to reconstruct where the sun is. That's because infrared radiation is part of the electromagnetic spectrum. And, in fact, it extends to all directions, infinity above and infinity below, with the higher wavelength going to the top and that going to the microwave and radio waves that we also can't see. And then beyond the gamma rays, going down into what are called x rays and gamma rays beyond the ultraviolet, rather.
- 3.33** Now, this spectrum of the sun, this actually is not the spectrum of the sun. This is a perfect rainbow. So you have pure white light. The sun is not pure white. You can tell it looks slightly yellow. It's actually more greenish than yellow. And our eyes have adapted to see it more yellow. But actually, if you look at the sun, you don't see a perfect rainbow. You actually see interruptions, absorption by different molecular transitions. Because hydrogen, the simplest element, the simplest atom, is made up of a proton, P, and an electron, E.
- 3.34** The electron is 1800 times less massive. And the electron in the Bohr model is orbiting around in a circular orbit. Of course, it doesn't actually do that. It's not like a little planet. But in the model of Niels Bohr, that was one convenient way to realize it. And then the electron can actually orbit at different quantized distances. And it does so by what's called the principal quantum number. So in its lowest energy state, it's called N equals 1. It can go out to N equals 2, and it could go all the way out to N equals infinity. Depending on how much energy is being shot at this hydrogen atom, the electron will absorb energy or release energy, and emit or absorb specific wavelengths. When it absorbs light, you see a gap in the rainbow spectrum that would otherwise be continuous.

- 3.35** This is what is called a chemical fingerprint. Just by looking at this, a spectroscopist could tell you, you're looking at hydrogen. It has a potentially infinite number, but the ratio, the distances between these different transitions, are related via a rule of quantum mechanics called this rule invented by Niels Bohr. So elements of different numbers of protons, which determines the chemical element number, are classified according to their absorption features. And once you have an absorption feature, you can do something incredible that was really first realized—this is a modern spectrum. So all these colors dispersed.
- 3.36** And you may wonder, well, do you use a prism when you look at the sun? Again, don't look at the sun. Professional astronomer here. Don't look at the sun with your naked eye. Only use a very, very powerful filter to do that, and a telescope, not your eyes. But the cool way that you can do this, that you can actually observe this, was actually discovered by this man. His name was Christian Doppler. And
- 3.37** what was discovered by Christian Doppler is familiar to anyone who's ever heard an ambulance. Hear an ambulance driving, its siren is at a certain pitch. As it comes to you, towards you, pitch gets higher. As it goes away, it gets lower. The higher the frequency is called a blue shift, blue meaning shorter wavelength, higher pitch, higher frequency. Red meaning lower. So here's an ambulance coming towards you in this direction. People over here will see a compressed series of peaks of waves that will correspond to shorter and shorter wavelengths. Those over there will hear a reduced frequency or pitch. Therefore they'll call it a redshift.
- 3.38** Now, to get the lights of an ambulance to change by this amount, you would have to be moving close to the speed of light, so you won't see it in ordinary circumstances. But in our universe, things do move at the speed of light or close to it, especially light. But even a star can move at an appreciable enough fraction of the speed of light that you will see a shift in the chemical fingerprint in the spectrum of a star. It will shift to the red or to the blue, or not at all, depending on its relative motion with respect to you. So here's a diagram. A star is moving. We're here, orbiting around the sun. This star moving towards us. We'll see a blue shift, higher frequencies. The characteristic chemical absorption features will be shifted to the blue.
- 3.39** This star moving away from us will be redshifted. It will have longer wavelength features than the one that's coming on the left towards you. And then one that's moving perpendicular to us, you won't see any shift the light that gets to us comes from the top to the sun, that's not shifted to higher or lower frequency. There'll be no redshift at all. So why is this surprising? What does this have to do with good old Vesto?
- 3.40** Well, what Vesto realized is that, every single galaxy he could see, they were all redshifted. Now, that's weird, right? That would be like, you're in a city and there are all these ambulances, and there's thousands of ambulances, and all of them are being redshifted. All their sirens are —what would that tell you? What would you assume, if you heard all these ambulances are having their siren sound extended to the red? What

would you assume, perhaps? That they're moving away from you. Maybe you were just at the scene of a great car crash or something. All these ambulances are leaving the scene, taking people to different hospitals. Exactly.

3.41 In the context of cosmology, there's no explanatory nature such as an ambulance or an accident. There had to be some reason why Vesto observed these things all moving away from each other. It's a great puzzle. So this is it depicted using the spectral technique. So, in the rest frame, in a glass of water, say, if you were to extract the hydrogen from this and you were to excite it just such that it would exhibit these different orbital behaviors of its electron, it would exhibit these lines here and these lines there.

3.42 If I took the glass of hydrogen, moved it away very rapidly, a good fraction of the speed of light, it would be shifted to the right or to the red. Redshifted. If I instead moved it towards you, it would get blue shifted, going shorter and shorter features. So you have a laboratory frame here, and you physically measure the amount of distance on a spectrometer, or spectrogram, of how much things have been shifted. The amount of physical shift in percentage is given by the velocity of the source divided by the speed of light. It's very similar in the Doppler shift of sound. But an ambulance can move approximately a much higher fraction of the speed of sound than any laboratory experiment can move as a fraction of the speed of light. So this equation tells you the redshift, symbolized by the symbol Z , Z equals the velocity divided by the speed of light.

3.43 And it's equal to a very simple measurement, the wavelength minus the wavelength you'd see in the laboratory at rest, divided by that rest wavelength. So it's a percentage. There's no units. And V divided by C is velocity divided by velocity. So all you need is an object that emits light, and you need a spectrograph. A spectrograph is a telescope connected to something like a prism that can disperse the colors of light. So this is a prism. This will have some properties, and we could all do this. This glass of vodka? No, this is just water, but it has certain prism-like properties.

3.44 But instead of that, astronomers and physicists have come up with a much more accurate way of measuring wavelength, and it's called a diffraction grating. I provided you all with diffraction gratings, so if you could take them out right now and we could dim the lights, we will play around with these and see what kinds of properties we can observe from the various light sources around us. These can be obtained for a few pennies or dollar or so, for a pack of 10 of these. And you can find sources for these online if you're watching at home. And printed on it, again, do not look directly at the sun. So if I can have one of the candles out here, we'll play around with that, and we'll look at some of the sources of light.

3.45 This source is very, very close to being white. I'll pass around this candle source and you'll see almost a perfect rainbow from it. So I'll pass this around. You guys can all look at it, and it'll pass another one around there. So take out your diffraction grating and tell me if you see a perfect rainbow from it. Almost perfectly continuous. You can't

really see these dark absorption lines here. So this is going to glow in just a bit. You'll take out your diffraction. This is going to glow mostly blue. This is a butane torch, so it's going to glow and it's mostly going to be blue. It's a very, quite hot flame, but it's also incandescent. It's not going to give off a single wavelength of light. The only things that give off single wavelengths of light are things like lasers. Lasers are highly monochromatic, we call it. If you look at this, this will basically just stay blue. You won't see a rainbow really very much at all.

- 3.46** OK, so this is very singular in color, but it's not quite perfectly singular, as you might imagine. And then the candle is also putting out pretty much continuous light. And these lights above us are also fairly continuous, although you can see, in certain cases, you'll see some very particular vertical, pronounced vertical lines. OK, if we can have the lights back up again.
- 3.47** So how can we explain the observation that every single galaxy we seem to see in the universe, with literally a handful of examples, exceptions, out of 100 billion galaxies, that all of them are being redshifted and almost none of them are being blue shifted? Well, here's a visualization made by one of my colleagues, Richard Behiel—an animation of an expanding universe. Imagine the galaxies in the universe are these little dots. As the universe expands, the light that they're emitting gets stretched to longer and longer wavelengths, just like the stretching of the sound waves in the example of the ambulance siren. The more distant the galaxy is, the more time has elapsed since it emitted its light, the more stretching of its wavelength has taken place.
- 3.48** So this was an example of a hypothesis that we can then compare to other pieces of evidence, namely whether or not the universe is actually expanding, or there could be some other explanation for the observation that seems to be quite puzzling, that all these galaxies, measured by Slipher and since, are redshifted, and almost none of them are blue shifted.
- 3.49** Another analogy that's frequently used is the expanding universe likened to the surface of a balloon, where you stick on little stickers representing galaxies, and you blow up the balloon and it expands. The separation between the galaxies increases, and any light on the galaxy's light trajectory from a galaxy to us will be stretched to the red, redshifted. That would explain it. Now, there's some problems with this analogy. Can anyone think of some problems with this analogy? I'm using it to explain the peculiar observation that the universe is expanding, not static, is dynamic and not forever unchanging.
- 3.50** Yeah, so a balloon has a center. So this is a flaw with this. There's also a finite amount of surface area. It's helpful. But what we have to conceive of, which makes this so difficult and does make analogies such as this helpful, is that we'll come to encounter space and time, and meld them together in what's called space-time. Space has three different dimensions you can move about in. Left-right, forward-backwards, up and down. Time is one-dimensional, and we don't believe that we can control how we

move through time. Lord knows, I would change some things about the past and maybe about the future, but we seem to be free to move about space, but not time.

- 3.51** So it's peculiar. We want to blend them together, but at any rate, even just restricting ourselves to the spatial component, what is the spatial component of the surface of a balloon? Is it three -dimensional? It's only two-dimensional. So if this were an accurate representation, the balloon would also feature expansion away like this. Now, there's something interesting about this analogy. The Earth is very closely to being spherical. It's not actually a sphere, by the way. Slightly misshapen, more like a pear. There is a very well-defined center of the Earth. The center of the Earth itself is the core of the Earth. And the center of the Earth in three dimensions is well-defined. But there is no center of the Earth in the two-dimensional surface, is there? There's no point that could say, "I am the center of the Earth."
- 3.52** A lot of countries throughout history have declared themselves to be the center of the surface of the Earth, but that's kind of farcical. There's no center on the surface. There's a center in three dimensions. So expanding in this format, at least on the center of the, from the perspective of the balloon's surface, there's no center. As you said, there is a center of the three-dimensional balloon. And so that's a failure of this analogy. But at least as a surface is concerned, restricting the three dimensions that we actually live into a two -dimensional analogy, it's helpful. Not without its problem.
- 3.53** Sometimes also shown in a more tasty version, as raisins in a raisin bread. That's a three-dimensional version. What's another problem with this analogy? What's the problem with this raisin bread analogy? What does the raisin bread have that perhaps our universe has, but perhaps it doesn't? An extent. Yeah. So it has some finite edge, the tasty crust. Right? So that is something that we do not yet know if our universe possesses. It may have an edge, a boundary at which, beyond which, there is no universe, but it may not. So all these things have problems, but they offer some solutions.
- 3.54** Does the expansion of the universe violate the Copernican principle we started this lecture off with? Does it violate the principle of lack of speciality or specialized nature of our position? No. In fact, the expanding universe, when it expands like this, with a linear relationship between the amount of expansion and distance, that's the only way a shape can expand and preserve its features. In other words, if things in the universe expanded, other than the way they do, where the distance between these objects is proportional to how much expansion they have, this object would not still remain triangular. It wouldn't have the same angles, rather. In other words, this angle between, say, the distance between point number one and two, and the angle between, and that line's angle between one and three, is maintained if I double the size of this triangle. Right? That's the principle of geometry.
- 3.55** But what if I said, instead of the triangle behaving like this at a future time, it will depend on the square of these distances. The angles would change. This angle would

expand less than, say, these two angles, this one between here and here. That's called a conformal transformation, and our universe is symmetric. It possesses a symmetry in time that, if tomorrow it's expanded by some amount, then two tomorrows, two days from now, it'll expand by twice that amount. That is the only type of expansion that results in a homogeneous and isotropic universe.

3.56 The math behind it is demonstrated here. And it turns out that, at a later time, you merely get a bigger triangle with the exact same angles, and any other expansion law would violate that. You still have a triangle, but the angles would change over time. That would be a very confusing universe to live in, and we don't live in it. So we seem to live in the only type of universe that possesses the symmetry that allows us to keep applying the laws of a homogenous and isotropic universe. That is, one that obeys the cosmological principle, which is one that also obeys the Copernican principle, of lack of anything special about where we are, who we are, or when we're doing this measurement.

3.57 And it turns out, if you go through and apply a very simple amount of calculus—this is, I think, the most calculus we'll use in this course—if you do what's called the time derivative and ask, "How much does the distance change over a tiny amount of time, dt ? How much does the distance change with time?", you get the time derivative, and you can construct these different ratios. And this leads us to the concept of what is now called the Hubble parameter, which is perhaps the most important number in all of cosmology. The Hubble parameter controls the size of the universe, the age of the universe, the rate at which the universe is expanding, and the rate at which the universe might accelerate into the future and rip us all apart. Coming soon, another cliff-hanger. Stay tuned for more.

3.58 This number, the Hubble constant, symbolized by capital H with a lowercase zero, is related to what's called the Hubble parameter. The Hubble parameter is related to what is called the scale factor. The scale factor is just a dimensionless number that tells you the following fact. How big was the universe when the light that we see now was emitted? I see a galaxy. It's at some great distance. If it's a million light years away, it emitted light a million years ago. I want to know, how big was the universe when it emitted that light?

3.59 If I tell you the universe is expanding, the only thing you know about the size of the universe back then is what? If it's expanding now, it must have been smaller back then. You don't know how small yet. That's the job of astronomers who measure this number, the Hubble parameter, today, or the Hubble constant. By definition, we say the size of the universe, the scale of the universe, A , t_0 is today. We say that the scale of the universe today is 1. Why is this interesting? Because it turns out that the scale factor, A , is exactly gotten from the redshift that we measure. In other words, we measure the redshift. If the redshift has changed from some position in the spectrum and it's gotten twice as long, the wavelength that emitted got stretched to be two

times as long, that means the universe has doubled in size since that photon was launched from that galaxy.

3.60 So, by measuring the redshift, you're measuring the size of the universe when that light was produced. It's extremely powerful. It allows us to control our measurements, calibrate our measurements, and measure something that's easy to measure. This redshift, using the spectroscopes that astronomers have used, have constructed—it's very easy nowadays to measure redshift. I'm going to show you measurements of millions of these galaxies. Every single one is shifted towards the red by different amounts. And we can use that to measure not the actual distance to the galaxy, but the size of the universe at the time that light was produced.

3.61 Because the relationship is, redshift is directly proportional. Z is how much the universe has changed in size. A is the ratio of the size of the universe when that light was produced. So this factor, $1 + Z$, is how much the universe has expanded. So if you see a redshift of doubled in redshift, that means the wavelength has increased by 2. We would say it's a redshift of 1. $1 + 1$ means the universe was 1 over $1+1$, or one-half its current size. We don't know what that size is, but we know the ratio between then and now. And it's completely consistent with the cosmological principle, which is itself completely consistent with the Copernican principle. So things are standing on firm philosophical foundations.

3.62 So it's powerful, extremely powerful to know the Hubble constant. It's very easy. It's not easy, but this is what astronomers can do. They can measure the redshift of a galaxy by looking at its spectrum, its library of light, its atomic fingerprint. These galaxies are all made of stars. Stars are made of hydrogen, primarily. Therefore, we know hydrogen's properties in the lab. We can compare it to what we see in this galaxy. Oh, we see it's been shifted. They're doubled in wavelength. That means they can compute the shift is 2 minus 1 divided by 1. We can compute the distance by knowing this one constant, h_0 . We still haven't said what h_0 is, what its units are, and how that relates to the size and age of the universe. And we're about to do that.

3.63 So this gives astronomers an easy way to measure the hardest possible thing. Remember, you don't have depth perception in cosmology. You see objects. You can't tell if it's twice as bright or if it's half as close or quarter as close. This is the easiest possible way to measure distances. And these aren't just distances on Earth, in the solar system, in our galaxy —these are distances in the entire observable universe. So this is called the Hubble Law, the Hubble-Leavitt Law. And this is how you do it.

3.64 If you know, if you measure the radial velocity, which comes from the wavelength redshift, you can compute the distance to the galaxy by dividing that wavelength change in the Doppler shift by this constant, h_0 . And this is a universal constant at any given time. Anywhere in the universe, this rate of expansion holds true. And Hubble realized this back in the 1920s. So he was working at Mount Wilson. As I said before, he observed this variable star here. He actually crossed out, he thought it

was a supernova or a nova. He wrote N originally, came back a couple days later, said, "Ah, no, this is one of Henrietta Leavitt's Cepheid variables." She told him what the distance was, or the luminosity was. He measured the flux. He computed the distance to it, showing it was 30 times the distance of the diameter of the Milky Way. That was in 1923.

- 3.65** And then he followed up with it by measuring things to every galaxy he could see. So I show this equation—let me skip over that. This is his very first plot. What Hubble does here, he takes a collection of galaxies whose redshifts were measured by Slipher, Vesto. Vesto said, "Here's how much they've been shifted." Almost all of them have been shifted towards the red, longer wavelengths, meaning they're moving away from us. There are a couple here. This is the velocity axis here. This is velocity moving away from us. This is velocity moving towards us. There are only three or so galaxies that he found were moving towards us. And many of them we already knew were things like the Hubble, were things like the Andromeda Galaxy.
- 3.66** And then he, like Leavitt herself, he could look up well, in this galaxy, the—Henrietta Leavitt had already measured the luminosity of a Cepheid variable. So she had told him that galaxy is this far away in parsecs, shown here. This is millions of parsecs. So these are megaparsecs. This is roughly 3 million light years away from the Earth. And then Slipher told him how much this thing had been shifted towards the red. And the farther you went out in space, the redder and redder these objects get, because they've had more time for the universe to expand, and the wavelengths have gotten stretched more and more towards the red.
- 3.67** So this is an incredible correlation, this relationship between distance and redshift. It allows you, by knowing the slope of this line, the slope of this line is that number, h_0 , the Hubble constant. So it turns out that when he fit this line through this graph, he actually got it pretty close to correct. He was only off by 800%, which sounds pretty bad. But for these days, remember, this is almost 100 years ago, he actually got this overestimation by a factor of 7 or 8. And he believed that a galaxy—I'll actually explain that. So this relationship, the properties of it, are correct. The farther away something is, the more it's been redshifted. The redshift amount is given by this constant of proportionality. That's still true, but the value he measured for that constant has decreased over time, and we'll see why that's important. Whereas he measured it with huge uncertainties, we now have about 1% errors on the measurement of the Hubble constant for every single galaxy we can see in the observable universe. It's outstanding.
- 3.68** So, to give you some context, 20 years ago, my friend and colleague Robert Kirshner made this compilation of what was known back 20 years ago. This is published in the Proceedings of the National Academy of Sciences, just like the great debate was 100 years earlier. And this is called the Hubble Diagram. This uses a variety of different techniques that we're going to concern ourselves with later. And you see this constant of proportionality is exactly what Hubble predicted—a straight line, a one to one

correlation between distance and redshift. That means you can use something that's easy to measure, redshift, just from looking at a light shift on a detector. You can use that to measure something extremely hard, distance, and it shows the universe is expanding.

- 3.69** It's pretty remarkable when you consider that the previous plot, this whole plot of these 20 or so galaxies, fits in a region of this current plot 20 years ago. Now it's even farther out. It's just that red square down there. So Hubble was pretty bold. He wasn't so egomaniacal to call the Hubble constant after himself. We did that later in life, but it's quite remarkable. And over time, the estimate of the Hubble constant has gotten better and better and converged upon values that we'll describe in upcoming talks, and that it's converged upon a value that we're going to use to do something very exciting.
- 3.70** Because, as I said, the implication of an expanding universe is tomorrow, a galaxy that we measure at a great distance, more than 100 million light years away, it'll be farther away. It'll be farther away by an amount we can predict exactly. Furthermore, it allows us to do something very, very unusual and perhaps unexpected. If the galaxies tomorrow are going to be farther away, and the galaxies yesterday were closer to us and closer to each other, if you think about the raisin bread expanding—tomorrow, if the galaxies are going to be farther apart, today and yesterday, they were closer together. Might there have been a time when all of them were actually on top of each other, at a single point, perhaps a singularity? That's the heart of the Big Bang concept.
- 3.71** And that was unavoidable once we realized the universe is expanding by observation of millions of galaxies now. The universe is expanding. Yesterday it was smaller. The day before that, even smaller. Now we want to unravel, at what point does our understanding of the laws of physics, do they break down? Do we not have an understanding of how time itself could have begun? That's where we're going to go next. As a prelude, I want to point out that these units in physics are very important. Velocities are measured in terms of change in distance divided by a change in time. So the velocity of something can be expressed as miles per second. It could be miles per nanosecond. It depends on the context. But something that has units of distance and something that has units of time.
- 3.72** In cosmology, the units that we use, the expansion rate of a galaxy is given by kilometers per second. The average galaxy nearby to the Earth is expanding 70 km away every second. Imagine something moving a kilometer per second, let alone 70 km per second? That's for a galaxy, a hypothetical galaxy. And actually, they're expanding much, much farther away than this, because the average galaxy nearby to the Milky Way is not actually expanding. It's actually collapsing towards us because of gravity, in a way I'll describe in an upcoming lecture.
- 3.73** So, actually, these units, it has units of distance divided by velocity. So it's distance divided by distance divided by time. That comes out to have something that has units of inverse seconds. And actually, if you convert the number of inverse seconds and you

take the reciprocal of the Hubble constant, that gives us something that's very closely related to the time that's elapsed after the Big Bang. It's given by approximately 1 over the Hubble constant. So the age of the universe, if you will, since all these galaxies were perhaps touching each other—it's not exactly the way we'll describe it—is given by something that looks like 1 over 4 times 10 to the -17 seconds. What is this 10 to the 17th? Remember I told you 30

3.74 years is about a billion seconds? This is rounding up about a billion billion seconds. What's a billion billion seconds? It's like a billion times 30 years. So it means 30 billion years. Now, it's half of that, so it's actually less than that. And if you go through and calculate it, you get a time that, hypothetically, all these galaxies were touching, is given by the inverse of the Hubble constant, which is about 1 over, which is about 70 km per second per megaparsec. If you do that, you get this reciprocal of this, 4 times 10 to the minus 10 to the plus 17 seconds, and you get a time estimate of about 14 billion years, which is remarkably close to the age we're going to calculate in just a few lectures, that we now believe the universe is about 13.8 billion years.

3.75 As I said, when I started teaching a billion seconds ago—sounds like a short time, but it's a very long time, 30 years ago—this age of the universe was somewhere between 10 billion and 20 billion, an uncertainty of 100%. It's like looking at somebody and saying, "You're about 27 years old, but you could be 54 years old." You say, "That's not so good. Why am I paying you all this money at this carnival?" But if I say you're 27 years old, you were born on May 4, 1990, what, 6 or something? Then it starts to become more and more impressive. The more accurate you are, how close you are to the true number, and the more precise the number of significant figures by which I can measure. How many microseconds you are old is much more impressive than how many decades you are.

3.76 So this age is actually independent of whatever galaxy you're looking at. It's a universal constant. It's fascinating. It holds throughout the universe, just like the law of universal gravitation does. Now, the question that you might be asking is, wait, there were galaxies 14 billion years ago? The answer is no. And we'll explain that in later lectures. But it gives you a sense of how space has changed over time. We're using galaxies as sort of the tips of the iceberg or the foam on the ocean. What's interesting to us are the ocean dynamics and the waves and the swells and so forth. But the only things we can see are the icebergs or the foam on the waves. We're using galaxies as tracers.

3.77 Remember I said we don't even care about stars anymore? And that's mostly true. We're not using stars. We're using galaxies as tracers. But what do you do when there's no galaxies around? You're going to have to use something else. We're going to come to that in the next lecture. But this tells you, really, the core concept of this class. The most important number in this class is the Hubble constant. We have different ways of measuring it. And the philosophical implication of the Hubble expansion is that the

universe is getting bigger. The separation between objects is increasing every day. It's increasing at a very precise rate known to the percent level.

3.78 And the most astonishing implication is, in the very distant past, billions of years ago, all the matter in the universe may have originated from a finite-sized volume. And when you have a near infinite amount of energy in a finite region of space, you're unavoidably going to encounter something called a singularity, where the laws of physics as we understand them today, break down. We're going to use the Hubble constant, as I said, to estimate the properties of the universe's past, when everything was compressed, and the deep future of the universe. Because how the Hubble constant changes with time, is it really a constant, like the speed of light, or does it change? That will be a key question.

3.79 But here we have the final resolution of the cliff-hanger from last lecture. In a finite aged universe, light from stars that are at a greater distance than the age of the universe, times the speed of light, have not had time to reach us yet. They may be there, infinitely far away from us, but their light has not had time to get to us. So there may be an infinite amount of space. There may even be an infinite amount of stars. We cannot disprove that, as we want to do in science—want to disprove things, not prove them.

3.80 This is a possible resolution of Olbers' paradox, that the stars are there, they just haven't reached us yet. So the night sky can be dark because there's only a finite number of galaxies that have a finite number of stars, within the age of the universe times the speed of light. That's what's called the causal horizon, the volume of space that we can have communication with or learn things about. That's the maximum distance permitted by physics.

3.81 So the final question you might be wondering is, is the universe infinitely big? Does it really extend forever? And the truth is, we don't know. But there will be implications of a universe that is finite—namely, does it have a boundary? Are there other universes external to our universe? That's a very, very fun thing to think about. And I'll just end with this lecture. As I said, the universe's most important number for cosmologists is the Hubble constant. There are many, many ways to measure this number. The fact that they agree very closely, but not exactly, is a sign of a healthy field, that we understand the physics of the expanding universe accurately.

3.82 It turns out that we have an estimate that's actually improved since this measurement here, by a team led by my friend Adam Riess, who works with the Hubble Space Telescope, that they've gotten to within, actually, a 1% level of uncertainty on the Hubble constant. Other colleagues of mine have worked using a spacecraft called the Planck satellite. It's different than the Hubble Space Telescope, and they get a different number with even smaller uncertainties that doesn't agree with the measurements made by the astronomers using the space telescope. So cosmologists using a space telescope called Planck have a difference between their age of the universe and the age

of the universe measured by the optical astronomers, and they differ by a billion years. And each one thinks that they understand their measurement at the 1% level. So this is called the Hubble tension.

3.83 It's much, much different, though. In the past, the optical astronomers thought the universe could be 10 billion years old, or 20 billion years old. The cosmologists had no idea what the universe's age was. We simply couldn't measure it accurately 30 years ago, and now they measure it—cosmology measurements are better than the precision of the optical astronomers, and both of them know them to within an equivalent, say, of measuring your height to within an inch or two. It's quite astounding if you're 6 ft tall.

3.84 And this is another animation by my friend Richard Behiel, showing measurement of distance versus redshift. These plots get redder and redder. This is a fit to the data that is essentially a model. There will always be some uncertainty, because even in a universe that's perfectly represented by Hubble's Expansion Law, you'll still get variations from point to point. Just like if you measure the temperature of this air in this room, and you measure the temperature of a small volume of this room, it may have a different air pressure, depending on how much hot air is coming out from certain regions of the room, namely near my mouth. It'll vary from point to point. So that's expected. And we average together and we get a more and more refined fit over time.

3.85 And here's the kind of amazing plot that shows all different galaxies that we've measured using multiple techniques, many different types of telescopes. They all agree that the universe is expanding. The rate of expansion, it depends on distance. It's a universal constant, as far as we can tell. It doesn't matter where you are in the universe, which direction you look, it's isotropic. It is the same Hubble constant measured in this direction as this direction. It's homogeneous, as far as we can tell. We can actually measure, if you like, moving our laboratory from different places on Earth to different galaxies in a way that we can describe in a more advanced description of cosmology called the Sunyaev-Zeldovich effect.

3.86 And we actually have been able to constraint, using all the optical data that we have, that this law is a universal truth, literally universal, throughout the universe. So we'll end here. In the next lecture, we'll talk about how far out can we measure and what does it say about the shape of space-time itself? And that will be a fascinating kind of distinction between philosophical treatments of space, time, and gravity.

Lecture 4: Forces and Fate

At a glance — Forces and the fate of the universe.

- 4.1 So, in lecture three, we described why it is we think that the universe is expanding, but we didn't describe how it is that the universe could be expanding. And that's a topic for today. Today, when we look out in space, what could possibly make the universe, almost anything, expand or contract? That's what we have to wrestle with. The observation of an astonishing fact often leads scientists to develop new insights into previously well-understood phenomena that actually really weren't that well understood.
- 4.2 In physics, which is really the underpinning behind cosmology—remember, cosmology is the laws of physics applied to the universe as a whole—the question is, which of the laws of nature are most important for our understanding of cosmology? And surprisingly, it's not really the one that's the strongest that has the biggest effect, and we'll see why that is. In fact, it's the weakest of all the forces. There are four known fundamental forces of nature—gravity, electromagnetism, and there are two nuclear forces, one creatively called the strong force, one called the weak force. And you can guess which one is stronger than the other, but you may not—well, I wouldn't understand necessarily why they're called that. We'll explain what they do in just a minute.
- 4.3 Electromagnetism is the force that governs the behavior of charged objects. Charges such as electrons or protons are really the only two objects that we would encounter, and their behavior, of both their properties when they're in motion and when they're stationary, that's the domain of electromagnetism. And we're all familiar with magnets. So magnets are one form of the electromagnetic force, but it's not clear. Why is it called electromagnetism? Why isn't it just called magnetism and electricity?
- 4.4 Well, in fact, they used to be thought to be separate phenomena, electricity and magnetism. I'll pass these magnets around. You can play around with them. Magnets and electricity don't seem to have much together in common whatsoever. In fact, magnets, as indicated here, they have a north pole and a south pole. Charges, on the other hand, can either be positive or negative, but they don't come necessarily as both. These are intrinsically dipolar. They have two different poles, north and south. Please pass those around. You can give them to the back when you're done. So, magnetism is the result of the motion, the dynamics, of charges. Electricity
- 4.5 is the static or constant current motion of charges, either positive or negative. And we can see manifest in both phenomena, like lightning storms or in these bar magnets or these horseshoe magnets that I'm passing around, the different phenomena of magnetism, but also electromagnetism is one of the primary forces that holds together atomic nuclei. In fact, the force that is primarily responsible for the behavior of

phenomena like fire is electromagnetism. These are chemical processes, chemical reactions, that are occurring thanks to a spark inside of here, which is electric phenomena, igniting a gas inside of here that then produces chemical transitions which give off infrared radiation, which Herschels described as heat.

- 4.6 Now, you couldn't make this thing burn, no matter how much or how big a magnetic force you put applied to it. It takes a tiny little spark, very high voltage, not unlike the lightning depicted there, to ignite it. So electromagnetism is familiar, and it was thought, though, to be two different phenomena. Electrical phenomena described by Benjamin Franklin, first in a quantitative fashion, and then the magnetic properties of bar magnets and other objects were thought to be two totally different things, until the 1800s when they were unified. And in physics, what we like to do is unify disparate multiple phenomena into a single phenomenon.
- 4.7 We like to take two different things and combine them together to have theoretical simplicity, which leads to elegance. And some say, beauty. And yes, believe it or not, mathematicians and scientists will look at an equation and say it's beautiful and then say it's probably right because it's beautiful. And in fact, a famous scientist, Paul Dirac, said, "It's more important that your equations be beautiful than they be right." And I, as an experimental physicist, take great issue with that.
- 4.8 There are two nuclear forces. One is called the weak force, although it's much stronger than the electromagnetic or gravity force, but it's weaker than the strong nuclear force. And in fact, the weak nuclear force was found to be unified in the same way that electricity was unified with magnetism. So too was the weak nuclear force unified in the '60s and '70s with electromagnetism. So now there's, where there were three separate forces, where there were three separate forces, now there's just one force called the electroweak theory. That subsumes within it electricity, magnetism, and this weak nuclear force, which is responsible for what's called beta decay, radioactive dating, things like that are related to this process, which is governed by sort of a version of electricity and magnetism on steroids.
- 4.9 And then there's finally the strongest force, speaking of steroids. Is a strong nuclear force, which does a surprisingly powerful thing. I said there are only, magnets only come in one flavor, one with a north pole and a south pole. If I were to take this magnet—hopefully this won't fall on my foot—if I were to take this magnet and break it apart, what would I get? Would I get a north pole over here and a south pole over here? No, I'd get a north pole and a south pole. This would be a south and a north. Then this would magically, this side would break into a north pole and a south pole. There's no way to break this and keep breaking it up and get a single north magnetic pole and a single south and a single south magnetic pole. So it's very difficult to do the following with electrostatic charges. It's much easier to do it with magnetic, bar magnets like this.

- 4.10 Electromagnetic phenomena exhibit the following property. Like charges repel, and opposites attract. We know that from the dating world, and it's true in electromagnetics. If I put the north pole next to the north pole here, and I try to get these to attach themselves to each other, you will feel a force that will repel them or should repel them. So this is actually mislabeled, and they will be repulsive. If you do the same with a south pole and a south pole, it'll also be repulsive. And if I flip it around north to south, they'll stick together. Opposites attract. Like charges repel. Like magnetic poles repel as well.
- 4.11 So how do you explain an atomic nucleus? Atomic nuclei are comprised of protons, which are positively charged, and neutrons, which are, as the name suggests, neutral, not negative, but neutral. Electrons are negatively charged, and they comprise an atom when an electron orbits around at least one proton and maybe more protons. Shown here is a compound nucleus of many, many protons. Maybe it is a carbon nucleus, for example. It has some number of protons, red with plus symbols, and some number of neutrons, slightly greater, perhaps. So how do you get these two objects, these two protons, to stick together? There must be a force stronger than the electromagnetic repulsion between two protons that binds them together. And in fact, there is.
- 4.12 There's another abstract form of charge called hypercharge that is intrinsic to each proton. The protons themselves are not elementary. They can be broken up into smaller things called quarks. Those quarks can bond together with themselves and with the quarks of another proton, such that a negative hypercharge quark can bond with a positive hypercharge quark. And that force of hypercharge attraction is stronger than the electrostatic repulsion, hence the name strong nuclear force. Do you get it? Two protons ordinarily would repel each other. Completely distasteful. But because of the mediating forces of quarks within them and their opposite charges in this quark abstract space, they actually stick together. That's the strong nuclear force.
- 4.13 And then lastly, most familiar is gravity. Gravity, despite it being incredibly weighty and incredibly resistant, as anyone who's ever been to the gym or thrown one of these—I will now do. No, I will not do that. This is extremely heavy and will cause a lot of damage. This is being pulled on by the entire mass of the earth, and it's all going downward towards the center of the earth. And if we go opposite side of the earth, it will also go upward there from our reference frame, but downward to them towards the center of the earth.
- 4.14 Gravity has a charge, too. It's given by its mass. The number of protons, neutrons, and electrons in here comprise the mass of this object. The number of protons and neutrons in the entire earth pull on it. But actually, with a relatively weak magnet, probably about 2 or 3000 of these could actually hold this object up, fighting against the entire combined mass of all the protons, neutrons, and electrons in the earth. So just a few thousand of these magnetic charged objects, namely just ordinary magnets, can resist the pull of the entire earth. That shows you how weak gravity is.

- 4.15 We think of it as strong because it's hard to go to the gym. Believe me, I know. But it's actually some more than 39 orders of magnitude weaker than electromagnetism, which itself is weaker than the nuclear forces. So gravity is the weakest of all the forces. And yet, and yet, it is the most important force in this class. We don't care about the quarks and so forth because they're only inside nuclei. We don't care about electrostatic forces because it's only inside lightning and fires and stuff like that. That's not relevant in space to objects that we care about. But gravity acts on the entire universe. In fact, it acts, in the most extreme cases, to have the longest range effects. And it can be so strong in long range because it's so weak and diffuse. And that's the mystery of gravity.
- 4.16 And so it may not be obvious to you, why are we talking about gravity in the class about cosmology? Except I just said that's related to the universe. But how? And why? And what in Newton's cogitations could it possibly have to do with the expansion of the universe? Gravity? What does it have to do with expansion? Doesn't gravity attract and suck in? No, we'll find it doesn't. So it has all these interesting properties. It's the longest range force. It's the weakest force. It's only attractive. There's no positive gravitational field object, a negative gravitational charged object.
- 4.17 The law of gravity was discovered in the 1700s by Isaac Newton, and he gave it a quantitative form, that the force of gravity is proportional to the product of the mass of the two objects experiencing gravitational force, inversely proportional to their distance squared. Always attractive—the presence of the negative sign means attractive. And modulated and mediated by a force of nature, a constant of nature, called capital G. And capital G is a constant that seems to hold throughout the entire universe. It's a universal constant, like the speed of light.
- 4.18 And gravity has entertained physicists for many, many years. Allegedly, Galileo dropped two different balls off of the Leaning Tower of Pisa to observe that they fell at the same speed, the same acceleration, to be precise. That means independent of their composition, independent of their weight. Now, why was that surprising? We take it for granted to this very day. But remember, the ancient Greeks thought that wasn't true. They thought this object, this magnet, because it's lighter than this object, would fall slower than this object, and they never did the experiment. It's kind of unusual.
- 4.19 They were so philosophically entrenched by this notion that the earth is the center of the universe, that things always fall downward, that the downward falling depends on the mass of the object and therefore the earth was perfect, so it was immovable, at the center of their cosmology. Maybe it set back science for thousands of years, although I can't say that I really could blame them, as, again, these facts are obvious to us with the benefit of hindsight. But they could have done, as Galileo supposed, they could have done a very simple experiment.
- 4.20 They could have taken, say, a magnet or any piece of any object, and said, "OK, you think that this object falls faster than something that's lighter?" Just take this object, break it in half, attach a string to it, the two pieces—so, you know, I'll break it in half. I'll

get a north pole and a south pole, a north pole and a south pole, just tie them together with a string. Now it's just one object. Shouldn't they fall at a different rate than the object as a whole? So each one individually? And they could have just done that experiment. They never did it. I don't know why. It took Galileo, allegedly, at the Leaning Tower of Pisa. So it's a great story.

4.21 So Galileo observed the phenomenon but didn't describe it mathematically because he lacked a very powerful tool called calculus, which not only Isaac Newton invented, but he used to prove the laws of Kepler, that planets move in elliptical orbits, and extend these throughout the universe. So gravity is a dominant force in the universe. We still need the other forces, or else we wouldn't exist. Atoms wouldn't exist. But gravity wins on the long distance nature of the universe.

4.22 Isaac, good old Isaac, did many, many things, including discovering three laws of motion that bear his name to this very day. The first law of motion is that an object in motion remains in motion or constant motion unless acted upon, as he called it, compelled to change its state, namely deflect its path, by an external force. This is Newton's first law of motion. The Greeks, including Aristotle, thought the natural state of something was rest. They couldn't comprehend, because the earth was the center—they couldn't comprehend the earth is actually in motion, and it's moving quite rapidly. And in fact it is rotating as it's moving around the sun. It's rotating on its axis, about 1,000 miles per hour at the equator. This seemed completely bizarre to them because the natural state of motion to them was rest. The natural true state is motion, unless a force, like friction, acts upon it.

4.23 Second law of motion shown here. There's Newton. Now, we know one thing about Newton. He never had any children. He never was married, never had any children. So this is not his daughter. But they're skating. And let's say she applies a force on him. They'll be, in his third law, an equal and opposite force. But if a constant force is applied, a push is applied to each one of these people, the less massive, the little girl, will move much, much more. She'll experience a much greater acceleration due to Newton's second law. The acceleration of an object is proportional directly to the force you push up, pull upon it with, and inversely proportional to its mass. We sometimes write that as F equals MA . But you could write it like this, so that a small mass will have, suffer a big acceleration for our constant force applied to it.

4.24 And the third law, as I already mentioned, when old Iky there pushes on his skate, the earth pushes against him, and he pushes against the earth, with an equal and opposite amount of force. So of course, he can only generate a tiny amount of force relative to how much the earth is. And the acceleration of the earth is minuscule because its mass is so many, almost infinitely larger than old Isaac's mass. But nevertheless, the earth physically does move a little bit every time you walk and push upon it, according to Newton's third law.

- 4.25 Now we come to the battle of the great haircuts in cosmology, and that is gravity. And its interpretation in the philosophy of Newton is very different than the way that Einstein and we modern cosmologists conceive of gravity. And there are different ways of thinking about gravity, which in turn changes the playing field, namely, how space behaves in the presence of matter. In terms of Newton's conception, space is static. It is not changing with time. It's neither expanding nor contracting, and it is flat. What does flat mean? Flat means that, no matter how big a triangle one were to make, whether you were to make one on a small, flat piece of paper, or you were to make one comprising of your position on earth and looking at two stars, one here and one there, this comprises a triangle.
- 4.26 And this triangle has the following properties—that its 3 interior angles, alpha, beta and gamma, when you add them up, they all sum to 180 degrees, no matter how far away the triangle is, the base of the triangle is, throughout all of space. These could be galaxies. They could just be little spots on the cosmic microwave background. And in fact, they will be, and we'll talk about that. Flat means the laws of Euclid hold, which means that no lines that are parallel ever meet. One of Euclid's postulates is on a flat plane, like this whiteboard, these two lines will never meet. On a curved surface, these lines could meet. And on a curved surface, a triangle can have an interior sum of angles much greater than 180 degrees.
- 4.27 But this is how you determine the curvature of space. Space can have three different types of curvature we'll see. And what Newton said is, we live in space that is flat. For any triangle you make, off of the surface of the earth, if you're doing it on a spherical surface, this won't be true. But any other triangles, no matter how far away, planets, three planets, three stars, three galaxies, you will always get an interior angle sum to 180. And also the Pythagorean theorem will hold as well, and we'll get into that in just a bit. This is way that you test whether or not you live in a flat space.
- 4.28 The second law, sorry, the first law, is that these objects were moving in a constant line or straight path, unless they're acted upon by an external force. So if you looked out at the ellipses, the elliptical motion of planets, as Kepler observed them and Tycho Brahe had observed before, they were moving in elliptical paths or circular paths, they thought originally. That's not straight. That's not a straight line. Something must have been acting on them. What was the force acting on them to deflect them from being totally straight? Gravity. Gravity is 100% attractive. The sun's gravity was attracting the planet to it, causing it to orbit in a circular, so they thought, but now we know elliptical, elliptical orbit.
- 4.29 That required a force. And in fact, the inverse square law exactly produces such behavior. And the force that acts on them is called gravity. And the amount of gravitational force depends exactly as Newton's equations postulated. So, in Newton's gravity, you have force equals mass times acceleration. What force is causing this thing to fall? The force of gravity. What acts on gravity? Another object of some mass. And

what Galileo and later experiments have shown is that the acceleration due to gravity is exactly as this mass would have you believe.

- 4.30** In other words, you measure the mass causing something to deflect its path and make an acceleration. That's called the inertial mass. The inertial mass is exactly identical to the gravitational mass to within a few parts in a trillion. It's one of the most accurately measured properties of the law of gravity. So if you abide by Newton, this is all you would need to know about gravity. Still doesn't tell you what's the origin, if you like, the ultimate origin of gravity. But it holds true, as was proven throughout as much of space as you can imagine.
- 4.31** Here's a video of NASA astronauts and the Apollo mission on the surface of the moon, demonstrating the equivalence of the mass of inertia and the mass of gravity. They took a feather and dropped it, and a hammer, too, and you'll see what happens in a minute. According to Newton and Galileo, they should fall at the same speed. There's no atmospheric air molecules. It's in a vacuum.
- 4.32** So it's pretty astounding that the laws that we can contemplate, postulate, here on earth, apply throughout the universe. And it gives us confidence, because if it weren't the case, you'd have to have a different set of laws here on earth, as you would on the moon, as you would on some other planet, as you would in other galaxies. That would be very, very confusing. So nobody in Newton's time really cared why the equivalence principle was true, that objects of different composition and different mass fall at the same rate. They just thought it was cool that someone finally did the experiment and could mathematically predict the rate at which these things would fall.
- 4.33** But Einstein wanted to understand things at a deeper level, and he stood on the shoulders of Newton, who himself had said he stood on the shoulders of giants. So Einstein did a thought experiment and he asked the following question. If you're on earth and you take this cannonball and you drop it on your foot—now I'm going to drop it on my foot—but you drop it, it'll fall at a rate that we can calculate. One G of acceleration. However, he wondered, what if you went into a rocket, deep, deep interstellar space, far from the earth, and the rocket was accelerating at one G, same exact acceleration, 9.8 meters/second. Every second, it increases its velocity. What would happen? Could you tell the difference?
- 4.34** Drop it inside of a rocket with no windows, accelerating at one G, or standing on the surface of the earth with one G of acceleration pulling down? You could not tell the difference. In fact, that was such an astounding observation to him that he later said it was the most exhilarating thought of his life, and an allied thought that he had, which as a tangent, as all professors like to go on tangents, we're told lately that artificial intelligence is a great danger to us, that artificial intelligence will subsume all of human knowledge and surpass us in what's called the singularity, a different kind of singularity than we'll discuss in a black hole or in the cosmos, but that it will become superhuman and make us all irrelevant, maybe turn us all into paper clips.

- 4.35 But I like to think that Einstein kind of gives us a little bit of hope, because he visualized the following situation. He said, "If I were in an elevator and the elevator cable broke and I was in free fall," he said, "I would not experience any gravitational force." Have you ever been in a roller coaster, on a car that's going too fast? You go over a bump. For a minute, you're kind of weightless, you're levitating, you're in a freely falling reference frame. And he said, "Such a person experiences no gravity." To him, he called this his happiest thought.
- 4.36 What does this have to do with AI? Well, I wonder how and to what extent an artificial intelligent computer could, A, have a happiest thought. What would it take to make a computer happy? I don't even want to think about it. And B, could it visualize the sensation as Einstein could in his famous thought experiment? I like to think not, and that there's a special, something special about the human mind that will always render us superior to our computer counterparts. But for future reference, I will submit to their will if they do eventually take over and cause the singularity to happen.
- 4.37 So what Einstein said is, not only is this experiment true for dropping massive particles and near the surface of the earth, it's also true if you shot a laser beam. In other words, if you shot a laser beam near the surface of the earth, the actual trajectory of the laser will be curved. On big enough distance scales, the laser will actually drop. So, too, if you shot a laser at the wall of your spaceship and the wall is accelerating upwards, the laser spot will also be down, if the spaceship is large enough that you could measure it. If it's wide enough that you could measure it.
- 4.38 So that means that even light is bent by gravity, the trajectory of the fastest propagating substance known to exist, light, is still not fast enough to overcome and resist the force of gravity. Of course, we've all seen this in the context of black holes, in movies like Interstellar, et cetera. So what Einstein synthesized was everything that Newton had done, but he interpreted it in the following way. He said that light doesn't always travel on straight lines. Space, therefore, isn't always flat. And it could be that triangles in space have angles that don't add up to 180 degrees. And you could detect curvature of the universe. Curvature produced by what? By mass. That mass tells space how to curve, and the curvature of space tells light how to move, or anything else how to move. It connects matter and energy to the curvature of space-time itself.
- 4.39 So the general theory of relativity was really Einstein's crowning achievement and probably his last great accomplishment. He achieved an incredible amount in just 10 years, between 1905 and 1915, as we'll see. The general theory of relativity, general means it applies in all situations. The special theory of relativity only applies where there's no gravity. We'll talk about that in just a bit.
- 4.40 So, when we talk about the universe, we can't ignore the fact that the universe is made of things like us that have mass and have matter. And because of that, we cannot construct a perfect coordinate system that's independent of time or space, as you would like to do in the cosmological principle. You could do so locally, just like you can

locally assume that a map of the earth is flat, even though you know the earth is actually a sphere. It will matter less on regions of the map that are much, much smaller than the radius of curvature of the earth.

- 4.41 Light takes the shortest path. Light is kind of lazy. It always takes the shortest path between two points. But that path may look like a curve to us in the presence of matter. And so space is not flat, and that tells the curvature of the trajectory of a planet or even a beam of laser light to deflect and take potentially highly curved trajectories. That is the heart of Einstein's theory of general relativity. So, to summarize the difference between Newton and Einstein, Newton believed that gravity was affected by how much mass there was. More mass, more gravity, and that told how much force to exert. The force then told the mass how to move, and it moved exactly as F equals MA . And the M and $F = MA$ is the same M in Newton's law of universal gravitation.
- 4.42 But in Einstein's conception, the amount of mass which, via E equals MC squared, means if you have something that has no mass, like photons, they have no mass, but they have energy. They too can affect the curvature of space-time. And then once you curve space-time, that affects the trajectories of how things move in that space-time. So, going back to the laws of Newton, if space were flat, then any triangle, no matter how big or small, would always have three interior angles that would sum to 180 degrees. Any two parallel lines would never, ever meet. Those are Euclid's postulates. So that's a two-dimensional space-time. It's flat, has zero curvature.
- 4.43 Now, if you add curvature, there's two different types of curvature. There's so-called positive curvature, which is familiar to you from a sphere or from a, or from the beach balls, balloons that we talked about before. On a sphere, you could have angles inside of a triangle that sum up to way more than 180 degrees. Imagine you're here, say, in Florida, and you walk all the way over to California. And from California, you walk due north to the North Pole. That's a 90-degree turn. You went from going pure west to pure north. That's a 90-degree turn. Then you get to the North Pole and you make another 90-degree turn, and you go way back down to Florida. And then you make another right turn and you go back to San Diego. Those are three 90-degree angles. What's $90 + 90 + 90$? 270 degrees.
- 4.44 In fact, you can have a triangle that has as many as 900 plus degrees of angle in its interior. And positive, curved, or in a positively curved space, parallel lines meet all the time. I just gave you an example starting from two different places, but starting off parallel, converging at the North Pole. And that would happen anywhere on the surface of a sphere. Parallel lines will always meet. There's the third type of curvature in contradistinction to flat and positive curvature, called negative curvature. You can demonstrate this on a saddle of a horse, or more deliciously so, with a can of Pringles, which I had asked for. They did not arrive, so my rider has been violated—no, it's fine to do that.

- 4.45 But if you had a little potato chip at home, you can do this, draw on it a tiny little triangle, and that angle sum will be much less than 180 degrees. This one has an interior angle that sums up to about 60 degrees. Parallel lines will not meet, but they will diverge in a negatively curved space. So you could determine if we lived in a negatively curved space or positively curved space by making triangles or seeing if parallel lines, like lasers, converge. You just have to have a big enough triangle. So we have done that in cosmology in order to determine what shape of the universe we have, and that determines how much mass there is in the universe, and that in turn determines what kind of gravitational forces we experience.
- 4.46 Here are two different observers going about it, as I've just described. One in flat space measures 180 degrees. One measures 200 plus degrees, 210, negatively curved would be even less than 180. These are curvature determining tools. It's using geometry to measure the curvature of space. Here's a picture of my hometown, San Diego, and it shows that flat approximation. This is really good. You don't need to have the full spherical calculation on Google Earth to get from Miami to Fort Lauderdale or from San Diego to Los Angeles, because the radius of curvature is much, much bigger than the radius between La Jolla or San Diego, or Miami and Fort Lauderdale.
- 4.47 But if you're trying to go from the longest possible path on earth, that you can go directly from the tip. This is the one path that you can go from America —this is now Alaska—and you can go to the west coast of India in a boat in a straight line. You set sail at this course from Alaska, and you will never have to turn your ship, not one degree. You just keep going in that direction and you will come back to exactly this point in India, which you couldn't reach by sea in a straight line any other way. And this is just an example, but this, you can't use it as a flat approximation. If you tried to do it flat, you'd have to make these radical turns. You could not do it without changing course, but take my word for it, or go off and do this.
- 4.48 There's actually even a way that you can walk from the southern tip of South Africa to the eastern tip of Siberia. That you can't actually—you can do it on foot. I think it takes like two or three years of constant walking, but it would also look like a curved line on this type of a plot. It's like 11,000 km or something. So it depends on how far you go, whether or not you can apply a flat approximation to an intrinsically curved space-time. Now, obviously, we're going out very far in space now. We talk about cosmology. We're always going to use this curve space laws, and not the flat space approximation.
- 4.49 Now, here's good old Pythagoras sketching out Pythagorean theorem on the sand somewhere in ancient Greece. You could do this with Legos. Here is his famous equation : $A^2 + B^2 = C^2$, where C squared is the square of the hypotenuse, which is the longest side of a triangle. And here you see a famous 3, 4, 5 set of Legos. So this is 3 by 3. 3 squared is 9, 4 by 4 is 16, equals 25. Check that out. That's a Lego proof. There's many ways to do it, to prove it. It's only true in flat space. It's not true for a curved positive or curved negative.

- 4.50 When Aristarchus, some 23 centuries ago, set out to prove that the earth was a sphere, he used the laws of geometry only, and he established, using the measurement of the shadow cast by certain sticks on the ground in Cairo and in Athens, that the earth had a shadow of a stick off of the equator, that it didn't have at the equator when the sun was directly overhead. And that was only possible if the earth had some radius of curvature bending away by an angle proportional to the distance between Athens and Cairo. So he measured not only that the earth is curved, but he measured its radius of curvature, or what we call now its diameter or circumference. And from that, he could actually measure the distance roughly to the sun and to the moon as well. So he's the first person to wake up in the cosmos and start to measure these extremely vast distances.
- 4.51 There are multiple coordinate systems that you can express your position in a curved space-time. One is by specifying the angle and the radius, and that's called polar coordinates. Another is by expressing two angles, latitude and longitude being the most familiar, and knowing the radius. So you could do this and calculate what's called these different trajectories. And the same thing can be true on a negatively curved surface. It's called a hyperboloid. So space could have been positively curved, negatively curved, or flat. The space of the cosmos, the space, the angles between two different, three different stars, the angles between three different galaxies, three different sources of gravitational waves or of neutrinos.
- 4.52 And we use these different objects to measure the intrinsic curvature of space, and relate that to how much mass and energy there is in the universe. By measuring the geometry of the universe, we're actually weighing the universe. Because of the equivalence between curvature and gravity, and the source of gravity and the source of mass energy. And we'll see how far out we can go in just a bit. Here's another example of ants, ant astronomers. And they're trying to determine the shape of the surface that they're on. So they started off. They were initially parallel to each other. One was here and one was here. They were walking north. Both were walking north. They'll converge because it's a positively curved surface. If they're on a Pringle, they would diverge.
- 4.53 So we can use parallax to determine this. Way back in lecture number two, we encountered parallax with our thumbs and blinking back and forth. We can also do this again, measuring the apparent angles between one star at two different points in the earth's orbit, separated by 180 degrees of rotation, in August and February, any month, separated by 6 months apart. Looking at a distant star, there'll be some angle. You can measure that angle. You actually get a very, very skinny triangle. Remember, this is about 93 million miles. And this, at minimum for the closest star, is 4 light years. Some hundreds of thousands of times bigger. So you have a triangle that's extremely skinny, which means that these angles are very close to 90 degrees, but not quite 90 degrees, or else these lines would never meet.

- 4.54 And you can then check, are these angles, this angle plus this angle, exactly what are they equal to? They should be equal. Adding them up, what's that angle between them? And you measure the curvature of space, in so doing. So, to any star, we can measure this, calculate the parallax angle in a positively closed universe. Again, the parallax angle will be less than the astronomical unit divided by the distance. And in a negatively curved universe, the opposite will hold true. The angles will sum up to less than 180 degrees, meaning that these angles will be greater than the astronomical unit divided by this huge distance to the star.
- 4.55 And you can go out and do it and measure. But again, you're limited by the precision of your instrument. You can't measure an angle that's infinitely small. In other words, there's some degree to which the Gaia satellite is capable of telling you—is this 90 degrees, or is it 89.99999 degrees? And that's very important because it will have some uncertainty. It's unavoidable. And any human-made instrument will always have some uncertainty. That will limit how far out you can go. It'll limit the maximum D you could measure to. And if you have a limit on your maximum D, you may not be able to determine if you're in a negatively curved space, positively curved space, because you can't distinguish them within your measurement uncertainties. Is that clear?
- 4.56 All instruments have irreducible noise you cannot get rid of, and it may be the universe's curve for real, but on distances much greater than the equivalent resolution of your instrumentation. So you have to go out farther. Maybe stars are not sufficient. Maybe we need to go to galaxies. So these measurements by Gaia, the most state-of-the-art instrument ever built to do this, satellite orbiting in space for a decade, measuring a billion stars, it can measure out such that it can measure fractions of a 10,000th of an arcsecond, where an arcsecond is one 3600th of a degree. You can't contemplate how accurate and amazing this instrument is, but it has its limit.
- 4.57 Its limit is that it can measure out to about 30,000 light years, which is across the galaxy, effectively. And in so doing, it can measure distances to the Magellanic Cloud, say, but no further. Does that mean that the whole universe isn't curved? Nope. It could be curved. We just don't know, because this type of instrumentation, this technology, can only take us so far, literally. So we need a bigger triangle with more accuracy to determine whether or not this angle leads us to declare that the universe's spatial curvature is flat, positive, or negative. Those are the only three options.
- 4.58 So, how the universe curves depends on what mass energy is within it. By measuring curvature, by measuring these triangles, applying these laws of gravity from Einstein and Newton, we can weigh the universe, determine what it's made of, and we come up with shocking discoveries, that the composition is not nearly what we thought, as we'll see in just a bit. To do this took us through, and will take us through, a lightning tour of relativity. I cannot teach a class on cosmology and not teach a little bit about relativity. I do hope someday to teach a class about all these extreme things, from relativity to

gravity to black holes, etcetera. But this is just one fraction of one lecture in our exploration of the universe. So necessarily it must be brief.

- 4.59** So, in what's called Galileo's math, you could have an object moving at some velocity. And if you're on a skateboard traveling at some velocity, you could throw this shot put. Of course, you'd move back a little bit because there's a lot of mass here. And I'm accelerating it so there's equal and opposite reaction. Maybe I'd slow down a little bit because of that acceleration, but I could do it. If it were really light, I could throw it. Say I'm moving at 10 meters/second, I could throw the shot put at 10 meters/second. I'd like to be able to do that. Then the shot will be going 20 meters/second.
- 4.60** But what happens if you're traveling at the speed of light and you turn on a flashlight? The speed of light is a constant. You're moving close to the speed of light or maybe even at the speed of light. So would you get twice the speed of light? But no, no, the speed of light is constant. It can't be that. Those were some of the concepts that led Isaac Newton, sorry, that led these gentlemen to invent the theory of relativity. It's a very quick lecture, subset, on the laws of electromagnetism that we started this lecture off at.
- 4.61** The laws of electromagnetism eventually lead us to a set of four equations. They're called Maxwell's equations. We're not going to get into them. They're called differential equations. And if you massage them and manipulate them, Maxwell found that it predicted that there'd be waves of electromagnetic force. He called these light waves. These light waves would travel at a universal speed, the speed of light. That speed of light would be in this reference frame, would be about 300,000 meters/second. Incredibly fast, 186,000 miles/second. You may be more familiar with it. And he didn't understand there to be a wave without some medium for the wave to be operating in. Just like I can make a wave of water here, the wave is a propagation of energy density, velocity, and it's propagating in the rest of the water itself.
- 4.62** So back then, they supposed that the entire space was filled with a medium called the ether. And the ether was kind of like the ocean through which, the medium through which the waves were being propagated, or air in this room propagates waves of sound. They didn't know you could have a wave in vacuum. And it's still quite astounding and mysterious. Again, a topic for an entire Peterson Academy lecture.
- 4.63** So they realized that there'd be something that they called the luminiferous ether. Luminiferous means supporting light, and ether meaning this ethereal substance that they conjectured to be the wave-like support for waves of electricity and magnetism. They did experiments to determine how fast the earth was moving with respect to the ether. So the ether had the property that the speed of light was constant within it. But then when the earth was moving with respect to the ether and the speed of light, then you should get a different velocity at different times of the year, depending on how the earth was moving with respect to the sun.

- 4.64 So there'd be some reference frame, the ether frame, everything—the speed of light is constant only in that frame. And then as the earth moved towards or away from where the ether was moving, you'd get some difference in the speed of light. That was their theory. So here it's shown here. So, at different times of the year, you'd either be moving with the ether, say, at this time of year, or away from the ether. And other times of the year, you'd be moving tangential to it, and you wouldn't get any increase or decrease. So by doing a measurement throughout the year, they would expect to see variations according to the annual modulation of the earth's orbit around the sun.
- 4.65 These gentlemen, Michelson and Morley, at my alma mater institution at Case Western Reserve University in Ohio, they did a measurement which is astounding in its precision, and actually led to Michelson being the first American to ever win the Nobel Prize. And this experiment, we can't actually find it anymore. It's depressing to people at my alma mater because they can't find it. They'd love to be able to find it. They just declared the whole building to be a historic landmark. What they found was a null result, not a negative result, not a disconfirmation, not a proof. They found they saw no effect. That never did it occur that the ether had a velocity that you could measure.
- 4.66 So they assumed, at that point, something was very strange, because they never could consider a wave waving without some medium for it to wave within, is very disconcerting to them. The way out of it is something called length contraction, which is mathematically described here. We now know it to be a property that's consistent with Einstein's theory of relativity, that when an object moves, it actually shrinks a little bit in the direction that it's moving. And that shrinkage is directly proportional to the velocity relative to the speed of light, actually squared with the square root factor. That's called Lorentz factor.
- 4.67 So when something is moving, it gets squished, and when something is stationary, it has a constant length. Therefore, when it was moving with respect to the ether wind, if there was such a thing, it would be contracting in one time and expanding in the other direction, and we didn't see any relative motion like that. The only way that could be resolved is if the speed of light was considered constant and both space gets affected by motion and time gets affected by motion. So time is not absolute. This is Einstein's great, great insight, that two different observers, two different people in relative motion, will observe two events that are the same event, say, a firecracker blowing up. They'll observe them at completely different times, lasting for different durations, depending on their relative velocity with respect to each other.
- 4.68 That's what relative in special relativity means. It doesn't mean everything's relative. We don't know what's going on. No, it means, very precisely, if there's a relative velocity between two observers, one will measure a slightly longer amount of time elapsed between, say, two fireworks going off, if they're moving in a direction, say, from one of the fireworks to the other one. And one will measure a slightly different distance, and the ratio between the change in distance and the change in time is always

the speed of light. The speed of light is always constant for any observer, no matter how fast they're moving.

4.69 And it was astounding. It really affected how people thought, because we thought time was absolute, that time doesn't depend on how you move, but it does. And even the length of objects depends on how fast you're moving. And it was actually proven that time is relative as well. Not only do things shrink in the direction they're moving, the time gets expanded in the direction that something is moving as well. And this was discovered over 100 years ago. But there are ways to test it today using a type of subatomic particle called a muon.

4.70 A muon lives for a few microseconds at rest. If you have a muon, it'll die and disintegrate in a few microseconds. But if you have one moving because it's been accelerated through space near the speed of light, it will actually last longer and live longer than the muon at rest in your laboratory. And we do these measurements for undergraduates. We can measure the lifetime of a muon that comes from space can be 100 times longer than a muon just sitting at rest in the lab. It's a way to get, like, life extension. But the problem is, you have to accelerate close to the speed of light to benefit from this behavior.

4.71 OK, so Einstein, as I said before, discovered this property that time is relative and that space is relative. And, in fact, what's called simultaneity. In a moving frame, one observer will see an event happen before another observer. In a frame where they're all at rest, they'll see a constant time frame. And in a frame where there's decelerating, the opposite behavior will occur. So you get all these different types of crazy behaviors called length contraction, time dilation, and eventually, Einstein derived his famous equation, which you recognize as $E = mc^2$. This is his actual handwriting. And here he's crossed out this symbol L . I don't know why he used L , except I think that was the Lorentz factor.

4.72 And the actual equation is energy equals mc^2 , which is the rest mass of a particle, times the speed of light squared, which is a huge number, divided by this dimensionless constant, $1 - \frac{v^2}{c^2}$, divided by the speed of light squared. That's called Lorentz factor. So an object at rest has an energy intrinsic to its mass, of its mass times the speed of light squared. This speed of light squared converts to energy at this huge, huge amount. And, of course, this is responsible for things like the atomic energy revolution and atomic bombs, that an incredible amount of energy can be released from a tiny amount of mass because of the factor c^2 is so enormous.

4.73 So we went from adding velocities together in Galileo's time to this transformation down here, given by this Lorentz factor. And when you go very, very slowly, in other words, when the velocity of the moving observer or the moving object is much, much less than the speed of light, this term here is very close to zero. $1 - 0$ is 1. Square root of 1 is 1. 1 divided by 1 is 1 , and therefore γ is equal to 1 . And the Lorentz transformation becomes the Galilean transformation. In other words, the laws

of Newton are the same as the laws of Einstein when the velocities are small. And they're actually true, too, when the masses are very small as well.

4.74 I'll skip over this. So where do we come to now? The final thing we're going to conclude with is how mass affects the curvature of space. So, Einstein, in 1915, 10 years after special relativity, came up with the general theory of relativity. General because it includes the force of gravity. When you have a massive object, that affects the gravitational force felt by that object, which then affects how the object can orbit or move around the massive object. The Pythagorean theorem in Einstein's theory, which is, by the way, the most mathematical we will get in this class—the Pythagorean theorem gets written to include time as well as space. And the time component, as well as space, leads to what we call space-time.

4.75 And as I started off lecture, lecture three with, we combine the symmetry of the expansion of the universe with the independence of where the measurement is done, who makes the measurement, how long the measurement takes. We get the Copernican principle, plus this time variation, leads to the cosmological principle. It's a symmetry principle, that the same principles and laws of physics are independent of where you are in the universe. So the universe is isotropic. It is homogeneous. It's the same for observer A as observer B. This leads to the cosmological principle.

4.76 And then the question was, is it independent of time? If the universe is expanding, what is causing the expansion? So I said that you can only use stars as long as they're within 10,000 parsecs, or roughly 30,000 light years. That's as far out as you can use parallax. Can you have bigger triangles in the universe? The answer is yes. And actually, we started off to discover that the universe can feature curved spaces on much, much smaller scales. Actually, our solar system features curved space-time. Now, what led Einstein historically to do this?

4.77 As I said, the laws of the orbits of the planets were found to be ellipses, and that is still true. But the ellipses have some peculiar properties. For example, it was known since the time of Isaac Newton that the orbit of the planet Mercury was an ellipse. But the point of closest approach to the sun was moving every year. Every year, the point at which the ellipse was tilted such that it was closest to the sun, the perihelion, that was changing with time. No one could explain it. And it's impossible to understand in the laws of just Isaac Newton. I think that this is a great mystery. It's a great opportunity for me to make a name for myself and to maybe, perhaps, predict why this would happen.

4.78 And he, in fact, retrodicted, meaning that he explained a phenomenon that had already been observed, and he did it using the laws of general relativity. And this is the historical interlude. It's kind of amazing to think about what happened. Einstein proposes this 7 years after the laws of special relativity, of the constancy of the speed of light, of the relativity of simultaneity, of length contraction and time dilation. He writes to the director of the biggest observatory in the world at that time, Mount Wilson, in 1913. He says that because of the presence of the sun, a star behind the sun

will appear shifted in position when it's behind the sun, versus 6 months later, when the sun is far away from that star. He predicted, due to the gravitational bending of space-time, that light will take a different trajectory sketched here.

4.79 How do you see a star that's near the sun? The answer is, you can't, unless there's a total eclipse of the sun. When there's a total eclipse of the sun, as there will be visible throughout most of North America on April 8, 2024, just a mere few months, depending on when you're watching this, or maybe it already happened. There will be a total eclipse. And if you've ever seen one, it's a magical experience, even for astronomers that are kind of grizzled, jaded old veterans like me. I will travel and I will travel. It'll be visible in Austin, Texas. I'm going to go there and visit it and watch it with my family, as we watched one in 2017. The next one, I don't believe, is until 2045. So you might want to get your tickets now to Austin, Texas. Join me there.

4.80 And there was an eclipse back in, only a year after Einstein made this first astounding prediction, that if you could observe the sun, a star near the sun, it would be in a slightly different position than it would have been if the sun weren't there, namely 6 months later when the sun, from our vantage point, has moved away. Luckily, our Einstein thought, "Hey, there's a total solar eclipse coming just a year from now. It's going to be in this region of eastern Europe, near Crimea." Now, to go and see an eclipse today is kind of hard. It's not that hard. You need equipment. You need filters. You need all sorts of things. Back then, they didn't have CCD cameras and all sorts of advanced television. They had film cameras, and they kind of have huge, huge expeditions on horseback to the middle of this place.

4.81 And back then, this region of eastern Russia or western Ukraine was contested. I don't know if it's still contested now. This is called Crimea, so hopefully, when people are watching this, it won't still be contested. But as of now, as it was 100 years ago, this was pretty contested. It was not a good time to try to do an astronomical observation from eastern Europe in 1914. There was a lot of incredibly horrific battles being fought not too far away. So actually the eclipse occurred, of course, but the Russians captured the European and English astronomers, and they were not able to collect any data. It turns out that was a good thing because Einstein actually made a math error. He got the calculation off by a factor of 2.

4.82 However, in 1919, a much, much better eclipse took place. And this one, the war was over, and they did an observation that returned with data that made Einstein perhaps the most famous person on earth. Here's, like, slice of the New York Times front page with a depiction, and the headline reads, "Lights all askew in the heavens. Men of science more or less agog over the eclipse observations." And I love this, it's so sexist down here. "A book for twelve wise men." And Einstein himself was pretty, was pretty egotistical about it. He was asked, "What would you have done if the results did not confirm you?" And he said the following. He said, "I would have felt sorry for God

because my equations are right and he would have been wrong." But of course, he wasn't wrong, because he had time to correct that factor of 2.

4.83 And what was observed was this tiny little shift in the positions of stars relative to where they would be if the sun weren't there. The sun is an enormous, massive object, hundreds and thousands and millions of times the mass of the earth. And because of its mass, attracts rays of light, deflects them from being parallel, if they were initially parallel, and it changes the position of the light on Earth. So it's been confirmed since then, many times.

4.84 And this is the picture that we see nowadays. The presence of mass curves space-time. Curved space-time changes the trajectories of everything that moves around it, from massive particles to light itself. And if you have enough mass, you get what's called a black hole, where even light propagating at the maximum possible speed, 300,000 meters/second, kilometers per second, cannot escape. Light effectively gets sucked into the gravitational curvature.

4.85 Now, that's just the space-time around the sun. What happens in the space around all the matter in the universe? That's our task for the remainder of the course, in a sense. To measure the curvature of space is a measurement of mass. You can invert these equations. You have a way of measuring mass. You have a scale that can measure the mass of objects you could never hope to even reach, like the sun, and weigh it based on merely measuring the angular displacement of light coming from stars much, much more distant. This is a way to measure the curvature of space locally, and it holds true throughout the entire observable universe.

4.86 So, to finish up, the question is, can we find a bigger triangle? The answer is yes. We will see. The biggest triangles we can find in the universe are the characteristic modeled pattern of fluctuations in the cosmic microwave background. We'll see where those come from in lecture five. But these fluctuations have a characteristic size scale. They make what's called a standard ruler, something of fixed length. And when you have something of fixed length, like the distance between your eyes or the distance of a ruler, all you have to do is measure the angle subtended between the edges of this ruler and you as the apex of the triangle.

4.87 And astronomers have done that, and we have measured the curvature of the universe to be exactly zero, to within the precision of our instruments. That means, as we will see, the universe has a certain amount of mass, a certain number of grams per cubic centimeter, a certain number of trillions of tons per cubic meter or cubic light year. And we'll see what that corresponds to. And why that's so important will be incredibly imperative to you. Because the amount of mass and the energy contained within that mass, that will affect the fate of the universe itself, whether it will expand forever, whether it will contract upon itself, or whether it will somehow coast to infinity. And those different possibilities will be covered in the next lecture. Thanks a lot.

Lecture 5: Cosmic Fossils

At a glance — Cosmic fossils — relics of the early universe.

- 5.1** When I was a kid, I always wanted to be Indiana Jones. And I'm lucky because childhood dreams come true. I get to study archeological relics that have been traveling through space and through time. And today, I'm going to take you on journey as well to investigate exactly how we've come upon these very, very precious fossils and what they can tell us about not only what the universe was like in the very, very distant past, but what's going to happen in the very, very far future. So come along in this journey back in time and back in space, as we uncover mysterious relics that look nothing like we see the world around us today.
- 5.2** So as we've discussed in previous lectures, when you look back in space, you're looking back in time. And when you look back in time, things don't look this way that they look as we look out, say, today. We look at something at our feet, it's much younger, newer, fresher, perhaps, than something that's deep down underneath the surface. And just like with archeologists, we can do the same, but the object we're looking for, none of them really exist in the same way that things exist today. So it's very different. There might be not dinosaurs around anymore, but there are lizards and things like that that are similar. But the things that we're going to be looking for, there are almost no resemblance to things we can see around us today.
- 5.3** So we'll be talking about three major types of fossils. These fossils have been traveling through space and time, and they reveal not only the conditions of what the universe was like when they were produced, ie., when they were born, if you like. But they also indicate everything that's happened since. They encode the properties of the universe from the time in which they were alive, if you like, walking around. Unlike dinosaurs, which we see them as a snapshot of when they actually died, you don't really use them to learn about the history of the plate tectonics on Earth or something like that. But with these fossils, you do. We'll be looking at three of them.
- 5.4** We've already encountered one. One is called the galaxies and stars within those galaxies that trace out the fact that we know the universe is expanding. Those are types of fossils. Those stars within the galaxies cause them to be tracers of the fact that the universe is expanding. We use properties of their light, and we discussed in previous lectures. Today, we're going to talk about a different kind of light, a light that's totally invisible to your eyes. And yet, it is the oldest form of light that exists that we can see in the universe. And that's called the cosmic microwave background radiation. It traces from an epoch, not of a few billion years ago, but perhaps right near the beginning of our universe's history, a mere 380,000 years after our universe came into existence.
- 5.5** The final piece of fossil evidence we're going to investigate today involves the chemical elements. How did they come to be here? How did we come to have a very, very ornate

periodic table of the elements? Well, some of those elements were made in future generations of stars, just like the stars that we've talked about. And some of those elements were made only during the first few minutes after the origin of our universe. And this type of fossil is a tracer of the first few seconds of the Big Bang's history. So it's incredibly precious. And these three fossils together, when assembled into one unified whole, provide consistent evidence for an evolutionary path that took us from extreme temperatures to very, very low temperatures that we see in the universe today.

5.6 The temperature and the age of the universe are intimately connected, as we'll see today. Those three pillars, those three types of fossils, are the quarries that we're going to be digging for today. We discussed the Hubble expansion when we described the expanding universe. We looked at the cosmic distance ladder. We saw how far things away were from us. We correlated that to a property called the redshift of those galaxies. Redshifts are easy to measure. You look at the color of the light coming from a galaxy. You compare it to what the color would be like if you were at rest with respect to that galaxy or in our own galaxy, which is not expanding. And the combination allows you to see what fraction of the speed of light the galaxy is moving away from us.

5.7 Using the law of Leavitt and Hubble, we can correlate the redshift to a distance. Therefore, we can trace how far things are away from us and how fast they're expanding away from us. Farther away, longer time elapsed, more expansion. If you run the movie backwards, the thought is, the universe was more compact, denser, and we'll see, hotter. The photons that are part of those galaxies are different than the photons we'll be talking about today. The cosmic microwave background radiation photons are not from stars. They're not from galaxies. They're not even from exotic phenomena like black holes. They're actually from a different type of process called the formation of the first atoms.

5.8 Recall that atoms are comprised of a central nucleus surrounded by one or more electrons. Those are the origin, that is the structure of the chemical elements. The nucleus itself, we're going to probe how the nucleus came into formation in the study, as we get to today, of Big Bang nucleosynthesis. So first we're going to go backwards in time. We started with galaxies, which are already full of chemicals and elements and stars. Now we're going back in time to the formation of those first elements, the first atoms. Then we're going to go at the end of today's lecture to the origin of the nucleus of those atoms. That's our travel itinerary today.

5.9 So just to remind you, we talked about in earlier lectures, the redshift of a galaxy tells you how much expansion has taken place since that light was emitted. We observe it today. We don't know the exact time it was emitted, but we know in an accurate sense how much the universe has changed in size. From its current size, backwards in time, it must have been smaller. Each unique redshift corresponds to a unique point in the universe's evolution, a unique time after the expansion started. Now, the early universe doesn't have these galaxies. There wasn't enough time for them to form until a few

hundred million years after the Big Bang. We're going to be looking at an era of the universe's history where there were no structures, there were no stars, there were no galaxies, there are certainly no planets, people or anything like that.

- 5.10** In the expanding universe analogy, we noted that, as a raisin bread expands or a balloon is inflated, the separation between different regions on the surface of that object or in the bulk of the raisin bread, it changes. The more far apart two raisins are here, the more expansion will also cause them to double and increase in size as the raisin bread or the universe keeps expanding. So again, when the bread starts off, it's more compact, it's more dense, and as it expands, it becomes more diffuse.
- 5.11** That also leads to a cooling phenomenon. We can correlate the time since the Big Bang, measured in seconds over here in this plot, and the universe's temperature. If you take something and you compress it, it necessarily gets hotter by the laws of thermodynamics. As you take something and expand it very rapidly, it cools off. And today, we have galaxy, stars, and planets. The universe is a relatively—it's actually, we can say it's an absolutely low temperature. It's very close to the lowest possible temperature that objects can have, which is called absolute zero. Our universe, we'll see in just a moment, is suffused with a type of temperature, a heat, that corresponds to a level of temperature a mere 3 degrees above absolute zero. Incredibly cold.
- 5.12** And that's very important because if the universe was extremely hot, the electrons and protons that make up atoms would be blasted apart by the energetic photons that were present. And in fact, that's what happened continuously in the extremely early universe, just a few nanoseconds after the Big Bang. So the lesson is, as time goes on, as the universe expands, it cools. As it cools, the characteristic wavelength of light is going to increase—redshift, if you will. And we'll see what that implies for the types of fossils that we're going to be digging for today.
- 5.13** So as I said earlier, there are things with different temperatures. The sun has a much higher temperature than the Earth, and the Earth has a much higher temperature than the planet—I almost said the planet, Pluto. Of course, has been demoted. So I'll say the planet Neptune, just to be politically correct, planetarily correct. So what does it mean to say the universe's temperature? When you put a thermometer, does that mean you're going to get a consistent reading? No. Different objects have different temperatures. But that wasn't always so.
- 5.14** If you go back to the extremely early universe, before there were any objects—stars, galaxies, planets, people, academies—the universe had a characteristic temperature. That was a type of energy associated with the characteristic wavelength of the light that was already bubbling throughout the universe. Where that light came from is a secondary question, and that will be the final fossil that has not been detected yet, and is actually the subject of the research that I do. There's another type of fossil that would indicate how the universe got the energy inserted in the first place. And that's a topic for the final lecture of this class. So stay tuned.

- 5.15** Very simple diversion into what's called statistical mechanics or thermodynamics of particles. That's that when you have a cool gas, the characteristic temperature is lower. The velocity on average of these molecules, in either the hot gas or the cold gas, the average velocity is the same. It's zero. You have as many atoms moving to the left as to the right, so their average velocity cancels. But what's called their root mean square, if you take their velocity and you average over the square of those velocities, you get a much, much higher level of value for the RMS of the hot gas than you do for the cold gas.
- 5.16** What we call heat is the random motion of trillions of molecules. That 's what we associate with heat. The more average velocity they have, the more energetic they are. The more energetic, the higher their temperature. Now, if you cool the gas down to absolute zero, all motion stops. But it takes an infinite amount of energy to cool something down to absolute zero. So things can tend towards low temperatures. And in fact, in my laboratory, we have instrumentation, refrigeration, that can cool down objects to just three 1,000ths of a degree above absolute zero. Incredibly cold. And that's nowhere near the record low temperature ever set,
- 5.17** which was not done certainly in a type of laboratory that studies cosmology, but studies the properties of atoms. And you can actually cool atoms by blasting them with lasers. And basically every single atom here, you could blast with an individual laser such that it cancels off its velocity as much as you can. And you'll start to cool from this kind of a state to this one. Then you blast these with a different wavelength of lasers. And eventually you make a very, very small collection of maybe million or so atoms, and they're at a billionth of a degree above absolute zero, a million times colder than what I can achieve with a half a million dollar dilution refrigerator.
- 5.18** So the temperature is the average kinetic energy or average velocity. When things are heated to different temperatures or cooled, if you like, they glow at different temperatures. The lights in this room have a characteristic temperature of a few thousand Kelvin. Kelvin is an absolute scale related to the Celsius scale. But unlike Celsius, you can have temperatures below zero in Celsius, but you can't have temperatures below zero absolute in Kelvin. The sun glows with yellowish color because it's at a temperature of about 5,500 Kelvin on its exterior. In its interior, it's at millions of degrees. That's the temperature associated with nuclear fusion, as we'll see as we go on in today's lecture.
- 5.19** Something that's glowing red-hot is about 1,000 Kelvin, and something that's yellow is about 2,000 Kelvin, and something that's blue is about 3,000 or 4,000 Kelvin, and so on. We can ignite this and you see the characteristic blue flame there is characteristic of its temperature. It's heating air molecules up. They're moving in random motion. They're emitting at a temperature associated with this characteristic temperature of several thousand degrees Kelvin. If you keep going up and up in temperature, the peak

wavelengths at which the so-called black-body radiation is peaking, is shifting to shorter and shorter wavelengths.

5.20 So a 12,000 Kelvin black body will emit, and a black body is just something that's perfectly absorbing and perfectly emitting. It will emit at a very short, beyond the violet or ultraviolet wavelength. And something that's cooler than, say, 1,000 Kelvin is still very hot, much hotter than room temperature, which is about 300 degrees Kelvin. It's still three times that. It peaks mainly in the infrared. You'll see it maybe, but you'll mostly feel the infrared radiation being emitted by it. So this is an inverse proportionality between temperature and wavelength. Higher temperature, shorter wavelength.

5.21 Now, what was present in the early universe? To answer this question, we're going back in time like that archeologist, like Indiana Jones. We're starting at things close to us. Those are galaxies and stars. We're going back to a time period before those stars and galaxies formed. At that age, the universe was much hotter than when galaxies formed. And in fact, we can go back to the time that it was just possible to have the formation of the simplest atom, hydrogen. Hydrogen is comprised of a single proton and a single electron. This object can be stable at temperatures below several thousand degrees Kelvin. If you heat up a hydrogen gas above a few thousand degrees Kelvin, it dissociates and breaks apart or ionizes.

5.22 Ionization means you now have a plasma instead of a neutral atom. You have a separate gas, if you like, of protons and a separate gas of electrons. Still electrically neutral, but ionized. Two different plasmas. The temperature being above 3,000 degrees will always lead to this ionization state. If we look back in time, if we go back to a time higher than 3,000 Kelvin, if the universe was higher than 3,000 Kelvin, we should expect to see a universe with no atoms in it, no hydrogen in the atomic form. In other words, all the protons and electrons will be bound together when the temperature is below 3,000 Kelvin. But if we go back in time, we're going higher in temperature, we'll see it completely dissociated, completely ionized.

5.23 And ionized gasses, as displayed here, are opaque. The sun is so hot that all the gas is in ionized form. It's still different elements, but they're completely ionized. And the fact that you cannot see into the center of the sun, even though it's a gas, relatively dilute, you can't see through it is a byproduct of the fact you can't see through a plasma. In fact, mirrors are like plasmas that are stable. They're not hot. The electrons in a mirror are extremely mobile, and that's why they can reflect light very easily. So you can't also see through a mirror. Mirrors are opaque, if you like.

5.24 And plasma that's ionized, or plasmas are always ionized, but plasmas of this kind are opaque. You cannot see through them. So what does this have to do with going back in time? Well, as we say, if the universe was denser yesterday and much, much denser billions of years ago, eventually you'll get to a time when the universe's matter, whatever matter would become the galaxies and stars that we see today, was in an

ionized plasma. It was too hot to be bound into protons with electrons to make hydrogen. And hydrogen is the ingredient, the only ingredient that stars need to start getting their fuel, to start producing nuclear fusion, to produce light and heat and eventually conglomerate into galaxies.

5.25 So knowing the fact that you can't see through an ionized plasma, knowing the fact that if you turn the temperature up on hydrogen, no matter what, it will eventually dissociate, and knowing that hydrogen was present in the early universe means that there must have been a time before which you couldn't see any light in the universe because it was opaque. And after that time, you could see things far away as you like, or far away there were light to be emitted.

5.26 And so the entire universe was effectively a cloud, a gas. And at one point it was completely ionized. And then after a certain point, it became neutral. Hydrogen formed from the fusion together of an electron with a proton. That time period, that epoch, is what produced the cosmic microwave background radiation. Although at that time, the wavelengths were not in the form of microwaves, today, because the universe has expanded and cooled, they are. So today we see ourselves around us surrounded by a glow of microwaves, as if we're inside of an oven. And thank goodness, the oven is at a very low temperature, only 3 degrees above absolute zero, or else we wouldn't be here studying the properties of it.

5.27 Now, as I said, this glow is not the way our universe looks. Today, we can see out literally billions of light years. So we don't live inside of a plasma anymore. The plasma has gone away. If you like, the fog has been lifted. So we actually see a very different universe today. Our job in the culmination of this class is to understand, how did the universe go from perfectly isotropic, perfectly homogeneous, as required by the cosmological principle, to have this type of structure, with galaxies and stars and dark matter and planets and academics?

5.28 So at one point in the universe's history, or actually it took several tens of thousands of years, but that's very small compared to several billions of years that the universe has existed for, the universe cooled down, the ionization stopped, and the atoms formed. I call that fusion of atoms because it's really bringing together a proton with an electron to fuse it together to make a hydrogen atom. That's the simplest form of fusion. And at that point, the universe became transparent, and you can see through a neutral gas of hydrogen atoms. That's perfectly a viable thing to do.

5.29 But that light, the leftover heat that was still present at the time of the formation of the first atoms, is still around us. And that's what we call the cosmic microwave background. And we can study that to get information about the epoch at which those fossils, those photons, were formed. That's our goal in the second big pillar of the study of cosmology, the cosmic microwave background. So we went looking like this today, or at relatively recent ages, and we can look all the way back in time to the formation of this cosmic background radiation.

- 5.30 Just this arrow here shows, as the universe expands, it cools. Here are some different characteristic size scales of the different wavelengths of light. We'll be talking today about microwaves. Microwaves are longer than visible light. They're longer than infrared wavelengths, and their characteristic wavelength is centimeters, so hundredths of meters. They are associated—I don't know why this is so useful, but they're the size of butterflies. I think that's cool. The cosmic microwave butterfly. They correspond to frequencies of several billions of cycles per second. We call that several gigahertz. And the cosmic background that we're going to see has a characteristic temperature on this very bottom scale of about 3 degrees Kelvin today. But when it was formed, it was a characteristic wavelength of this light visible in ultraviolet.
- 5.31 And since the universe has been expanding since the time when the universe was hot enough to keep hydrogen ionized, it's cooled off by a thousand times. From 3,000 Kelvin to 3 Kelvin. That means the universe has expanded by a thousand times. Or the CMB, cosmic microwave background, is produced at a redshift of 1,000. The wavelengths stretched by a thousand times since then. So optically, things look like this with galaxies and stars. But actually, surrounding us, there are way more photons, way more particles of light, in the form of radio waves, microwaves, and their kind. So it's actually static.
- 5.32 If you have an old-fashioned TV, which I bet almost none of you have, you used to be able to tune it to a channel and get this pattern. You won't see this on YouTube. You won't see static like that. And it was discovered in this telescope. This doesn't look like a telescope, but it is. It's a radio telescope in New Jersey. And there were two astronomers in the early 1960s, and they were using this telescope. Now, why would astronomers be using a weird telescope like this? They actually weren't looking for anything astronomical. They were working at a facility called Bell Laboratories. And Bell Laboratories is kind of like Google or something like that way back in the 1960s. It was a cutting-edge technology lab where they would do some things based on the very first satellite that ever had been launched.
- 5.33 So the very first satellite was launched by the Soviet Union, 1957, called Sputnik. And the United States got very nervous because Sputnik was actually made from a nuclear warhead that had been decommissioned of its nuclear payload, and they had put on a tiny little radio transmitter on it, and it would fly over America. And so the Americans were naturally worried that, well, they could actually next time put the nuclear warhead back in and drop it over Washington, DC, or something. So America was very embarrassed at that time. And then later, the Soviets launched the first astronaut, and so the US was beaten again. And so the US wanted to have some level of national pride. And one of the things they could have pride in is making the very first communications satellite.
- 5.34 So today we take for granted. Every week, there's a launch of a Starlink satellite or a NASA communications satellite or a defense satellite. But back then, this was the first

or second time it ever happened. And what they did back in the '60s is they would realize that you could actually communicate with things beyond the horizon. Over here in California, there's a NASA facility called the Jet Propulsion Laboratory, where not far from where I live. And then in New Jersey was Bell Labs. And they realized they could communicate signals to this object if they were to shoot radio waves and bounce them off of a satellite floating around up here. That is what that thing is supposed to represent.

5.35 They would broadcast from a dish down here. They would broadcast radio waves, and it would bounce off of this, and it would come down here, and you could communicate beyond the line of sight beyond the horizon. So this was the first satellite. This was called the Echo and Telstar satellites. And so we actually won this race. So it gave us some national pride. But these two astronomers were working at Bell Labs because they had discovered something very peculiar. They knew how much signal was being shot at them, bounced off the satellite, because there were so many watts of energy being transmitted from the Jet Propulsion Laboratory. They knew it was so many watts of energy, and then it was getting reflected, and they were seeing the signal that was being transmitted.

5.36 But for some reason, they were getting a ton of extra noise. They couldn't understand why the noise was so much higher than the signal that they were seeking. They could see it, they could hear it, if you like. And so it was the very first telecommunication satellite. This is 1961. We take these things for granted now. And the astronomers, they could not figure out. And it was very embarrassing because they thought their system should be many, many times better than it was. So be like paying for one gigabit internet and getting like one megabit because there's so much noise, you're not getting enough signal. And so they wanted to get what they paid for.

5.37 So they studied this thing and they found that, not only do they get a certain amount of signal, so this black lines represent the signal, but they were also getting a tremendous amount of noise. These red lines are like noise waves coming in. And they found no matter where they pointed it, even if they didn't point at the satellite, if they pointed 180 degrees away, they were also getting this noise. Now, I'm from New York and not far from New Jersey, but not too close, because New Yorkers don't like New Jersey very much. But if you looked in a different direction, you also got the same exact noise, but not the same exact signal. You got no signal when you looked away from, say, the Jet Propulsion Laboratory. But you got the equal amount of noise. They couldn't understand it.

5.38 Little do they know that a few miles away from Holmdel, New Jersey, was a place called Princeton University, where my PhD advisor's PhD advisor, my grand-advisor, David Wilkinson, was working with his colleagues and friends, including Jim Peebles and Robert Dicke. They had been looking for a signal that would account for the noise. So they were seeing, the astronomers realized, thanks to talking to Dicke and my late

grand-advisor, that they were actually seeing not only the satellite signal, but noise, heat, microwave energy in the exact same wavelength range as was being transmitted, but it was coming in all directions, at all times of year, without change. It was a background hiss, a hum of noise.

- 5.39** They ended up winning the Nobel Prize. My grand-advisor didn't, sadly. One of the four Princeton colleagues eventually did win it. So they discovered this background. Background means it comes in all directions. It doesn't depend on time of year, time of day, season, who you are, where you are. So it's exactly consistent with the cosmological principle. It's an equal amount of noise, independent of time, season, day, person, wherever it is being done. And what they realized they were seeing is depicted here. In the analogy that we like to use, this is another space-time diagram.
- 5.40** This person is looking out in space, he's looking back in time. Here at the present, we're at the apex here. What we analogize this to is looking out and seeing galaxies and so forth, the earliest light that we can see before these galaxies are much, much younger than this CMB last scattering surface. It's exactly analogous to looking out—we have this often in California. It goes by the name of May Gray or June Gloom. I don't know if you get it in Florida or not, but it's called the marine layer. And what it is, is the fact that clouds scatter light very, very efficiently. And just like you can see up to the surface of the cloud, and no more, with the cosmic microwave background, it's an ionized plasma. We're not seeing it at the temperature at which it was a plasma. We're seeing that plasma redshifted, cooled off by 1,000 times.
- 5.41** So this traces to when the universe was 1/1000th of its current size. It was a 1,000 times hotter because of that factor. And that means we're seeing microwaves that are the very first light that could ever be seen. And that's very important. Humans are extremely visual. We have something like 50% of our brain is dedicated to vision. So it's very important to make use of all the light that humans can see or detect. And this is the oldest light, not only that we can detect with our technology, it's the oldest light that ever could possibly be seen.
- 5.42** In the final lecture, I'll introduce you to a fourth type of fossil called a gravitational wave. A gravitational wave is not a form of light. It's a form of radiation, but not light, not electromagnetic, not nuclear. It's a form of the ripples in time and space that we talked about in lecture four. And that phenomenon may allow us to see with a different form other than light, things that happened earlier than the surface of last scattering, this epoch of a 380,000 years after the Big Bang. And the analogy I like to use is, yes, this person can't see an airplane up here, but he could hear it. So using a different type of sensation, not light, in that case, sound, this person could learn something about what's happening beyond what he can see visibly. And we'll see that in order to get to earlier and earlier epochs, closer and closer to time zero, we need to use a different form of radiation. That we'll save for lecture number 8.

- 5.43 We look out today, we see on average, no matter where we look, when we look, who we are, we see 3 Kelvin. Actually, it's 2.7262 8 degrees Kelvin. And it seems perfectly uniform. No fluctuations, no variations. And you might wonder, as long ago as Isaac Newton had wondered, in a perfectly isotropic universe that 's completely homogeneous, how does anything know where to form? What breaks the degeneracy and allows a galaxy to form over here in a giant void with nothing in it to form over there?
- 5.44 That's a great mystery, because in truth, it is the case that a perfectly uniform gas, you will have no fluctuations, no variations, no breaking of isotropy or of homogeneity. So it was a great mystery to see how could we go from perfectly uniform to extremely non-uniform, as we see today. And the answer is, it's not perfectly uniform. Our goal, again, in lecture 8, is to have a plausible mechanism, consistent with all the things that we've studied up until now, that can account for the tiny variations needed to make little seeds of gravity grow, to turn into galaxy, stars, planets, and people.
- 5.45 So just to recap, the leftover light from the formation, from the fusion of proton with electron to make hydrogen, there's heat still left over. Just like nuclear fusion, which we'll encounter very briefly, allows for the combination of two protons together, accompanied with a vast release of heat. That's why nuclear fusion produces the light from the sun. It's leftover heat from the fusion process. It's not 100% efficient. So too, when atoms fuse from proton and electron into a neutral hydrogen atom, there's leftover heat. That's what the CMB is.
- 5.46 Over the years, we've studied it more and more from that perfectly uniform orange image that Penzias and Wilson would make. In the early 1990s, we made a map that revealed tiny fluctuations, tiny deviations from perfect isotropy at the level of one part in 100,000. So one part in 100,000. I don't know if any of you bowl. I bowl, but not very good. A bowling ball, if you looked at a bowling ball surface, you would see, under extreme magnification, you would see it 's actually not perfectly round or smooth. It has some variation in its surface. That's natural byproduct of the fact it's made up of atoms and so forth. And imagine this is the radius of the bowling ball, 10 centimeters. So I'll call this 0.1 meters. These fluctuations are on average a few 10ths of a millimeter, or 10^{-4} meters. So this is 10^{-1} meters.
- 5.47 So the fluctuations on average compared to the radius are 1,000 times smaller. So your bowling ball looks very smooth. Actually, if you measured it, it's about a fluctuation of about 10^{-3} of the radius. These fluctuations are one part in 100,000. That's 10^{-5} . That's 100 times more smooth than a bowling ball. It 's incredible how these tiny fluctuations lead to everything that we see. Without the fluctuations at this level, too big, there would have been really massive fluctuations in the early universe that would have formed only black holes. We wouldn't be here. Too small, we wouldn't be here. The universe would be too uniform to form.

- 5.48 And over time, my colleagues have made exquisite measurements. These are three different spacecraft launched from the late '80s up until 2009. And my colleagues and I nowadays are even looking at deeper and deeper images than ever before, using tools and technology at the South Pole, Antarctica, and in Chile, where most of my research takes place. This telescope, again, it doesn't really look like a telescope, but it is. It certainly doesn't look like an optical telescope. This heat is the leftover relic of the Big Bang, the oldest photons in the universe, the oldest light fossils we can see.
- 5.49 So how did it go from being visible light, optical light, infrared, ultraviolet, to microwaves? The hint is on the screen. It really is the result of the fact that the universe has expanded. Since the time those first atoms formed, we don't see atoms forming around us today. They only occur at a great, great distance from us, or equivalently, looking back in time, a great redshift, when the universe was 1,000 times smaller than it is today. So it's due to the expansion. The universe is expanded by a factor of 1,000. The wavelengths went from optical to microwave, one millimeter to one micron, or the other way around, one micron to one millimeter.
- 5.50 So if you were around back then, you might have been able to feel the heat. It wouldn't have been a very interesting place to live. It would have been just protons and electrons, and all of a sudden, they coalesce and condense, and there's a lot of heat left over in the form of infrared. And then over time, almost nothing would have happened for several hundred million years to a billion years, until the very first stars formed. I'll talk about that briefly.
- 5.51 So if you look at all the different types of energy and matter that we know about in the universe, there are protons, there are neutrons, there are electrons, neutrinos, photons, and there may be dark matter. We'll see about that in lecture 6, cliff-hanger coming up. And the dark matter is completely unlike the other objects on this table. And of course, the photons are very familiar. We know we live in a universe with light because we can see each other. The energy of a photon depends on its wavelength or, inversely if you like, on its frequency, its wavelength. The energy is proportional to the frequency, so inverse to its wavelength. The higher the frequency, the longer, the more the energy. And the energy comes in discrete quanta called photons. And those are very simple objects to understand.
- 5.52 So we look at it, we're seeing an image of the baby universe, the oldest fossil light we can imagine. It corresponds to an age of the universe of less than a half a million years after the formation of our universe. So the universe was still quite warm, 1,000 times higher temperature, about 3 Kelvin. And this fits in to a theoretical paradigm that is consistent. Even though it is completely different than the Hubble expansion of galaxies, the actual conclusion that you would draw, if there were only observations of galaxies, you could understand and predict that the universe was smaller in the past and has been expanding. If there were only microwaves, and astronomers only had

microwave telescopes, you would also come to a conclusion that the universe was smaller, denser, hotter in the past.

- 5.53** Two completely different types of technology, two completely different theoretical paradigms, but actually synthesized into one. Those two different things are actually just manifestations of the same model that the universe is expanding, which is the heart of the Big Bang model. You would see redshifts from galaxies, that redshift that you would see would agree with the redshifts that you see from the CMB. But there's no cosmic distance ladder that relates the distance that you can measure using the cosmic microwave background's temperature and the distance ladder that we use to measure the distance to galaxies. So there's no overlapping rung. We'll see there is an overlapping, there is an overlapping angular way to measure distances that will lead us to believe the properties of the universe are even more consistent with an origin in a Big Bang.
- 5.54** So this wasn't really discovered until 30 years after Hubble observed expanding galaxies, the expanding universe via the redshifted galaxies. So it's been measured exquisitely accurately. It fits in as a consequence of the fact the universe was hotter and denser in the past. It doesn't yet explain where the universe came from, doesn't yet explain if there are other universes, but the paradigm is now completely consistent between observations in two wildly different fields, microwave astronomy and optical astronomy.
- 5.55** And the next thing we're going to do in the next major pillar and fossil that we're going to study is completely different from either one of those two. It doesn't even involve light. It doesn't even involve microwaves or light, which are just different forms of the electromagnetic spectrum. The paradigm is the naturally and a hot universe that started expanding, it cooled off, eventually reaching a temperature where hydrogen was stable and could form before which it wasn't. Then since that time, you could see all the way back from the formation of those first atoms. We call that light, the cosmic microwave background.
- 5.56** It corresponds to a redshift at a given temperature. You can uniquely correlate the temperature of the universe today at any time in its past and compare it to the size of the universe via redshift, as we did in previous lecture. We can actually relate thermodynamically. We can consider a very simple problem. If you have a region of the universe that is a box, and that box has a link that we call A , and today A equals 1. If you have a box, and the box has some dimension, today A changes with time. It's a scale factor. Today, A of T , where we say T is now, we call that A_0 equals 1. So by definition, going back in time goes to smaller and smaller A . The volume of that box is a cube of A , a^3 .
- 5.57** You ask at earlier and earlier times, what was the volume of the box? It was A^3 as well, but A was smaller. So the volume was smaller by the cube of the scale factor. The temperature of a box that's filled with photons is very different than the properties of a

box filled with protons, even though they sound the same. And very different than a box filled with croutons, my favorite particle. So if you have some number of particles, be they photons or protons, you say that there's N of them in a volume, V . And you ask, as A increases over time, the volume gets bigger, the density, symbolized by ρ —this is the density—the density is going to go up as you go to smaller and smaller A , and it's going to go down as you go to bigger and bigger A because the volume is increasing with time as the cube of the scale factor.

5.58 The number of protons is not changing with the increase in the volume. If I have a box of 10 shoestrings and I double the size of the box in each dimension, the volume is up by 8. Did I change how many shoestrings are inside the box? No. So the density dilutes as the volume increases. But protons and photons are different, because when you stretch the box—the box is stretching because the universe is expanding—something happens to photons that does not happen to protons. Can anyone guess what that is? What happens to the proton, or doesn't happen to a proton, that happens to a photon? If you remember, we described the amount of energy in a photon. We said that the energy of a photon, let me say this, first, the energy of a proton is the mass of the proton, m_p , times the speed of light squared.

5.59 You can't apply this equation to photons because they have no mass. But obviously, they have energy. The energy of a photon, and that's symbolized by γ , is not its mass, it's its frequency. The frequency of the photon times this constant $\hbar$, the Planck's constant. So for a photon, this is the relationship. And then we said that the frequency of a photon is inversely proportional to its wavelength. So the longer the wavelength, the lower the frequency, the lower the frequency, the lower the energy. So what do you think happens to a photon as the universe expands? What happens to its wavelength? What happens to the wavelength? It goes up. What happens to the frequency when a wavelength goes up? The wavelength gets bigger, the frequency gets smaller, the frequency gets smaller, the energy gets smaller. So the number density doesn't change in a box of expanding photons, but the energy density changes. And that's actually what we measure. It's energy density.

5.60 So a universe that has photons in it, it has a change of energy that's radically different than the change in energy of a purely matter-filled universe. So the universe starts, becomes neutral at this time scale, as we've discussed several times. Transitions, becomes transparent, and the energy is dropping dramatically as the fourth power, not just the third power—it gets one power of decreasing energy because the wavelength goes up by A . So the energy density goes up by the fourth power of the scale factor, as you go to smaller and smaller sky scales.

5.61 So again, as I said, I like to think of hydrogen as forming in a fusion process where the electron combines with the proton, fuses together, makes an atom. And we know from basic chemistry explorations on Earth that if you zap a hydrogen atom with 13.6 electronvolts worth of energy—that's a unit of energy, which is in the form of, say, an

ultraviolet X-ray or gamma ray —it will zap apart the proton from the electron. Again, so this characteristic energy scale is characteristic of a hydrogen atom being dissociated. And so at some earliest point in the universe's history, the universe was hot and filled with photons greater than this ionization energy of 13.6 electronvolts. So the universe is completely a plasma. No neutral atoms possible. And then as it cooled, hydrogen forms.

5.62 The photons that we see are the fossil relic of that epoch of the formation of structure, of the formation of the CMB. So we've covered now the oldest light in the universe, the oldest galaxies in the universe. Those are the galaxies that we see at the highest possible redshift. They both indicate the light that we see from the CMB, the light that we see from galaxies. Radically different forms of light. Just like if you go and take a picture with your camera, it's a very different type of technology than when you to the dentist and get an X-ray.

5.63 And yet if there was some way to look at your tooth with optical light, and you go into the dentist and say, "I've got a toothache." "Let me look at it." Your dentist, she looks at your tooth. "Oh, yeah, there's a crack." Visibly, she can see it. "Let me confirm that with an X-ray," totally different. They both hang together. The same hypothesis holds. You have a crack in your tooth or a cavity or there's a piece of popcorn in there. I don't know. I'm not responsible for dental hygiene. Now we're going to see, though, that just like light, CMB and galaxies tell the same story of a hot, dense earlier portion of the universe. So too, in very dramatic fashion, that 's something that has nothing to do with visible light or X-rays or any form of light whatsoever. That 's the formation of the chemical elements.

5.64 We'll see that that fossil, classification of fossils tells the exact same tale. Just as there are multiple types of galaxies, and we see the same expansion or redshift for each type of galaxy, a spiral galaxy, an elliptical galaxy, and we see even within those galaxies, we see different stars, supernova, Cepheid variables—they all tell the same story. While there's only one type of microwaves that we can see, there are many different types of chemical elements. And furthermore, each one of those chemical elements that we can detect today trace their origin to the Big Bang at a much, much earlier epoch. We started with galaxies billions of years ago. Now we're going billions of years after the Big Bang. We've gone to an exploration of the CMB, 380,000 years after the Big Bang. Huge increase in time scales.

5.65 Now we're going to trace the first seconds and minutes of the Big Bang's history. We're going to do that in a self-consistent fashion that ties together the properties of all these different types of chemicals. Each one is an independent tracer of the physics of the Big Bang that we claim to be happening. It gives us great confidence. In science, you should have a theory that explains many things with few assumptions as possible. And so later in the final lecture, we 're going to encounter gravitational waves, which are not light and they're not matter. They're waves of pure gravity. And that will also potentially tell

us about things that happened even earlier than the formation of the chemical elements to which we now turn.

- 5.66** So we're going to estimate what the conditions were in the universe in order to make matter. The matter that we're going to talk about is really simple matter, but it's still quite complex. Furthermore, different types of elements have different numbers of what are called isotopes. Isotopes, the most familiar that you have probably heard about, are different levels of neutrons for a given number of protons in the nucleus of an atom. The most familiar one may be carbon-14, which is a radioactively unstable, and therefore decays, isotope of carbon. Carbon-12 is what most carbon is. And a real archeologist, not fake astronomer-archeologist wannabes, use carbon-14 to date things that are tens of thousands of years in age.
- 5.67** We're going to go back billions of years. We have to use different types. And we're going to see that the story told by each isotope is consistent with the overall picture that the universe was extremely hot and dense when the first atom, atomic nuclei formed. We're going to use just really these 4 or 5 different nuclei. We're going to use hydrogen, and it has isotopes. It has two isotopes. We're going to use helium, which has—sorry, hydrogen has three isotopes, helium has two isotopes, and another element called lithium is also sometimes included in the mix, although we won't talk much about it.
- 5.68** So to recap, putting the pieces together, galaxies tell us about the universe as it was at billions of years in age. The CMB tells us about the universe a few hundred thousand years after the Big Bang. How do we go back earlier in time if we can't use light to look at things? There's nothing to see because galaxies hadn't formed yet. And we can't use light because the CMB is an opaque screen or fog that surrounds us beyond a certain time scale, going backwards in time. We could use things called neutrinos, but they're extremely hard to detect. In practice, they're almost impossible to focus and use as a telescope, although there are astronomers that do use them, but not for cosmological purposes as yet.
- 5.69** We'll find it's much easier to use the abundances of the elements because the lightest elements can only be formed at the Big Bang—hydrogen, helium, and those isotopes I said before. Just to remind you, Einstein's famous equation, $E = MC^2$, is a special case, the lowest possible energy state of a massive particle like a proton or electron. It can be more energetic if you accelerate it, give it more energy, give it more momentum. Momentum is symbolized by P in physics. The minimum energy it has is known as its rest mass. That's the energy you could convert to pure light or heat, if you were able to do so. The total energy, if it has no momentum, if it's at rest, P is zero, and E^2 equals M^2 squared C^4 to the fourth, take the square root, you get $E = MC^2$.
- 5.70** So if an object is at rest, P is zero, this radical is 1, and you get the famous, most famous equation in all of science. But if it's moving low velocity, we can do what's called a Taylor expansion and reduce this fraction to this number here. You get a perhaps

familiar expression, maybe from high school physics, that the energy of a massive particle is its rest mass energy, which is enormous, plus a tiny bit of what's called kinetic energy, which is the energy due to motion or momentum, which is one-half the mass times the velocity squared. But we assume the velocity is very small compared to the speed of light. So this is a tiny, tiny correction.

- 5.71** But physicists strive for precision, and it'll become important. And again, I showed you this when we talked about special relativity. There's an equivalence between the rest mass and energy. And this is Einstein's own handwriting. And you can gain mass by absorbing kinetic energy, gain energy as well. And we talk about these strange units before, called electronvolts, because we're dealing with particles, either protons, neutrons, or electrons. And the least massive of those three is the electron. And so it's the easiest one to accelerate, according to Newton's law, $F=MA$. Remember, you can accelerate something, give it a high velocity for a given force, much more so if the mass of that object is small, and the mass of an electron is 1,800 times smaller than a proton or a neutron.
- 5.72** So we typically describe energy at the atomic level, not in terms of joules or calories, but in terms of electronvolts. An electronvolt is the amount of energy that an electron would gather and acquire if you had a one-volt electric potential to accelerate it. Electrons are charged. You put a one -volt battery on it, it would require one electronvolt of energy. And photons can be converted to energy using the Planck formula, energy equals h times the frequency. And so you can ask what wavelength or color does light have of a given energy in electronvolts? That's what this process is describing here.
- 5.73** The temperature of the universe follows this behavior. When the temperature of the universe was 10,000 Kelvin, the average energy of the photons in the universe was 3 electronvolts. In that case, it is below the 13.6 electronvolt level needed to ionize hydrogen. So no individual photon could have ionized hydrogen, even though the temperature is higher than 3,000 Kelvin. But the saving grace or the fact that it does permit the universe to have this ionized plasma, even at much lower temperatures at 3,000 Kelvin, is because there's so many photons, and each one has this level of energy, and they're all crashing into a hydrogen atom at the same time. That gives it enough energy cumulatively to dissociate it and make these plasmas that I discussed.
- 5.74** So here's a calendar, if you like. This is time after the Big Bang. This is the characteristic temperature of the universe, and that's the characteristic energy. Going back in time from 30,000 years to 12 days to 1 second to 10 to the minus 6 or 1 microsecond. This is the level of temperature, the corresponding energy. Now, you will ask, what temperature does it take to break apart an atomic nucleus? Because once you get below that temperature, you can't break apart atomic nuclei. They're stable. Above that temperature, you can break apart a nucleus into quarks and its constituent particles.

- 5.75 And so, we can ask, as we go back in time, the era in which we could make atoms is above this time, after this time, 380,000 years. Before that time, they were just nuclei. If you go higher and higher in temperature, you actually blast apart the nuclei into its constituent particles, the protons, neutrons, and electrons. Then you could actually break those apart. So out of the whole 112 different elements on the periodic table, to cosmologists, we only really care about 3. And in truth, we only discussed 2 today, and that's hydrogen and helium. Everything else is just, well, they're there.
- 5.76 Now, you may wonder, how does all this stuff get here? We'll talk about that later on, and that will involve your wonderful parting gifts or gifts I give away on my website, those meteorites that I got you. Those come from much, much later processes called supernovae that occur when the universe is billions of years old. They're not relevant to the Big Bang. They're relevant to astronomers for sure, and they're one of the handful of things that you can actually collect on Earth that represents something that happened in the universe. Remember, we can't do an experiment. We can't take the universe and change its temperature versus time curve, or change how many neutrinos were there versus neutrons, or protons versus electrons. We cannot do a proper experiment. We have to make do to what comes through our telescopes or what lands on Earth. Everything else is basically not a part of cosmology. So it's very challenging.
- 5.77 Luckily, as I say, each one of these, hydrogen and helium, these two elements, have between them five different isotopes. And each one of those five could be used to determine the temperature versus time history of the universe. That 's what we're trying to do. We're trying to see, is there a consistent story between the temperature and the size scale of the universe, as told by the cosmic microwave background? As told by the observation of galaxies? And now, as told by the abundance of these different elements and their isotopes? We can actually prove consistency with just one of these isotopes, but we have five. And actually, there's many more that professionals use. We won't get into. But hopefully, I'll impress upon you that all five telling the same story is incredible, powerful evidence for the veracity of the Big Bang model.
- 5.78 So the most abundant element in the universe is hydrogen. Hydrogen makes up about three-quarters of all the elements in the entire universe. There's a remaining 25% And 24% of that 100% is helium. And that means there's 1% of everything else, everything that we're made of—the carbon, the nitrogen, oxygen, phosphorus, sulfur, everything else that's in our DNA, that's in our bones, calcium. That is a minuscule part of the budget of the number of atoms there are in the universe. It's really astounding. Obviously, that type of matter is very important to us. But it wouldn't be here without the hydrogen and helium that we're going to be discussing today.
- 5.79 We use hydrogen, helium, and the five isotopes that there are of them to see if this story holds together. Can you trace the temperature time history the universe using elements, matter, not photons, not galaxies, actual honest-to-goodness matter? It turns out you can, and it becomes one of the most solid pillars that gives us credence in the

Big Bang theory. What are the different isotopes? There's three isotopes of hydrogen. Ordinary hydrogen is a proton, single proton, and a single electron. Atoms, by the way, I should have mentioned, are always electrically neutral. They always have the same number of positive charges as negative charges. Positive charge comes from the proton, negative from the electron. Neutrons are neutral, and they're important when we build up more complex nuclei with more than one proton, because, as we showed, we described the electromagnetic force. Like charges repel.

- 5.80** So you need something stronger than the force of repulsion naturally incumbent upon proton-to-proton reactions. And you get that from what's called the strong nuclear force, which provides a different type of charge, if you like, that's more powerful, stronger than the electrostatic repulsion. So the simplest compound nucleus is called deuterium. Deuterium. "Deu" is like two, and it has two nucleii, one proton and one neutron. And protons and neutrons bind together relatively naturally. And to break apart, this nucleus is present without an electron. It would have one electron to be electrically neutral. But of course, the universe is way too hot a few seconds after the Big Bang to have an electron bound to the proton-neutron nucleus.
- 5.81** So the question is, how much energy do you have to zap? If you have to zap a hydrogen atom with 13.6 electronvolts of energy, how much energy do you have to zap this thing with to blow it apart into a proton and a neutron? So going backwards in time, that would mark the point at which these things become unstable. Or if you like, that when the fusion occurs between a proton and a neutron, after that period of time, now you have stable compound nuclei. You can use this to trace that, as a thermometer, to trace the temperature time dependence of our universe after the Big Bang. So this requires millions of electronvolts, not 10, not 13.6—1 million electron volts, 2 million, 2.2 million, to zap it apart, to break it into a proton and a neutron.
- 5.82** So we go back to the temperature time curve and we ask, how old was the universe when it had a characteristic energy of 2 million electronvolts? Plugging in the equation we saw before, the temperature that corresponds to is about 50 million—sorry. Yeah, 50 million—sorry, 500 million Kelvin. It's incredible. 480 million Kelvin. And that, if you convert this to time, occurred 7 minutes after the Big Bang. What we're doing is we're going backwards in time. We're taking a look at a curve of temperature versus time. This is the temperature of the universe, and this is the temperature of the universe, and this is time. And we're going from today, which corresponds to a scale factor. I could also have a scale factor up here. A is one, here. This is the scale factor, A of T . A of zero would mean the universe has zero size. We're not confident in the behavior of space-time at zero size scale, so we won't discuss that.
- 5.83** And then we're asking, how does temperature change with time and what different things are occurring along this graph? Galaxies can form, and as I said, the CMB can form. Nuclei can form. This is the way we're going backwards in time. We're asking, how big was the universe at those different stages in time? And do they hold together

where they could find some overlap, or are they on the same curve that's uniquely determined by the Big Bang theory? This temperature time relationship is uniquely determined.

- 5.84** So the process by which you would break apart or form a deuterium nucleus is shown here. It's almost like the chemistry of forming nuclei. This is actually a nuclear fusion process, not an atomic fusion process. Proton combines with neutron, you get a deuterium nucleus, plus you get leftover heat. This fusion process releases energy. The energy that's released is characteristic of the fusion energy scales. When you calculate that, you find certain number of deuterium nuclei can form, given an equal amount of protons and neutrons. So if you don't have a neutron to match for every proton, they're lonely, and you can't make a deuterium atom. Now you may think, well, there might be equal amounts of protons and neutrons, but there's no reason for that to be true. There's no reason that the number of protons has to equal the number of neutrons. We do believe
- 5.85** that the number of protons has to equal the number of electrons. That's called a symmetry property. That's called charge conservation. We don't see in the universe large -scale distributions of pure positive charge with no negative charge. Every element we see is electrically neutral. Plasmas are not, but even the amount of plasma that we get is always equal, and opposite plasmas are formed at the same time.
- 5.86** So deuterium forms, heat is left over. You could actually use this deuterium water. You can actually make dihydrogen oxide or water from heavy hydrogen, from hydrogen that has an extra neutron. So there 's another isotope of hydrogen called tritium. Tritium is made, again, it has to only have one proton to be hydrogen, and it has to have one electron to be chemically neutral, but it can have two neutrons and one proton. Or it could be the combination of deuterium with a neutron. That makes tritium. That also is a fusion reaction. It takes energy to smush them together. That releases energy in the form of leftover binding energy. That's the gamma ray that's left over.
- 5.87** Each one of these compound nuclei is a separate thermometer. It's a separate time-dependent thermometer. It fuses with a different amount of energy. There's a different rest mass and a different amount of photons that are released in this process. That happens at a slightly different time. Now you've got two different proxies for temperature versus time, just from the three isotopes of hydrogen. Then there are two isotopes of helium, ordinary helium, and then helium with one fewer neutron called helium-3. Those processes occur, and they keep occurring, at heavier and heavier elements up until the formation of element number 5. We're not going to get into that. But each one of these makes up a different proxy, a different thermometer, time-dependent thermometer, and they all tell the same story. The universe's temperature versus time is extremely well-predicted in the Big Bang theory.
- 5.88** The hydrogen is not the only one. There's also a substantial amount of helium. There's one story I didn't tell you, kind of sad, is that when we wanted to make hydrogen into

deuterium, we needed an extra ingredient called the neutron. When we wanted to make deuterium into tritium, we need another neutron. Protons are eternal, even more than diamonds. I mean, you need, diamonds are carbon. It's pretty simple. Carbon has a lot of protons. But neutrons are not eternal. Neutrons die. Neutrons are the lonely hearts of the cosmos. Because if a neutron is not bound up with a proton partner in the nucleus of a compound nucleus, it will die. It will decay in less than 15 minutes. If we had a neutron right here, on average, it will live for about 800 seconds or so.

- 5.89** And that has very important implications because any neutrons that aren't bound to protons in the early universe are no longer around. The only ones that survived were bound to protons in these different isotopes of hydrogen and helium. It's incumbent upon them to find their partners. And so that even if there were an equal amount at the very beginning of time, of protons and neutrons, over time, there'll be nothing but protons, unless the neutrons partner with a proton or more than one proton to make a compound nucleus.
- 5.90** So the question is, before the nucleus processes started forming, there were, on average, two neutrons for every 14 protons. So these are indicated by a little sticker, gingerbread men and women. Imagine you have a universe such that, for every 14 protons, there's two neutrons. At most, what could you possibly make from these two neutrons? These two neutrons could bind with two or more protons. And then you have to ask, well, what elements would be left over? And then you compare those elements that you predict, given these ingredients, given the lifetime of the neutron, and the time dependence of temperature. How much fusion energy was produced, and what are the byproduct fossils that still remain to this very day? Because these things cannot be made after the Big Bang.
- 5.91** You can get two protons and two neutrons, and that's the nucleus of ordinary helium, like you get in a helium balloon. What would you expect to see in this universe? You'd expect to see one helium nucleus, two protons, two neutrons, and you'd have left over 12 free protons. You'd have a 12 to 1 ratio of hydrogen to helium. Do we agree with that? And by number. So now, 25% of the initial protons and neutrons should be in helium. The rest will be in hydrogen. Pure hydrogen will be left over. We asked the question, do we see primordial hydrogen abundance of about 25%? And spoiler alert, I already told you it's about 24%. And we can actually do that calculation more precisely. And we find it actually should be 24%, not 25%.
- 5.92** And again, thank god, because without it, we know there's 75% hydrogen. If there were 25 % exactly, there'd be no elements like carbon, nitrogen, oxygen, phosphorus, sulfur, calcium, and all the elements that we need to make up our body. That wouldn't exist. You do that same exercise, counting protons, neutrons, accounting for the lifetime of the unstable nucleus, accounting for the heat left over from the fusion reaction of the earliest lightest elements. You do that for every single element that could possibly have

been made in the first hours after the Big Bang. And there's a couple of dozen isotopes and elements. They all agree. Every single one would have been completely sufficient to prove the reasonableness of the Big Bang theory. But we have 12 of them. Would have been sufficient to have one, and yet we have 12. This is incredible prediction.

5.93 Completely different science, right? Astronomers using telescopes, seeing a universe that was smaller in the past, denser. Astronomers using microwave telescopes, seeing a universe that was hotter, denser, according to the thermodynamic properties of photons. And then nuclear physicists, studying the properties of the constituents of atoms. And that also hangs consistently, all telling the same story. All these fossils are tying together the same story, just like our theories of archeology can be combined with theories of the early Earth's climate or the early Earth's tectonic plates.

5.94 Why do we find dinosaurs' skeletons in Antarctica? These things tell the same story, that the Earth was a dynamic place, that the Earth has changed and shifted over time. Archeologists, paleontologists, climatologists, they all tell the same story. So too were nuclear physicists, microwave astronomers, and optical astronomers, all seeing the same story. We're going to see the same thing holds true for X-rays, for gamma rays, for everything we can possibly see. So these matter particles are very tangible.

5.95 You can literally, inside of this glass of water, we can count how much deuterium is in there, how much tritium is in there. We can ask, this is not pure water with just protons in the nucleus. There's a lot of deuterium. Ordinary water is actually comprised of a lot of heavy water, not a lot compared to how much ordinary water, but percentage, fractions of a percent. Then even the tritium nucleus, with two neutrons and one proton that's inside this glass—the only place it could have come from is from the Big Bang. You can't make it in other processes later on in stars. It's impossible. These stories are very consistent and it's very tangible, almost visceral. You can actually drink evidence for the Big Bang. That's pretty cool.

5.96 Now we have determined three pillars of the Big Bang that takes us all the way back to a few seconds, to billions of degrees, millions of degrees above absolute zero, all the way down to three degrees above absolute zero. We're talking nine orders of magnitude. But cosmologists are greedy. We want to go back to much, much higher temperatures, much, much earlier time scales. Telling the story over nine orders of magnitude is impressive. But we want to go back even to higher temperatures, even to earlier time scales. Not negative seconds, perhaps, but fractions of a second, fractions of a nanosecond. We'll find we may be able to go back to one trillionth of one trillionth of one trillionth of a second after the Big Bang, not using particles of matter, not using galaxies, not using microwaves, but using waves of gravity.

5.97 We've tied together this story using matter. It's incredibly tangible. We can actually go up to heavier and heavier elements. They all tell the same story. Their isotopes tell the same story. It links together with our understanding of the evolution of time and temperature. That links together with the formation of the first atoms, of hydrogen,

with binding together with electrons in an atomic fusion process. Then finally, connecting that epoch to the formation of galaxies, all are telling a very consistent story.

- 5.98** We then want to examine the question of where do the heavier elements come from? I've brought samples that you may have of actual meteorites. Those are samples of stars and gives us credence in the formation process of the tracers of galaxies and their redshift. Namely, how do these stars get to produce heat and light that we can see? They fuse together the hydrogen and the helium and the tritium and the deuterium and the light hydrogen called helium, light helium, called helium-3. They fuse all those together, and they keep fusing and fusing and fusing, making heavier and heavier elements. Until they get to this nucleus called iron. When they try to make iron, they can make it, but iron takes in more energy than it releases during fusion process.
- 5.99** And that causes the star to eventually not have enough pressure to keep it inflated. And the star collapses on itself, explodes in what's called a supernova, and blasts out into the universe the last thing it was making up, which is this heavy metal called iron. And it actually has other things in it, like cobalt and nickel. And you'll find when you take those meteorites home, they're magnetic because iron is magnetic, and so is cobalt, nickel. You can actually prove this abundance ratio and prove that these atoms were not made during the early moments of the Big Bang. They had to be made in stars. Everything above lithium and its isotopes was made in reactions long after the Big Bang, millions, hundreds of millions, and even billions of years after the Big Bang. We can't use that to trace the the Big Bang itself.
- 5.100** Now, you may wonder, how do those stars get there? Do you just have stars in the universe that just form out of hydrogen? We're going to see how the hydrogen had to have been accompanied by another form of matter. That's much more mysterious, called dark matter. We talk about hydrogen gas. We're actually talking about molecular hydrogen. And molecular hydrogen is the combination of two hydrogen atoms. They share a bond. They share these two electrons between them. And that makes molecular hydrogen, which is actually what makes up gas in nebula. And the nebula gas is what makes up the ingredients for stars to start forming in the extremely early universe.
- 5.101** So most of the hydrogen in the universe is in the form of molecular hydrogen, or H_2 . And those are very relevant, again, not for cosmologists, but for astronomers studying how stars themselves form and how gas is distributed in the universe. And once you get high enough pressure and high enough density of hydrogen, it can fuse and make a star. And here's just a very simple schematic of a star. Again, our sun is about a million kilometers in diameter. It's huge. And that is comprised of an inner core of about 10 or 15% of the volume of the star is actually producing nuclear reactions, fusion of hydrogen into helium, helium into beryllium and other things, and then eventually

making up as much as possible, and hopefully for a very long time, before it runs out of fuel and stops producing energy.

5.102 Our star is not big enough to make into a supernova for technical reasons, so we don't have to worry about that. But the schematic diagram here is just to complete some curiosity gap or fill in some gap that you might have in how stars play into this. Again, very interesting, very important certainly, for us to have a sun and for astronomers to study stars, but not relevant to cosmology. The processes of the star do not impact our understanding of the early universe. So the question is, what is remarkable is how far we've been able to come in just these 5 lectures so far. And how exciting and important the upcoming lectures will be to understand how much farther we can see into the early universe. The ideal goal would be, we have some evidence we can look at the Big Bang itself. Can we do that?

5.103 But so far, we've already disproven theories, which is the goal of a scientist, not to prove a theory. To disprove every other theory and what's left should be, no matter how impossible as Sherlock Holmes said, the truth. So up until the 1960s discovery of the cosmic microwave background, most physicists believed in what was called the steady-state universe, that the universe had always been here. It was eternal. There wasn't an expansion at all. And so this combination of observations of the cosmic microwave background, with the abundance of the nuclei, which were discovered slightly afterwards, was actually the death knell for this rival theory. It doesn't prove the Big Bang, but so far we've got dozens and dozens of pieces of evidence.

5.104 As I said, every isotope of an element doesn't prove, but it gives us credence in the Big Bang. The photons of the CMB give us credence. Every single galaxy that we see. Remember, it only takes one observation to kill a theory. If we saw all these—we saw many, many galaxies that are at a great distance and they were coming towards us, that would rule out the Big Bang. We haven't seen that. It doesn't mean that the Big Bang is perfectly proven. If we saw one of these elements and it could only be formed if the universe was trillions of Kelvin, just making it up, that would kill the Big Bang. But it doesn't. We haven't seen that. It doesn't prove the Big Bang. Gives us confidence in it.

5.105 So the steady-state universe had a lot of different things that were theoretically virtuous, that it didn't require a universe to come from nothing. We'll see some of the problems in the Big Bang as we go on. It also explained a universe that didn't change its properties, was consistent over time for all time, because time really was irrelevant. So it was a perfect instantiation of the Copernican principle. It didn't matter. Tomorrow, the universe wouldn't be any bigger than it was today. It wasn't even changing in time. But of course, it was ruled out by this.

5.106 And there's always been, as I was asked about in previous lectures, there is always an unavoidable association between theology or philosophy and cosmology. It's unlike, say, studying a superconductor or studying fusion in a reactor. You don't think about, "Well, is there a God?" But there were many, many astronomers who preferred, almost

as if it was a political or religious urge, that the steady-state was right because, as one of the most famous physicists of the last 100 years, Steven Weinberg, said, he said that astronomers and scientists love the steady-state theory because, quote, "the steady-state theory is philosophically the most attractive because it least resembles the account given in Genesis," in the Bible, in the Old Testament. He 's a Nobel Prize winner, one of the most renowned physicists of all time, certainly the last 100 years. And he's saying, "I love this theory, many love it—because we're atheists."

- 5.107** Now, it's impossible to decouple one's political, social, sociological views from doing science because science is done by scientists. Scientists are human beings. But it's important to be aware of your biases and to change your mind when new evidence comes in, as he did. I like to put it, when I gave a talk in the UK, I said, "If there's no Big Bang, you don't need a Big Banger." You don't need any entity, any god, any force external to science. And so it was a dominant theory. There's me in front of it. What I love about the CMB is how it killed off this entire class of models that suggested for thousands of years, the universe was eternal, static, and unchanging. It did so in a paper, which you would not realize that this paper, given its title, was so significant. It said, the title of the paper by Penzias and Wilson, "A measurement of excess antenna temperature at 4,080 megacycles per second."
- 5.108** That humble title led to a revolution, not proving a model, but falsifying many, many alternatives that have been cherished for decades. A very famous agnostic scientist, Robert Jastrow, said the following after this. He said, "For the scientist, like the atheist who's lived by the faith in the power of reason, the story ends like a bad dream. He has scaled to the heights of mountains of ignorance. He is about to conquer its highest peak. As he pulls himself over the final rock, he is greeted by a band of theologians who have been there sitting for centuries."
- 5.109** Meaning, the CMB seemed to agree with the origin of the universe in the Biblical sense, "Let there be light." Now, of course, this isn't a philosophy class. This isn't a theology class, certainly. I think this is a little overdone myself. But it's interesting to note, just as a side note, science is done by scientists, and you have to be aware of the biases and the predilections of individual scientists.
- 5.110** The final thing you may be wondering today is, we started with a universe that was extremely hot, extremely dense, made of a couple of different types of matter. We got to a universe that looks like this, just looking through the Hubble telescope or James Webb telescope, how did the universe get to be so anisotropic, so inhomogeneous, if it started off so pure, uniform, Copernican? And that will, of course, take us as we zoom in, into our universe, into the cosmic microwave background. We have to tie together the formation of elements, the formation of stars from those elements, the formation and leftover heat of atom formation, and eventually, the formation of planets and people. And that will be the subject of our next lecture. Thank you very much.

Lecture 6: Unseen Universe and Galactic Motions

At a glance — The unseen universe and galactic motions.

- 6.1** I hope I've convinced you that doing astronomy is incredibly difficult, not only because we only have a very limited amount of objects that we can actually study on earth in the laboratory, including particles, meteorites, and photons, particles of light. But cosmology is even harder because we don't even have multiple examples of other universes we can study. And yet it's actually been easy going up until now, because at least we could have access to, we could experiment on the things that do come into our laboratories.
- 6.2** Now we're going to cover a type of matter which may not even be detectable by any known means of apparatus, any type of technology that we could ever hope to invent. And that would be quite stunning, because we would be forced to believe in something for which there is absolutely no evidence. And that makes scientists justifiably uncomfortable. So today we're going to talk about the dark side of the universe and encounter the first missing component of the universe not accounted for by the standard Big Bang model. And yet it seems to be an—and yet it seems to be an essential ingredient of our understanding of the universe.
- 6.3** It's called dark matter, and it is an incredibly fascinating subject that is actually linked to the topic of our previous lecture, which was the oldest fossil light in the universe, the cosmic microwave background radiation. We called it a background because it seemed to come in all directions, at all times of year, everywhere in the cosmos. And for the first 30 years after its discovery, it seemed to be completely uniform, featureless, identical, homogeneous, and isotropic, coming in all directions with equal strength, no matter where or where the experiment was being done. And then in the 1990s, it was found, actually, there are slight perturbations, variations, tiny, tiny fraction of extra temperature, extra heat coming from some locations relative to others.
- 6.4** But actually, I didn't tell you that there's a much more big effect, much larger effect, rather, that was known even before the 1990s, and it has to do with today's topic, dark matter, in a surprising way. It will be related to the topic that we've already discussed, the Doppler shift. Doppler shift, not of light, of galaxies, but of the cosmic microwave background itself. Now, we've described the CMB as a glow coming in all directions, which is true, but we have a peculiar motion with respect to these photons that are coming towards us. It's as if we're in a car driving in a rainstorm. The raindrops are coming from straight overhead, but we see them as coming from in front of us because we're moving, or the Doppler shift, if you like, will result in a slightly increased frequency in the direction of travel of the car.
- 6.5** Well, the earth is not a car, but it is in motion in the cosmos, and the question is, what is causing that motion? Where are we going? And what is the nature of the dark matter

that is imputed to be the source of that very motion? So I told you the key properties of the cosmic microwave background in the previous lecture. The background is uniform. It peaks in the microwave. It seems to be very, very homogeneous and very, very isotropic. The characteristic wavelength is a few millimeters and the characteristic temperature is 2.7 Kelvin.

6.6 Now, imagine you're in a room and the room is a very low temperature oven. If you are stationary with respect to the room, to the walls of the oven, it's producing three Kelvin radiation. But if all of a sudden you start to sprint at a large velocity of the speed of light, you would notice a redshift away from you with lower energy photons, and you'd increase the frequency of the photons in the direction you're traveling. That is called a dipole shift. Dipole because it looks as if it's a big cosine wave in the sky. The direction that we're moving is an increase by an amount of our velocity divided by the speed of light.

6.7 The velocity of the earth and the spacecraft that orbit the earth inside our galaxy is about 371 km per second, a vast, vast increase over what we've talked about so far. And we seem to be moving in a direction that's near the Virgo constellation. And the question of why we should be moving towards the Virgo constellation will be answered very shortly. But the fact is, we are moving at an appreciable percentage of the speed of light towards this constellation. Now, what's in this constellation? I happen to be a Virgo, as I've told you in previous lectures. So it's very interesting and exciting for me. And away from me is the opposite direction of where we're coming from. What does that mean—coming from, going to? What's pulling us? What's causing this to occur?

6.8 If you go out and look at the optical properties of galaxies in the Virgo constellation, you will see an excess of galaxies in that location. The so-called local group of galaxies, of which the Milky Way, Andromeda, and the two Magellanic Clouds that we discussed in earlier lectures, is a part. Our local group of galaxies is moving towards this huge cluster of galaxies, of thousands and thousands of galaxies. Each one is massive, or more so than the Milky Way. And we're traveling towards that cluster of galaxies. The question is, why? What is it about those galaxies that's pulling us towards it?

6.9 Well, a lot of it is due to the number of stars in those galaxies. Stars are massive. Stars that have massive properties attract gravitationally all other matter in the universe, as we've discussed since lecture two. We call this a peculiar velocity, a motion of the local group of galaxies, inside of which is the Milky Way, inside of which is our solar system, inside of which is the earth, and satellites and telescopes on it. The combination of this motion, being pulled towards the Virgo supercluster of galaxies, is about one-fifth of 1% of the speed of light, which is very, very fast, much faster than any velocities we've considered before.

6.10 So the question is, what is causing us to be pulled in that direction? There are only two possibilities. There's either excess matter in that region of galaxies, or there's some change in the law of gravity—and this would be not Einstein's laws of gravity, but

actually just simple Newton's laws of gravity—that would affect the force versus mass and acceleration equation, Newton's second law. So this cartoon is representing a tug-of-war between dark matter forces and forces that attempt to reconstruct Newton's theory of gravity. We'll consider both of them. You have to, as a good scientist, consider alternatives, even if you feel your personal predilection is towards one or the other. Otherwise, you leave yourself open to bias

- 6.11 and perhaps confirmation of a false effect. And that happens extremely often in science, unfortunately. Again, science is done by scientists, and we're not flawless automatons, at least not yet, practicing our craft without any influence, impediment, or bias from sociological or non-scientific forces. So we will see that the paradigm that cosmologists favor today is the dark matter paradigm, that there is matter that is not emitting or absorbing light. Remember, if it emits light, it's not dark. If it absorbs light, it eventually will emit light, because all objects that absorb light, by the laws of thermodynamics, have to emit it, or else they would get to infinite temperatures, because light has energy. We know several things about dark matter, but it's much, much less well understood, certainly, than ordinary matter that we can experiment with on Earth. Because it doesn't interact with light. It's very difficult to see, obviously, and it's very difficult to manipulate using tools of instrumentation, because instruments are made of atoms. Atoms communicate forces using electromagnetism. And anything
- 6.12 that is not interacting with light will also not be charged. All charged particles interact with light, all electrons, all protons.
- 6.13 So these dark matter particles, if they exist, must be electrically neutral, like neutrons. We've already said, neutrons are kind of strange and depressing in a way, because they die if left alone and not bound up with a partner. But it's even more difficult to detect dark matter if it exists. And in fact, it has never been detected. And yet it appears to be there, much more difficult to detect than, say, the reportedly 90% of an iceberg that is underneath the surface. You can see the effects of the iceberg much more easily than you can see the effects of dark matter.
- 6.14 And despite the fact that it far outnumbers the amount of ordinary matter, the protons and neutrons, my favorite particle, the crouton—despite the fact that dark matter is far more abundant than those, it has proven impossible so far to detect. Now, how was it discovered? How are we led, as cosmologists, in just the last 60 years, to adopt a paradigm that is central to the Big Bang theory? That without dark matter or something like it, we wouldn't be here asking questions about the paucity of dark matter?
- 6.15 And it actually traces to the 1700s, that we've already discussed, the discovery of Kepler's laws of planetary motion. In our solar system, 99.8% of the mass of our solar system is contained in the one source of luminous energy in our solar system, namely the sun. So the sun is responsible for all the illumination in the solar system. It's a very good approximation. And its mass is nearly 1,000 times more than all the other planets, asteroids, meteoroids, and stuff combined. So it's a very, very good approximation to

say that when you see a star, you're seeing almost all the mass of that particular solar system.

- 6.16** Now, galaxies are nothing but stars, or so we thought. Maybe there's some gas, some dust, some meteors, and things like that. But if it were true that a galaxy is merely a large collection, 100 billion stars, then the mass to light ratio of a galaxy should be the same as the mass to light ratio of the star. It should just be 1,000 masses. I mean, sorry, it should just be 100 billion masses equal to the sun, divided by 100 billion luminosities of the sun. Or the mass to luminosity ratio of a galaxy should be one, as equal to the mass to luminosity ratio of our sun.
- 6.17** We don't see that. It's very different. In fact, we can impute, based on the velocities and Kepler's laws, you can predict and, in fact, astronomers do this in what are called extrasolar planets. When they see a planet around another sun, they can measure the mass of that star by looking at the orbital period of these planets and vice versa, and the size of the planets and the size of the star. In our solar system, here is a plot of the velocity, the orbital velocity of each planet in our sun, in our solar system, including Pluto, versus the distance. It follows this $1/R$ type dependence. That's exactly computed from Newton's law of universal gravitation.
- 6.18** The gravitational force is equal to constant capital G times the mass of the sun times the mass of the planet, inversely proportional to the square of the orbital distance, the radius of that planet to the sun. And then you can calculate the so-called centrifugal velocity. And you can equate those, and you get the actual speed. You can predict what the velocity is of a given planet, and it exactly matches Kepler's laws as derived from Isaac Newton's laws. But when you look at a galaxy which is spiraling, you might think that the stars should trace with some velocity that we can measure from their Doppler shifts. And the observations of starlight compared to the observation of planetary velocities are as radically different as you could imagine.
- 6.19** In fact, in our solar system, the velocities decrease uniformly with distance from the sun. The central source of luminosity is the central source of gravitational mass. In a galaxy, it increases completely, radically, violent disagreement with the laws of Isaac Newton, with the laws of Kepler. What's going on to make the equivalent of planets in a galaxy, that is, stars, orbit at extremely high velocities, as if there's missing mass not accounted for by the luminous structure of the galaxy's light source? That's the dark matter problem.
- 6.20** Actually, a very titanic astronomer named Vera Rubin spent some time at my institution, UC San Diego. She spent a better part of the '70s looking and plotting out the equivalent of Kepler's laws, but applied to galaxies. She observed dozens of galaxies, and she saw for each and every one of them, as opposed to falling off with distance in this very steep fashion that our solar system does, where all the light is concentrated, where all the mass is, to very high approximation, the opposite was true. The velocity went up and saturated and was maintained at a very high level for very

large radii. And only after extremely large distance away from the central light source of the galaxy did it start to finally decay.

- 6.21** So the picture that became self evident was that there must be unaccounted-for mass, mass that did not produce light or heat. That was called dark matter. It should really have been called invisible matter or shadow matter or something like that. But I wasn't around during the naming process. So the picture that we have now, in the so-called dark matter paradigm, which is by no means proven, but is extremely consistent with later phenomena that we will see in the upcoming lectures, is that our galaxy's luminous stars are like the foam on the ocean surface.
- 6.22** And the vast, vast majority of the mass, the honest-to-goodness gravitational force-producing mass, is invisible, and it surrounds, at great distances, far beyond the visible light of the spiral galaxies or any other galaxy structure. It's as if our sun was a pitiful, minuscule amount of matter and there was some matter in our solar system that was extending far beyond the orbit of Pluto. It would be shocking to us. And, in fact, there are connections between this. So, to recapitulate, the mass, predominance of mass is not contained in the luminous sources of the galaxy. In fact, those are a relatively small fraction.
- 6.23** Now, the stars are extremely important because the stars trace the presence of the invisible matter. We can only see the effects of invisible matter because there are stars to look at. If the whole galaxy were dark matter, we wouldn't be here, because we're made of ordinary matter, not dark matter. We interact with matter. Matter bounces off of us. We absorb heat, light, et cetera. So it's a radical discovery, and it was first made in the early 1960s and has been further and further improved upon by studies ranging from wildly different fields of astronomy, from optical astronomy to the microwave astronomy that I do, and even beyond, as we'll see when we encounter high energy cosmology experiments, like X-ray astronomy satellites.
- 6.24** So the paradigm that we have come to is that a galaxy, the visible light, is a tiny fraction of the mass-producing and gravitational force -producing contents, instead of falling off, like planets do, around a single bright mass and light source, as 1 over the square root of the radius from the star or from the galaxy center, it actually becomes flat, like this, out to many, many times the luminous radius distance. So this is an image you could think about, surrounded a bubble of dark matter, a sphere of dark matter, of relatively constant density, more or less symmetrical shape, and then a very irregular and anisotropic planar galaxy of ordinary matter.
- 6.25** And the question is, can our model of the Big Bang be reconciled with this? And because we can't interact with it, it doesn't interact with light, it doesn't interact with ordinary matter. How can we possibly detect or prove or falsify these models? It seems like an impossible challenge, and it's quite difficult. So to forestall as much as possible the hard work of going out and detecting dark matter as a particle, astronomers said, "Well, are we really sure it couldn't be chunks of matter that just don't give off light,

chunks of matter being giant planets, perhaps even other stars that aren't that bright? Could that account for the mass that's needed to produce the rotation of stars within these galaxies?"

6.26 So astronomers did a heroic thing that's actually probably the final—well, there'll be one more connection to theology and to philosophy, but I like this connection. There's a quote in Genesis, in the book of Genesis, in the Old Testament, where God is having a conversation with this person, Abram, who later becomes Abraham, the patriarch of monotheistic Judeo-Christian religion and Islam. And he says to him this phrase that has nothing to do with cosmology. But he takes Abram outside and he says to him, "Look up to the sky and count the stars. Count them." That's weird. Why is he giving them, like, astronomical homework? I should be giving it to you guys. He's telling it. And actually in Hebrew, it's interesting. Hebrew has a command form.

6.27 You may have heard of mitzvahs. Mitzvah means commandment. It doesn't mean a good deed. It means, "I'm commanding you to do something." So there's ten Commandments, or actually the ten mitzvot, or commandments, and he's telling them, "Go outside and do it." English doesn't have such a form of verb form. And he goes, "Go out. I'm commanding you. Go out and count the stars." So I like to think this as the very first astronomical homework. And he says, "Go out and count. That's how many descendants you will have." Of course, he didn't have any kids at this time. And he goes, go out. And he keeps talking about how many stars there are, and he also compares them to the dust on the seashore, the sand, avak, or sand, or the grains of dust on the seashore.

6.28 So I made a calculation once. Is it possible to actually count the number of stars there are? Well, this is something that astronomers have tried to do, because if you could account for the stars, or you could say there's actually dust that's making up this matter that we can't see, perhaps that could be an explanation that wouldn't require invisible missing matter, like dark matter. So astronomers have been doing that for a long time. They don't only use optical light, visible light. They use X-rays, gamma rays, ultraviolet, microwave, and so far, the amount of visible light and visible ordinary matter is completely inadequate. It cannot explain how much mass there must be inside of each galaxy and inside of each cluster of galaxies, because, as I said, if a star, if a galaxy is just a compendium of 100 billion stars, then a galaxy cluster is just a trillion or more galaxies or stars put together to make a giant supercluster of galaxies. Now, what about sand grains? So what about the grains of sand? There's a beach in San Diego. Just come visit. It's very different

6.29 than Florida. How many grains of sand are there on all the beaches on all the earth? Could you actually count it? So you can't count how many stars there are. There's simply too many. And it turns out there's roughly 3 billion billion grains of sand. And that's greater than the number of stars in the Milky Way, which is about 100 billion. So

it's a billion times more than that. But it's much, much less than all the galaxies and all the observable universe.

- 6.30** So it's actually, if you wanted to count, how long would it take? I counted that it would take, if you could count one star or one galaxy, one star per second, it would still take you 300 lifetimes, average lifetimes, to count up just the stars in the galaxy that we call home, the Milky Way. OK, Biblical side tour over. So now let's do an examination. Let's zoom through a galaxy, let's look at it, and then let's put these galaxies together and superimpose them and ask, what are they made of? And what is the cluster made of? Maybe there's some difference when you put a galaxy inside of a cluster, just like there might be some difference when you put a star inside of a galaxy versus, say, a rogue star.
- 6.31** So this is structure of galaxies. They're made of gas and dust and obviously hundreds of billions of stars. But then surrounding them and suffusing them is stuff you can't see, the invisible matter that we call dark matter. And then when you put these clusters, when you assemble a cluster of galaxies, you're assembling not just the pancake pinwheel galaxies that are shown here, but each one is a sphere that surrounds it. So it's a very different structure, when you combine galaxies together. You're combining the visible light will be these pancake-shaped, two-dimensional, almost two-dimensional structures. Each one is embedded in a three-dimensional sphere, if you like, of dark matter.
- 6.32** Putting those together results in a very, very complicated fractal shape that cosmologists have been able to reproduce almost exactly, only if and only if they include dark matter in their computer simulations. Remember I told you three of the tools we use are a human brain, telescopes of varieties of types, and computer simulations. So it turns out, just counting the amount of stars in the universe, just the stars themselves, not the planets, you get about 1% of the total mass energy of our universe. So that means 99% of the universe is not in the form of the mass of the universe. Mass energy of the universe is not in the form of stars, let alone what we're made up of.
- 6.33** Remember the mass that we talked about in the universe in terms of atoms, 75% in terms of hydrogen, 24% helium, less than 1% atoms like carbon, nitrogen, oxygen, that we're made of. Now that, all of that matter, is just not 1% of all the mass energy in the universe. So we're talking about 1% of 1%, 1/10,000th of the mass energy in the universe is atoms of the type that we're made of. It's incredible. Obviously, it's very important or we wouldn't be here. And it's the only matter that's conscious of itself. That's pretty cool.
- 6.34** So counting up stars, we find it incredibly inadequate, cannot account for the amount of energy there is. So there are two implications. There's some kind of dark matter that's at least 40 times the amount of stars, accounting precisely for the observations across all domains of wavelengths of light that we study. It can't be in what we call

baryonic form. I'll explain that in just a bit. Or it could also be that we don't understand gravity. Remember, all this was predicated on the fact that galaxies are like big solar systems and big solar solar systems, small solar systems—it's funny to talk about solar system as small. It's obviously quite enormous—takes light, you know, hours and hours to get to the orbit of Pluto.

- 6.35** But they're small compared to galaxies. Galaxies are small compared to clusters, and clusters are small compared to the universe, as I said. So maybe Newton is right, but only on the scale of our solar system. He might be wrong, even on the scale of a galaxy. And it might be, once you get beyond some critical distance, the inverse square law doesn't hold. Some other law takes over. And that law explains the apparent need for missing matter. But now you don't need missing matter. You just didn't understand your theory of gravity right.
- 6.36** That would make a lot of people more comfortable, because you wouldn't have to inject matter into the universe that's undetectable, perhaps. Philosophically, that's a very important thing to science. You should not inject things that cannot be proven wrong. If you say, "This matter, here it is. I've got my invisible purple dragon here. He helps me give these lectures. You can't detect him, you can't interact with him," it's not very satisfying, even though I could blame him if I get some facts wrong.
- 6.37** So stars provide these great lampposts to track things. There's an old joke of a policeman. He sees a guy drunk on the street. He says, "What are you doing underneath the lamppost here?" He goes, "I lost my keys to my car." And the police guy goes, "Oh, that's great. Do you want some help?" They search for another hour. They can't find the keys. And the policeman's like, "Are you sure that you dropped your keys by the lamppost? And the drunk says, "No, no, no. That's just where the light is." So we tend to look at things and be biased by the availability bias of what we can see and what we can detect. But that doesn't mean that there's actually something there where we're looking. So we have to be very careful. But in the case of stars, stars do provide nice lampposts under which to look for.
- 6.38** It actually goes back over 70 years, to the time of this astronomer cosmologist at Caltech named Fritz Zwicky. He was the first person to coin the term dark matter in German, "dunkle materie." And he said he could not explain this, the behavior that he witnessed, which was due to the gravitational behavior of galaxies within a cluster of galaxies, called the coma cluster. And he found actually a much, much greater—remember, the ratio of mass to light in our solar system is one. There's one solar luminosity, and there's one amount, one solar mass unit of gravity in our solar system. In our galaxy, it's something like 10. And in clusters of galaxies, it's more like 100. There's 100 times more, 200 times more, invisible matter than the visible matter in terms of stars. The solar mass to light ratio is less than half a percent. It's incredible.
- 6.39** So this was startling to him. He was a character. He wasn't very well-liked. People called him all sorts of names. And he said about his colleagues that they were—I'll clean up—

he called them spherical jerks, because no matter how he looked at them, they were always a jerk, utilizing the symmetry properties of the sphere. So there's my cartoon of spherical jerks. So a lot of people didn't accept his theory, and this is important in science. Again, science is done not by soulless, heartless, humorless, automaton robots. They're done by people. And people sometimes don't get along with other people. And because he was such a curmudgeon and a difficult character, a lot of people didn't accept his ideas and theories.

- 6.40** Then when Vera Rubin came along in the 1970s, 40 years later, and showed also, it appears, not only in clusters of galaxies, but in individual galaxies, we need some kind of dark matter, people started to believe it. She was much more reasonable, agreeable, thorough, and interactive. And when I was writing my first book, *Losing the Nobel Prize*, I actually got a letter from Zwicky's daughter that said, "How dare you give credit to Vera Rubin when it was my father did everything?" And I realized sometimes the apple doesn't fall far from the tree. She said it was a big conspiracy, due to Vera Rubin having been at Vassar College. And I was thinking, Vassar College has this huge power to create conspiracies among astronomers over 50 years? Anyway, that's just a side note of my interaction with Zwicky's daughter, not really pertinent for what we're talking about.
- 6.41** So clusters contain even more dark matter, apparently, than ordinary matter and certainly than dark matter in our own galaxy. And so how do we detect whether or not a galaxy or cluster of galaxies has dark matter? You can't see it, so you have to have some other way of weighing it. We can use a variety of techniques to weigh and determine how much mass there is in a galaxy. And even though there's some inaccuracy, it's much more accurate to count how much light there is, because you can actually see the light. No matter what you do, the most conservative estimates for how much mass there could be are still dwarfed by the amount of missing mass that we can't see.
- 6.42** All sorts of evidence from the '30s, when Zwicky started and coined the term dark matter to a local cluster of galaxies called our local group, with Andromeda and so forth. Then the Virgo cluster and beyond, galaxy independent, single galaxy rotation curves, which use what's called radio astronomy, 21 cm, and optical astronomy, and then upper limits from the cosmic background that we'll get into in just a bit. All these pointed to the same piece of evidence, that there's missing matter, or that we don't understand gravity. We'll get to that in just a moment.
- 6.43** Now, how do you weigh a cluster of galaxies that's a billion light years away? What kind of scale do you put it on? Well, you use what we discussed back in chapter four, in lecture number four. General relativity predicts the deflection of trajectories of light, or matter particles like planets, due to the gravitational warping of space-time. Mass tells the universe's space-time how to curve. The curvature of space-time affects how photons, particles of light, travel. So using what's called gravitational lensing, that these

objects act like lenses, like this glass of water distorts the image, and it makes a different image of you, and it makes a different image of me, depending on how far away it is, what it's made of, how much water is in it, and the properties of the water, which is called its refractive index, how much it bends light—gravity bends light and is transparent. It's very much like this.

- 6.44** And so we can use the properties of the distant light sources and how they appear skewed when we observe them, deflected when we observe them. And it's exactly consistent with the predictions of Einstein from lecture four. And it allows us to have a very accurate scale. We can weigh all these galaxy clusters, that is, determine their mass. With optical telescopes, we can count out how much light they have, and we can see how much energy is missing.
- 6.45** And as we will see, that we actually find, the atoms in stars and so forth, when combined with gas and dust in galaxy clusters, makes up about 4.5% of all the matter and mass energy in our universe. There's a mysterious substance called dark energy, about which we know even less than dark matter. And we'll come to that in lecture seven. And the implication is that, of the remaining 24%, there must either be dark matter or some change to our understanding of gravity, which I will also explain.
- 6.46** Now, we can actually count very easily ordinary matter. It's called baryonic matter, just a fancy name to say normal matter, like protons, neutrons, and croutons. And astronomers have come up and counted it, and they find how much is there. There is a component, a large wedge of this pie, that's missing, that we don't understand where it is, but we know it can't be bigger than 100% of all this matter that's missing. So all of this ordinary matter, the matter we're made of, makes up at most 4.6% ordinary matter. And that's in the form of galaxies, cluster, cold gas, intercluster material, ionized things called Lyman-alpha forest, and then hot and warm, what's called interstellar media. This has been very accurately surveyed. Of course, it's not perfect. We don't understand it. But we can make a very conservative estimate and say it's no more than four and a half percent, 4.6%.
- 6.47** So what is the dark matter? That's a very interesting question. There's a couple of different alternatives. Each one has virtues and vices. The key as a scientist is to study them and see, can we determine which one is the most likely one to be a candidate to explain that huge amount, 25% or so, of the mass that seems to be missing in the universe's energy budget? Globally speaking, physicists break matter into either baryonic matter, which is ordinary matter, the kind that we're made up of, or non-baryonic matter, of which there's only one form that we've detected, called a neutrino.
- 6.48** A neutrino has mass. It's very low mass, but it's not a baryon. It's not a proton or a neutron, even though it sounds like it. They're almost massless. They travel close to the speed of light. They interact only via the weak nuclear force. They're neutral. They're not charged. You cannot affect them with a magnet. You cannot charge them with a static electricity, and you can't see them with light. So they're only weakly interacting.

- 6.49 We counted up in the previous slide all the different baryonic matter candidates. There's one interesting one called a MACHO. And I like this because my colleague at UC San Diego, Professor Kim Griest, came up with the term MACHO, which is kind of fun. And I'll explain how he came up with it and what he and his team did to prove that these cannot explain by any stretch of the imagination how much dark matter there is. So imagine what we're doing.
- 6.50 We're trying to discover why these galaxies seem to rotate so fast, why these clusters of galaxy are buzzing like a hive of bees around a central hive that's a gravitational central source of force, and you'd like to do it using known properties, known particles, known forces, and we'll find we can't do it. So Kim Griest and his colleagues came up with some solutions that might involve things you can't see because they don't give off light, they don't absorb light, and they might produce enough matter, and then proposed a methodology to detect them. Spoiler alert, the detection method found some of these objects, but way too far few to explain the missing gap between the matter that we need to match up with the matter that we see.
- 6.51 And so these could be a variety of different things. Black holes, what are called brown dwarfs, which are basically failed stars, planets. Planets don't give off light, really. They give up some heat, infrared light. But primarily they reflect light. So these are called massive compact halo objects. Massive, you need matter. You need mass to explain the missing matter that we're seeing in the universe. Compact, because they would be objects like planets, balls of gas, dust, and planetary material. Halo, because they can't be in the central part of the galaxy to explain the rotation curves of those galaxies.
- 6.52 Remember, the picture is that the dark matter, which could be MACHOs—we're going to rule it out, but it could have been —has to extend to many, many times, that is, out into what's called the galactic halo. Many times the visible light radius of the galaxy. So the funny name is, it's a MACHO. I mean, masculine or whatever. And so the technique to use to detect it was, if these MACHOs were in our galaxy or in another galaxy — here's the halo surrounding it—they'd have to be far outside. They'd have to be quite massive. And then when one of them moved between a distant star and the earth, the Hubble Space Telescope, when one of those objects moved through, it would very briefly focus the light intensely using gravitational lenses.
- 6.53 So this object, passing in front between you and me, eventually the starlight would get much, much brighter and then would get much dimmer, and it would never happen again. It wouldn't be a supernova explosion that would be incredibly luminous. It would be a magnification followed by a diminution of the light. And that would happen many, many times for these many, many objects. But it would only happen once for a given object. This is sort of a random motion. So they had to look for the following type of behavior, that the light would increase and then decrease. And as a cross-check, the light should not be changed in its color. Gravitational lensing doesn't change color. Whereas if it was a piece of dust or a rock or something that was just dimming or

brightening, it would have a different color characteristic. You would get a different brightening curve and dimming curve at different frequencies.

- 6.54** So they could prove that it was actually, truly, there were these objects. They are either mini black holes or they're chunks of planets. And they discovered many of these. I shouldn't say that many. I think it was only about, yes, 20. So over the course of 10 years or so, they looked at 10 million stars. Those were the background. And then they saw brightening and decreasing diminution. Exactly consistent with the presence of MACHOs in the halo of the Milky Way Galaxy. And they only saw 20, and they would have needed to have seen something like 20,000. And they calculated very accurately all the ways that they could have missed seeing true events, made a false negative, ruled out the dark matter hypothesis as being composed of MACHOs.
- 6.55** So this is a heroic experiment. It's really important because what does it do? Did it prove dark matter exists? Nope. It falsified one of the candidates to explain the preponderance of dark matter in our universe. Very important thing to do. Disappointing, obviously, because you'd like to have the answer, so to speak. But this was an important step. These are just some of the star fields they looked at. So MACHOs can't do it. So astronomers, being very clever people, what's the opposite of a MACHO? A WIMP. So what is a WIMP?
- 6.56** A WIMP is a weakly interactive massive particle. It only interacts weakly, meaning it's not charged, it's massive, or else it doesn't explain the missing mass. It has to be massive. And it's a particle. It's some form of elementary or fundamental particle. It has to last for a very long time, maybe has an infinite lifetime, like a photon, like an electron. But it had to be present in the early moments of the Big Bang, as we'll see. And it had to be present today because of the effects of galaxies, including the Milky Way. So dark matter, if it exists, is still around today. It was produced during the Big Bang. It must have a lifetime of at least 13.8 billion years.
- 6.57** Now, there are other examples. There are other things that could possibly explain these phenomena that we'll turn to, including a modification of Newton's laws. All of these measurements that we've talked about so far rely on Newton's laws being correct on galactic and cluster scales. There's no proof of that. We can't do such an experiment, by definition, on earth because we're inside of a single galaxy.
- 6.58** So a particle physicist will look at our universe and say it is made up of a periodic table similar to the periodic table of the elements. But now these are the periodic table that make up the nuclei. So molecules are made of atoms. The atoms appear on the periodic table of the elements, but the individual nucleons, the protons and neutrons themselves, are made up of just two quarks, called up quarks and down quarks. And then there are other forms of matter that have mass that are present in the universe. You may have heard of the Higgs boson. And then you've heard, certainly, of the electron. That has mass. But again, these are all baryonic or fermionic types of matter, or bosons, which aren't present because they can interact with light.

- 6.59** So none of these particles could be the actual form of weakly interacting massive particles. The neutrinos are weakly interacting. They are massive. However, they don't have enough mass to account for the amount of dark matter we seem to observe. These are charged, they can't be it. These are charged, they can't be it. These have no charge. But they either, in the case of the photon, have no mass, or in the case of the W and Z bosons, would interact too strongly, and we wouldn't see them either. And the Higgs boson also is insufficient to make up this type of missing matter.
- 6.60** But physicists are clever. Maybe there's some other type of matter that forms a different periodic table. After all, we have more than one element. We have more than one elementary particle. There's 17 here. Maybe there's a dark matter periodic table, too. And in there, astronomers and physicists have come up with some ideas. We really don't know them, so we just call them, hypothetically, WIMPs, weakly interacting massive particles. And they could be detected via interactions that are basically mechanical. They're basically, a WIMP coming in will not see an atomic nuclei's, an element's, electrons, because it's not charged, but it will come in and interact weakly with the proton and the neutron, the quarks within the nucleus itself.
- 6.61** That will cause a slight recoil. And you can actually measure the temperature and the change in momentum of the nucleus because of an impact with a WIMP coming from the galaxy or coming from a cluster far away. The probability of interaction is extremely small. And this type of interaction does occur with the interactions of neutrinos. Neutrinos are a form of WIMP. They're weakly interacting, they're massive, but their individual masses are not enough in their abundance to account for how much dark matter we see.
- 6.62** So there are experiments. One is led in part out of UC San Diego. My friend Kaixuan Ni and Elena Aprile at Columbia are some of the leaders. And these are deep underground, almost like mining experiments. In fact, they're in mines because you want to block out as much natural radiation, neutrinos, muons, and other particles from around the universe, from contaminating your experiment and falsely leading you to believe that you've detected these massive dark matter particles. So they're buried under thousands and thousands of feet of rock to screen out.
- 6.63** They use very, very pure materials like liquid xenon, which is a noble gas, so it doesn't interact with itself or other elements. It's like a heavy, heavy, super heavy version of liquid, of helium, which is also inert. And these are very, very ambitious and sensitive detectors that so far have come up empty-handed. These detectors have only set limits on how massive the neutrinos. Sorry, how massive the WIMPs could be, never detecting it. So it's extremely difficult to detect something that only interacts using the weak nuclear force or gravity.
- 6.64** The worst possible scenario, worst in the sense of being able to ever detect, would be if these massive particles only interact gravitationally. Then you almost have no hope of detecting them. If they're WIMPs, they interact weakly and gravitationally to account

for the matter. But if they're gimps or if they're only gravitational, could almost have zero chance of detecting them because you couldn't collect enough of them. There's no chamber that would enclose them. There's no way to cool them down or filter them out. They would just be this unavoidable fact of cosmology.

- 6.65** So, as of now, we have no convincing detections. There's candidates, there's other ideas. Photons, a version of a photon could be slightly massive. And that would be possible. We don't have any evidence for that yet. It doesn't stop us from looking. Just because you have no evidence doesn't mean it doesn't exist. So it's part of our job, my job and my experiments that I work on, to see what kinds of detections or limits can we put on these new proposed hypothetical explanations for dark matter.
- 6.66** But there's another option that we should, as good scientists, keep in mind, that it may be our understanding of gravity. After all, it was really Newtonian gravity and Kepler 400 plus years ago that suggested that we look for this behavior of rotation of galaxies, rotation of planets originally, to determine the mass of the central gravitating force. But why trust someone who's 400 years old, right? So maybe we should modify gravity, and you don't have to modify Einstein's gravity. Einstein was pretty smart. Maybe it's more challenging to overcome some theory of Einstein than a theory of Newton, who is much older, as I said.
- 6.67** And this goes by the name of modified Newtonian dynamics, sometimes called MOND. And MOND was proposed 40 years ago as a way to avoid having some hypothetical weird particle that doesn't interact with light. That was found to be distasteful. Why do you have to hypothesize a particle? Why couldn't it be that we just cannot test gravity on scales bigger than our solar system? So maybe Newton is wrong. And at large distances, that is, the size of galaxies and clusters, gravity is different.
- 6.68** Now, there's a precedent for this, and this actually goes back to the discovery of early objects in our solar system. Let's see here. OK. Here it is. So, in the 1700s, there was a claim that there might indeed be dark matter in our solar system. 1700. What are you talking about? Dark matter? Yes. It was observed in the planet Uranus, that the planet Uranus had strange behaviors of its orbit, as if there was another chunk of matter beyond the orbit of Uranus that was sometimes pulling on it to accelerate it faster in its orbit, and at other times, pulling from behind to decelerate it.
- 6.69** And this astronomer, again, cool name, cool hair, came up with a theory that there's a chunk of dark matter beyond the orbit of Uranus that was later called Neptune. Neptune was true. They detected dark matter exactly where he predicted. Of course, it's not perfectly dark. It reflects light from the sun, which is why, with even a moderate telescope, you can see it on a dark night. So at these two different positions, it afflicted and affected the orbit of Uranus in an exact, precise way, as predicted by Newton's gravity. Very cool.

- 6.70** So, because of this, this created a great sensation that Newton's laws were quite important. And when you see strange anomalies in gravity and orbits, you should first look for chunks of matter, not some weird change to the force of gravity. And then this guy, like many of my colleagues, got a little bit cocky because, as I told you in a previous lecture, there was observed, in the early 1900s and before, anomalies in the orbit of the planet Mercury. Mercury is the closest planet to the sun.
- 6.71** And it was shown that Mercury's orbit was elliptical, as Kepler and Isaac Newton had predicted. But the point of closest approach to the sun, the perihelion, was changing from year to year. One year it would be over here, a year later, it would be over there, and so on. And actually, the effect is miniscule. It's very small, but astronomers, even in the 1800s, could measure it. And it's completely unexplainable, according to Newton's law, with the only mass being provided by the sun.
- 6.72** So what do you think this guy Le Verrier did? What would he have proposed? Another planet. And, in fact, shown there, exactly, as Cooper is saying, a planet hypothesized to exist inside the orbit of Mercury, because this time, it has to be inside of it, or else we would see it, called Vulcan. And Vulcan was much, much closer, three times closer to the sun than Mercury, even. And so it was so close, you couldn't see it. So it was missing matter because you were blind to see it. But it was there. It was pulling on Mercury, and it was causing its perihelion to advance.
- 6.73** That was a time when he should have quit while he was ahead, because it turns out that the anomalous perihelion advance of Mercury has nothing to do with a planet whatsoever. It has to do with the curvature of space-time induced by gravitational mass, the type of curvature described by Einstein in his famous general relativity equations, which, written here—this is actually $F=MA$. You wouldn't recognize it. It's called tensor notation. It's for a future course, perhaps. But massive objects distort space-time. The more massive they are, the deeper the depression gets. The trajectories of particles, like a billiard ball rolling around here, will get deflected. And so, too, with an even more massive object like the sun, will it affect the curvature of space-time, causing, by a slight amount, the perihelion of Mercury to advance every single year, exactly as predicted by Einstein.
- 6.74** So this was a case where dark matter was falsified, and a new theory was needed. But who's to say that now on the scale of galaxies, that the laws of gravity shouldn't be changed? In other words, just as it worked for Le Verrier to propose missing matter—Neptune—and just as it worked for Einstein to propose a change of gravity and no new matter, so, too, may it be that the laws of Isaac Newton's gravity need to be modified, such that force doesn't equal mass times acceleration, but it equals mass times acceleration, times this new constant, this new function, μ , which is a modification of acceleration that you get that only occurs at small accelerations. So, because galaxies are so big and they're so old, you only need a small acceleration over billions of years to

accumulate a large velocity that we see in the stars at great distances from the central gravitating luminosity.

- 6.75** And so if you go through the calculation, you find the velocity becomes more or less flat. It's different. And actually, we haven't seen evidence of this in data beyond galaxies. It does appear to explain the results of galaxies like the Milky Way, like the clusters of gal—like the galaxies that Vera Rubin studied. But it cannot explain the behavior of galaxy clusters, for example, and collisions between these galaxy clusters, seemingly.
- 6.76** So, what is dark matter? Where we started this lecture, honestly, we don't know. But wait, it gets better or worse, because as we explain dark energy, we'll know even less. So we should take some comfort that dark matter is at least perhaps a little bit more tangible, a little bit more visceral, than what we're going to encounter next. And I will give you a bit of an allusion to what's going to happen in the future. Dark energy is going to become much more important to the future of the universe. Dark matter, it was very important in the past history of the universe. But the future of the universe is almost entirely dictated by how much dark energy there is and the property of dark energy.
- 6.77** We have evidence for something like dark energy. We have no real good models for what it is. In other words, there's no analogous form of WIMPs, MACHOs, or MOND for dark energy. It's completely unknown what it is. It's known that it is, but we don't know what it's made of. That's a preview for lecture number seven.

Lecture 7: Dark Energy and Destiny

At a glance — Dark energy and cosmic destiny.

- 7.1 So, in the beginning of the course, we asked ourselves to consider waking up in a universe, an unfamiliar place, full of light, matter, unseen forces acting upon us, and wondering what sense we could make of it. And I said at that time, one of our goals is to understand, what is the ultimate fate of the universe? And another key goal is to understand what the universe was like, perhaps at the very beginning, and maybe if there are other universes. But before we get there, we have to take a detour and understand the very far future in order to understand the very distant past.
- 7.2 We left off last time talking about dark matter and what role it plays in the geometry, the shape, the size, the structure, the dynamics of our universe. And we made an honest admission. We don't know what it is. We seem to think that it is there. The alternatives to it have been mostly ruled out, and ordinary matter can't explain it. So what is it? We don't really know. But take heart, I said, because we know much more about dark matter than dark energy, which is the ultimate governor of what the future of our universe will be like. Today we want to understand what dark energy does and how we can possibly go about understanding the future fate of our universe based on the properties of dark energy.
- 7.3 So the last lecture, we concluded that there must be some unseen form of matter, some type of mass or energy or modification to Newton's laws, that affect and afflict the way that gravity affects structures of clusters of galaxies, individual galaxies, stars, etc. We looked at ways to detect the presence of dark matter or the effects of unseen forces and modifications to Newton's laws. And the inescapable conclusion, even though we don't know what dark matter is or that it exists, is that there must be some 20% to 30% of mass energy in our universe that is unaccounted for by visible matter alone.
- 7.4 So we did estimations, and we come to find the universe has this missing component, which we call dark matter. So we had to ask, what could the dark matter be, and what other types of mass or energy are there in the universe? We are made of matter. So even for all of eternity, people have known that we exist. Matter exists. We know that light exists, heat, and so forth. In recent centuries, we've known about subatomic particles that exist that make up the ordinary matter, and even strange matter like neutrinos that only interact via gravity and the weak nuclear force.
- 7.5 So is there something else in addition to ordinary matter? To dark matter? Yes. The answer seems to be there is incontrovertible evidence for the existence of a strange form of all pervasive energy, pure energy that fills and suffuses the cosmos, that does not seem to clump the way dark matter does, that seems uniformly distributed in accordance with the cosmological principle, but nevertheless is perniciously difficult to

attempt, to even detect, its true nature. So we have to go as far as we can with our current tools—telescopes, particle accelerators, etc.—in observations of the distant universe that led to this notion that there is this missing energy, which, as I say, we call dark energy.

- 7.6 And it's only been known about in one true sense. Although it was originally predicted by Einstein, none other than Albert Einstein in 1917, still we've only known that it actually exists for the last 25 years. So in the late 1990s, two different rival teams of astronomers were looking for the following seemingly obvious fact. Because there's matter in the universe, it should gravitationally attract all the other matter in the universe, and eventually the expansion of the universe would not continue. The gravitational pull, like the pull of the Earth on a baseball thrown from its surface, would eventually cause the universe to collapse on itself, to decelerate, to cause the universe to stop expanding after some period of time. Whether that was deep, deep in the future or not was a question that remained to be seen.
- 7.7 Their presumption, their bias, was that they would find that galaxies that are far away from us would eventually start coming towards us, which is not in violation of the Big Bang model. It's just a natural consequence of a universe that is filled with matter. Matter only attracts gravitationally, unlike electrostatic charges, which could be positive or negative. So they had a prejudice, that the universe should be eventually contracting once you look out on far enough time scales. They found exactly the opposite. They found the universe was not only not slowing down, but its rate of expansion was increasing. The universe is accelerating.
- 7.8 And they did this by comparing the theoretical models that describe the different universes possible in Einstein's theory of general relativity, with the observations of the known amount of matter, of light energy, of dark matter, that was predicted. And they found a gap between what was observed and what was actually seeming to be present in the universe. And this has some of the most startling conclusions, not only for the composition of the universe today, but the composition of the universe tomorrow, and that determines its fate. The composition, we like to say, the density of the universe, determines its destiny, and it will determine the far, far future state of our cosmos.
- 7.9 So the Hubble equation from which we derive the expansion of the universe and can predict the behavior of the dilution of temperature with time, that we described in lecture five, in full form is written as the Friedmann equation, which is shown here, which shows that the rate of expansion, the Hubble constant, the Hubble parameter, depends on the density of matter, ρ_m , plus the density of radiation. Actually, radiation gravitates. It pulls. Radiation acts like a form of matter. It only acts gravitationally, attractively, as surprising as that sounds. And you can actually add a term to this equation that represents how much energy there is in the universe.
- 7.10 And astronomers thought they had it all figured out, and all they had to do is measure how much luminous matter and dark matter there was. And that would tell them, when

is the universe going to end its fate in a collapse, in a crunch, due to the decelerating nature that they expected? So this is quite surprising in that, because of the presence of an ultra-luminous class of stellar explosions called a type one-A supernova, astronomers revolutionized their conception. It overthrew and proved them all wrong, that what they expected to see was in the opposite of what they actually observed.

7.11 And in some sense, that's a very, very good thing, because the opposite of that is when you confirm what you already preconceived to set out to detect. That leaves you liable to what's called confirmation bias, which is a phenomenon where scientists, because we're people, tend to exclude data that disagrees with their favorite hypothesis, and they tend to only include data that seems to agree with it. It's a form of cognitive bias, psychological prejudice, that afflicts all people, and scientists are not immune from it. So it's more pure to make a discovery that completely obliterates the hypothesis that they had set out to achieve. And this will have two ramifications, because it will actually give us more credulity in the theory, based on the new evidence that contradicted what everybody thought before. And it's not some big conspiracy. And furthermore, we'll see that it ties into a mistake by the great Albert Einstein, which even his mistakes turned out to be incredible successes in another way. We'll explain in a moment.

7.12 So astronomers in the 1990s were looking under the lamppost, looking where the light is. They were looking at ultra-luminous beacons called type one-A supernovae. When one of these objects blows up, it's caused by the coalescence of what's called a white dwarf with a larger star exceeding a certain mass threshold, causes a gravitational atomic bomb to go off. And these objects, as you can see on here, this is one from the 1990s, was shown down here in this galaxy, this Sombrero Galaxy, it's called. This galaxy's entire light of 100 billion stars will be out-shown, outshined by the single, single star that blew up. The entire luminous output of an entire galaxy dwarfed by a single star blowing up.

7.13 The catch is, the galaxy will go on for billions of years. That star only puts out this luminous amount of light, this incredibly ultra-luminous burst, for a few dozen days. And in that time period, astronomers have figured out how to use this as a type of standard candle, as we discussed before. These are gravitational time bones. They blow up because a white dwarf, or WD, accretes mass from a companion. It actually turns out that our sun is unusual in that it's a solo star. There's no other star nearby our sun, but most stars, slightly the majority of stars in our galaxy, are actually binary stars. There's one star or more than one star, and they can even be dozens in a cluster.

7.14 When a white dwarf sucks matter in as a very compact, dense object, it has very strong gravitational tidal forces that can pull mass from an ordinary star, just like our sun, perhaps. When the combined mass of this star system is greater than a certain threshold known as the Chandrasekhar mass, the star blows up, and that's only 40% more than the mass of the sun. So it takes slightly more than the mass of our sun to accrete onto another star, called a white dwarf, to cause it to detonate and blow up

and convert this mass of the star, 1.4 solar masses. Get instantly, not instantly, but very rapidly, converted to energy.

7.15 So Einstein's equation again, $E = MC^2$. C^2 is enormous. This just has to be 1.4 solar masses. When that occurs instantly or very rapidly, the star system explodes, detonates, and releases in the form of energy photons, exactly 1.4 times the mass of our sun, no more, no less. If it's less, it can't blow up. If it's more, it would already have blown up. That gives us an opportunity to have a source of known energy that detonates and explodes over a finite time, or that gives us a standard candle, a source of known luminosity, a source of energy that is produced over a finite period of time—the definition of luminosity. It's as if it's a light bulb. The light bulb is not in watts, very easy to express. It's trillions of watts, and that's the conversion of that solar mass energy, 1.4 solar masses, to pure energy.

7.16 So it's a standard candle, and once you have a standard candle that's this bright, you can detect it from clear across the universe, from distances of billions of light years away, which give us a chance to trace back the history of the cosmos over a longer distance ladder rung than we could do with the Cepheid variables, and certainly much farther than we could do with the parallax method. Remember those rungs on the cosmic distance ladder? Radar, parallax, Cepheids, then the Hubble flow, and then this type of detonation can be seen literally across the entire universe. It's that bright.

7.17 We use the standard candle formula to compute the luminosity distance. Shown here, these three standard candles. I couldn't bring in the 1.4 solar luminosity version of the mass version of it. But these can also tell us distance. By measuring and knowing the written-down luminosity of each candle, you can measure the distance. It's a way of making a ruler by just knowing the geometry. So when this was done, the standard candle nature allowed the intrinsic brightness to be roughly known, which allowed the calculation based on the observed flux, to calculate via the luminosity distance formula we discussed several lectures ago. So, once you know the luminosity, we got the distance from the telescopes that observed it via the flux that they observed.

7.18 What they found was astonishing. It doesn't—again, astronomers and physicists, as I showed you with the paper of the cosmic microwave background that really obliterated the standard model that the universe had been static and eternal for all time, that was entitled "A measurement of excess attendant temperature at 4,080 megacycles per second"—not exactly something you'd really think had the dramatic impact that it did. These plots here from two teams in the late 1990s demonstrated once and for all that our universe couldn't be made of just matter and radiation. You needed a third component, a missing component. Not matter, not dark matter, not light, not neutrinos, not gravitational waves. Something else.

7.19 And what they found in these technical plots here, the farther you go back in time, the more the light had been diminished, as if the stars that were producing those stellar explosions called supernovae, they had been more jam-packed closer together, farther

in the past. And now, as time went on, they were getting more and more diffuse at a rate greater than just a linear expansion of the universe, that the universe, someone was pressing on the cosmic accelerator pedal, that the rate of change between the galaxy's distance was getting bigger and bigger and bigger, day after day after day. Not just the distance, the rate of change of distance, the velocity. Therefore, when velocity changes, you have acceleration. That's the definition.

- 7.20** The only way this could be included was that there was a new form of energy that was inflating the universe, expanding it faster and faster every day. And so it was very different than what they expected. They expected to see this. They expected to see that time, over time, the universe was decelerating. It was still expanding for a while. We'll see what maximum expansion it could achieve in just a bit. But the thought was that the universe was going to expand from the Big Bang eventually, where we are today at the present day, and then it would either stay constantly the same size, or eventually it would start to collapse and compress, sort of a Big Bang in reverse.
- 7.21** Instead, we got the following situation, that in the past, these two plots are more or less consistent before now, although not exactly consistent. And then in the future, things are going to get more and more widely separated at a faster and faster rate. The Hubble constant is not a constant. It was changing, and changing at an accelerating rate. When something is changing, it can either increase or decrease. They thought it would be decreasing. They found it was increasing—again, very pure, because this was counter to their expectations. So you couldn't say, "Oh, you were just biased to find what you wanted to find. Of course you found it."
- 7.22** The same thing happened, by the way, with Penzias and Wilson discovering the CMB. They went out to look for a source of excess noise around a satellite. They weren't saying, "Let me go find evidence that refutes the steady-state theory." Again, that's a serendipitous discovery. Those are very powerful in science, rather than discoveries that you go out and seek to discover, which can be valid as well. Once you know a phenomenon exists, you're going to try to confirm and learn more about it. Same with the double helix structure of DNA. People had different ideas that it could be a triple helix or a single, and then it was found to be a double helix. Slightly, in a similar vein, a serendipitous type of discovery. Therefore, it carries more weight.
- 7.23** So we know now, since the late 1990s, the universe is not just expanding, it's expanding at an accelerating rate. That means it's consistent with the Big Bang origin, because this type of energy was not important until a few billion years ago, and now it's the dominant form of energy in the universe. And the behavior, as we said before, the behavior of energy depends on what substance you're talking about. So the density, we'll call it N , the energy density of a substance depends on what that substance is made of. So the energy density for matter would depend on the number of matter particles per unit volume. The volume, in the case of matter, or in the case of the universe, expands as the scale factor to the third power.

- 7.24 Now, the energy—so this is the ordinary energy density of $E=MC^2$ type particle energy. But for photons, we said it was the number density times the energy times energy, E , per photon. And the energy per photon depends on the frequency or inversely proportional to the wavelength. So this phenomenon leads to the fact that the number density times the energy per photon dilutes as not just 1 over A cubed, that it goes as 1 over the number density, goes as 1 over A cubed. But then the energy per photon goes as 1 over A . Because the wavelength increases, the frequency decreases, the energy per photon decreases. The combined effect is that this decreases as the fourth power of the size of the universe.
- 7.25 So, in the early, early universe, in the late universe, if you plotted energy density, if you plot the energy density versus component of the universe, there's three different behaviors. The energy density of radiation— so I'm going to plot script E , depending on its constituent. And this is versus scale factor or time. As the universe gets bigger, the energy density of photons drops. The energy density of matter drops as well, but it drops slightly more slowly. So this is going as 1 over A cubed. This is going as 1 over A to the fourth. And then if you have another source of energy which is constant in time, its energy does not dilute. It stays constant. And this is the energy of what we're going to call a cosmological constant, or a form of dark energy.
- 7.26 So, at various points in time, the universe either had more energy in the form of photons, in the early time in the universe's history. Then it had more energy in the form of matter, and currently, it has more energy density in the form of dark energy. The question is, how does that affect the observable quantities and parameters that govern the universe's vital signs, as I call them? So the mass energy density in our universe can have an infinite variety of different possible values. It could be above a certain value or below a certain value, or exactly a given value. And it turns out, our universe, of all the three different types of curvature that it could have, of all triangles you could make in the universe with galaxies, clusters of galaxies, stars, planets, what have you, it seems to be the one universe that has zero curvature, or therefore Euclidean curvature —flat.
- 7.27 Any parallel lines you shoot out in the universe away from other local sources of matter density, like the sun or a black hole, those two lines will stay parallel forever. That means we live in what's called a critical density universe. A universe that has flatness is a very special universe. And in such a universe, we can ask the question of the total amount of energy in that universe is made up of ordinary matter, us. Radiation, we can see things. We see light, and we see heat of the CMB, and then dark energy, and we can add those up. But they have to sum up to 1 , what's called the critical density.
- 7.28 And for also about 25 years, we've known, based on the distribution of the pattern of fluctuations of the cosmic microwave background, that the total amount of mass energy density in the universe is this critical value. It's the only value that can make triangles always add up to 180 degrees. That's called a flat universe. So the flat universe

implies that the sum of the energy density for matter plus the sum of energy density from radiation plus the sum of energy density from dark energy equal to 1.

7.29 We can measure very accurately all three of those, and they also agree. So you can do geometrical. You can make a geometrical measurement of the triangle patterns and see that they add up to 180 degrees. And when you do that, you add up the total amount of mass and energy today, and you find there's about 30% in the form of ordinary matter. There's almost no energy density in the form of radiation because the CMB photons are so low energy today, and there's 70% of the critical density in the form of dark energy. So it's exactly a flat universe where Euclid's laws hold, and it is dominated by dark energy.

7.30 Now, the question is, is this dark energy truly constantly, or does it depend on time? Are there other sources that could cause the dark energy not to be constant with time? And it actually has a long history in cosmology, dating back to Einstein himself. If it's constant in time, if the dark energy doesn't change, then we call it a cosmological constant. And this was actually a term first conceived of by none other than Albert Einstein, because what did he want, at that time, before the 1965 discovery of the cosmic microwave background? The prevailing paradigm was a static universe. But I already told you that a universe cannot have perfect stability in it as long as it has matter and radiation, because those are only attractive. Even photons, even light, is attractive gravitationally.

7.31 So Einstein came up with a concept that you could insert into the vacuum an all-pervasive fluid, if you like, a form of energy that he called the cosmological term or vacuum energy. We don't know what it is, but we see that it exists. And in a sense, this is kind of, this is kind of remarkable that Einstein had the scientific magnanimity to accept that his term was actually wrong. So what happened was Einstein predicted that the universe should be static based on all the evidence of that time. It didn't look like things—the universe was expanding or contracting in the 1950s.

7.32 He didn't even know, as nobody knew, that the Milky Way wasn't the entire universe. Remember, it took the work of Vesto Slipher, Henrietta Leavitt, and Hubble into the 1920s to discover that Andromeda was located 30 times the visible radius of the Milky Way. So it couldn't be in our Milky Way. It had to be another galaxy. So certainly by 1917, when Einstein was working on this theory, there was good reason to say the universe is static. After 1929, though, when Hubble showed Einstein at Mount Wilson—a famous picture of them together—did Einstein say, "I made the biggest blunder of my life," and as I always say, it's too bad, because he could have had a good career. No, he actually had a great career, although general relativity really was the last major accomplishment. Of course, it has so much incredible discoveries and predictions from it.

7.33 So he actually was a great scientist because he was able to say, "I was wrong. Even though the data seemed to confirm my hypothesis, when I was shown new evidence

that contradicts me, I accepted it." Now, what's ironic is that he threw out the cosmological constant. He rejected it because of the expanding universe. But he didn't realize that 100 years after this, astronomers would discover that actually the universe is not just expanding, it's accelerating. And the only way to have acceleration is to put back in this term that he discarded as the greatest blunder of his life. So it's kind of ironic. Things really went full circle. I like to think about talking to Einstein after his death in 1955 and saying, "Just wait a couple of years." He should have kept this in place.

- 7.34 So once you know there is mysterious substance, we don't know what it is. We just see its effects on the space-time geometry and curvature of the universe. Once you know what it is, now there are several dramatic—there are several dramatic scenarios that can take place in the far future of our universe. So there are typically several different classes of what can happen to a universe that has dark energy and ordinary matter. If the dark energy is too low compared to the matter, it will still, the universe will still accelerate for a while, but eventually the acceleration will meet a maximum. There'll be a maximum distance between all galaxies, and eventually they'll start to come together, sort of in a reverse of the Big Bang, or Big Crunch.
- 7.35 You could also have a very, very slowly decelerating universe that, over any realistic timescale, while there are still matter particles in the universe, that it could basically stop accelerating and just stay at a constant size for all time. That is possibility as well, when you have matter in the universe and radiation as well. You could have a slowly expanding universe that doesn't have acceleration, that just has expansion forever. That's a possibility. That's called a coasting universe. And lastly, you can have an accelerating universe. And even with dark energy and dark matter, all these possibilities are still in play, because there's nothing to say that matter exists forever.
- 7.36 If matter can annihilate, as we know matter and antimatter can, then it's possible for the matter energy density to eventually convert to radiation energy density, and that will change how fast the universe expands. Similarly, dark energy may not be constant. It may slowly change, and just over the length scale of our current observable universe, it hasn't changed enough to become measurable. That doesn't mean it won't be measurable very soon with telescopes that I talked about in the very first lecture of the class, the Nancy Roman Space Telescope.
- 7.37 So, if you want to know what your life expectancy is, you go see a doctor, and the doctor will take some tests, some blood tests, some, you know, different, you know, maybe, I don't know, saliva. I don't know what doctors do, but they'll take your past history. How did you, how did your parents live? Now, the universe doesn't really have a parent in the sense, except if it came from another universe, from a cycle of crunches and bangs, you might think that it does have a past history, and that could be relevant, as we'll discuss.

7.38 So, knowing what the universe is made of, knowing how old it is, right? So you have a different number of years left to go. If you get this consultation with your doctor when you're 15, versus when you're 70, there's a certain difference in expectation, just by virtue of the fact that you've lived longer than a 15-year-old. And so you can make predictions about the various possible fates of a patient. So we have to look at what are the vital signs? Well, the analog of age for the universe is the reciprocal of the Hubble constant. As we said, the units of the Hubble constant are kilometers per second per megaparsec. Kilometer is a unit of distance. Megaparsec is a unit of distance, about 3 million light years. And therefore, you have distance divided by distance divided by time. So it's a reciprocal of time.

7.39 Therefore, if you take the reciprocal of that, you get an age. So when Hubble first made the estimation in 1929, as I showed in lecture three, that led to an overestimate of the age of the universe, an underestimate of the age of the universe, because the Hubble constant he measured is about 7 times bigger than we believe it to be today. So, therefore, he got an age of the universe that's 7 times lower than the age of the universe we know today. We believe the universe is about 14 billion years, 13.8 billion years today. Therefore, he measured an age of the universe about 2 billion years, which was highly embarrassing and actually caused questions about the validity of the Big Bang theory, because there were already things on the radioactive isotopes and dating of elements and so forth on Earth that indicated the Earth was 3 billion years old. So if you went to a doctor and he says that you're older than your parents, you might not go back to that doctor again, right? So it was embarrassing to know that there were things that were older than the universe, inside of the universe. That was not acceptable. But as scientists realized, the measurements got better and better. They came down to be, to be consistent with the observations of local observations of Earth

7.40 and things in the Earth's history, which are the oldest things we have easy access to.

7.41 Now, when cosmologists were thinking about how to increase the age of the universe, one way was to add a cosmological constant, this lambda term that Einstein had put in originally. And a lot of people were uncomfortable with that, until the vital sign of measuring the universe's composition was actually accomplished. So this is going back to our survey of the universe, where we talked about Cepheid variables and the so-called cosmic distance ladder. These tools are still used today. So parallax is used for local things in our galaxy. We can find things like Cepheid variables. There haven't been any type 1 supernovae in our galaxy in a long time, maybe not in recorded history, although before, a couple hundred years ago, we wouldn't have noticed it, but now we would certainly notice it. And depending on how close or far away it is, it could be quite a bad event for our galactic future.

7.42 But be that as it may, we overlap in cosmic distance scales from parallax to Cepheids, which are fine, and they won't harm us. And then Cepheids then, we can use to calibrate galaxies that have both Cepheids and type one-A supernova. Those overlap in

the distance ladder rungs. So we can measure out, effectively, billions of light years across the observable universe, as long as there were galaxies producing these types of explosions. And the reason that these types of explosions are used, as I said, is because they lead to very close to standard candles, candles whose luminosity is written on them, in effect, because they can't be brighter, they can't be more massive than 1.4 solar masses, or else they blow up before that time when they start to accrete.

- 7.43** And so you have to once determine that these are type one-A supernova, which astronomers are very good at doing, based on their chemical fingerprint called the spectrum. Type one-A are very different than other types. Type two that make the meteorites that I've produced and given to you guys, I didn't produce, but the universe did. But you have them as your parting gifts. So the other vital sign, your doctor might put you on the scale and weigh you. And as I said, the density of the universe determines its destiny, its future, in the sense that overmassive will cause the universe to collapse, even if there was a large amount of dark energy as well.
- 7.44** We believe we live in a flat universe, again, where all triangles add up to exactly 180 degrees. We get that startlingly important number from observations of the cosmic microwave background. And so how does this determine the fate of the universe, knowing its geometry? Well, an analogy is helpful. So there's a phenomenon called escape velocity, which is that, if you have a projectile and you give it enough momentum, mass velocity, enough velocity, there's a question of whether or not it will continue accelerating and reach into space. It's not how rockets work, but it is a phenomenon known as escape velocity.
- 7.45** So, depending on how fast you throw an object or give velocity to it, it can actually escape the Earth's gravitational potential and go to infinity, never come back to Earth again. If it has less than that, it can get to a very, very high distance, far distance. But it will eventually come back, collapsing, if you will, to the Earth's surface. Larger speed, longer, farther distance, smaller speed, shorter distance. And there's a critical velocity called the escape velocity, at which it will go to infinity, no matter what. And what determines the escape velocity are the properties of the gravitational mass that you're resisting and you're throwing out to a distance.
- 7.46** So the escape velocity depends on the density, which is the amount of mass per cubic meter, and the radius—the closer you are to a gravitating mass of a given density, the more escape velocity will be required. And from the Earth's surface, it's about 11 km/second, so if you throw a baseball at 11 km/second, it'll guarantee to reach and never come back to the Earth's surface. If you throw it with 10.9 km/second, it will come back down to the Earth. It's pretty hard to get a baseball to go 11 km/second, but that corresponds to 25,000 mph. Trust me, it's much worse on the surface of the sun because the sun is more massive and much higher gravitational potential than is the Earth.

- 7.47 So now imagine in our universe, you're looking at a region of the universe that's a spherical region that contains all the matter, all the gas, all the protons, all the neutrons, all the electrons, all of the dark matter and all of the dark energy. And now imagine you're at the edge of this sphere. This is analogy to the escape velocity in the Earth. If you throw something, or if the surface of the sphere is traveling faster than the escape velocity based on the radius and density of that sphere's mass energy, the sphere will escape and will expand forever. If it's less than that, it will contract.
- 7.48 So we're looking at the whole universe and dividing into little spheres and asking, are those spheres expanding? Yes. Will they always expand? Depends on the amount of mass energy inside of it. When you calculate what that is, you get an effective version of the escape velocity and you can solve for whatever that escape velocity is in the exact same way you did it on the surface of the Earth, just using Newton's laws. And when you do that, you come up with a critical density of mass energy in the universe that's equal to the velocity divided by the radius times the 8 times pi times Newton's G, gravitational constant. This does not involve Einstein whatsoever. So a small sphere that's rapidly expanding would need a large mass energy density to cause it to collapse or to stall and stop expanding.
- 7.49 This turns out to be exactly equal to the Hubble constant. Hubble constant is basically the velocity of an object, is the Hubble constant times the radius or distance to that object. That's how fast a galaxy at a distance R is traveling. Its velocity away from us is Hubble's constant times the distance. So therefore, the velocity divided by the distance is the Hubble constant. And this is just the velocity divided by the distance squared. Therefore, the critical energy density of our universe is $\frac{3}{8\pi G}$ times the Hubble constant squared.
- 7.50 Why is that useful? Because we're now measuring the mass of the universe, or the density. We've now constructed a tool to measure the density of the entire universe, and it involves just knowing two numbers. One is G, which we can get on the Earth very accurately because it's a universal constant. It holds here, it holds in the Andromeda Galaxy, and it holds in the farthest reaches of our universe. And then the Hubble constant we have to measure. But astronomers are good at doing that using the cosmic distance ladder, using standard candles.
- 7.51 And when we do that, we get a value for the critical energy density, above which the universe will re-collapse on itself, below which it will expand forever. So we know those two values, so we can plug those in, and we get a value for the density of the universe. If you go out into the universe and you have a meter-sized cubic box, and you ask, what is the total mass energy inside that box? It's equivalent to a mass of about 10^{-26} kg, which, no matter what, sounds very, very low. And it is.
- 7.52 It's literally a billion, billion, billion times smaller than the density of air in this room. So 1 cubic meter of air, which is not that massive, has a billion billion billion times the amount of mass in it than the equivalent region of the universe, once you're far away

from a planet or a galaxy or a star, intergalactic space. The average density. We can convert this to more appropriate-size scales, and we get that the equivalent mass energy in a box, it's the equivalent of about 5 protons. Imagine you have a box in the middle of nowhere in space. Inside that box, there's only 5 objects, 5 protons. Their mass energy equals MC^2 , converts to this, 10^{-26} kg/m^3 .

- 7.53** That's an incredibly, incredibly dilute, almost completely, I mean, if there were 5 fewer, there'd be nothing. So changing that value by 1, adding 1, would make the universe collapse on itself. One proton worth of energy per cubic meter in the universe. Now, of course, there's a lot of cubic meters in the universe. We're talking about billions of light years. Convert that to meters and then cube them. Incredible.
- 7.54** Now, when we look out in the universe, we see things like the Virgo cluster, these other clusters, it's not really clear. You have to think that the sun is the only star for 4 parsecs in any direction. The closest one is a little bit over 4 parsecs. And 4 parsecs, convert it to cubic meters, and then you take the sun's mass and divide that, and it comes out to be —it's not actually clear if the average density of the universe near the sun is actually above or below the critical density. But when you survey on big enough size scales, you find that the universe is exactly equal to the critical density. This is a vital sign, like the weight of a patient.
- 7.55** It's not completely predictive, but it's important, and it helps to determine the ultimate fate of that person. So the separation between galaxies will increase over time, if the universe is at the critical density of these 5 protons per cubic meter, average mass energy. Or if it's too light, there's not enough gravity to keep the escape from occurring, when you throw a projectile, or if you like to keep the expansion slowing down. It will expand forever if the density of the universe is less than the critical density. But if it's greater than the critical density, if ω —this parameter ω , called the energy density parameter—if that's greater than 1, the universe will collapse on itself after a finite amount of time that we can calculate.
- 7.56** These are possible fates. These two involve expansion forever, like escape of a projectile to infinite distance. And this involves the re-collapse or the collision of the projectile with the Earth. So there's a couple of different scenarios. They have different names. We call one the Big Crunch. If the universe is over-dense, it will collapse on itself, becoming hotter and hotter with time, if the density is greater than that 10^{-26} kg/m^3 volume. Or if it's less than or equal to critical, it will dilute forever. The universe will get larger. The scale factors will increase. The energy densities will decrease. The universe will dilute forever.
- 7.57** Its temperature will become colder and colder, never exactly reaching absolute zero, but getting colder and colder. And this goes by the name of the heat death of the universe, which is ironic because its heat is, we think of as warm. But it actually means that you're not able to do anything with the universe's energy anymore. Its become so highly disorganized, so spread out. So in a Big Crunch, you could have a single Big

Crunch, and then that's it. You could have had a previous Big Crunch where we're living in this phase where the universe is expanding, but eventually it may reach a maximum size scale and then collapse.

- 7.58** And you don't know where you are in this phase diagram or in the ultimate timeline. But you know, you're sort of at one of these points where the universe is expanding. It's definitely not contracting. And in fact, we don't believe we live in this type of a universe. This type of universe has been falsified, of a collapsing universe. As far as we know, our best observable evidence leads to that. So we have three different possibilities for curvature. I've already said those. In a negative curvature universe, it's open. It will be infinite, like an infinite saddle or Pringles chip, expanding forever.
- 7.59** If it's compact, it's closed, it's finite. The universe has a finite expansion that will reach a finite size. You could actually take a trajectory. Light could follow a trajectory that would bring it back to where it started in such a universe. And that's a—that was a possibility, with a concomitant collapse, Big Crunch, or oscillating universe that I showed on a previous slide. Or it could just sort of coast forever, expanding after an infinite amount of time, for an infinite amount of time, maintaining this perfect spatial flatness.
- 7.60** And these are the different possibilities. And then within them, there are subcategories of possibilities where there are variations in the amount and property of the mass energy that can cause differences in the behavior of the matter, if it were to decay, or the dark energy if it were to decay as well. So astronomers are looking very carefully, along with particle physicists, to see, do protons decay? Are they like neutrons, just with much, much longer lifespans? Are they more comfortable being alone than a neutron, which dies in 800 seconds, if not bound to a proton? Maybe a proton will only live for 14 billion and 1 year. You know, in a couple hundred million years, it will be gone. We don't know. So scientists are setting limits on it and actually have extended it way beyond billions of years to perhaps trillions of years.
- 7.61** So the curvature universe of our universe is an important parameter. It's related to its destiny and its density. We've calculated how much matter density there is. So of the critical energy density, only about 5% is in the form of matter like us or stars like the sun, or gas like the plasma that produce the cosmic microwave background. There's about 25% dark matter in the universe that we don't account for. At least, it seems to be our best guess. We don't know what it is. And that means that 25% plus 5% is 30% of the critical energy density. 70% remains. And that's all in the form of dark energy.
- 7.62** The question is, is the dark energy a constant, what's called a cosmological constant? So, we already looked at this plot in the previous lecture. In the previous lecture, we looked at the composition of all the atoms in the universe. So here's the overall cosmic pie chart. Dark energy is slightly over 70%. Dark matter is about 23%. Ordinary atoms are 4.6%. And then if you break down the atoms and the dark matter in this chart here, you can divide that into how much these different atoms make up the periodic table.

And actually, people have done it for the atoms that are necessary for life, for human life. So we don't need, say, argon or nickel or aluminum in our body, but we do need calcium. And so these amounts, the amount of calcium, for example, makes up our bones.

7.63 It's on a relative scale. You can't even see it here. Hydrogen makes up 7,500 ten-thousandths of this pie chart here. And then calcium would make a 0.7. So you're talking 10,000 times less than the amount of hydrogen in the universe. So the stuff that's important to us is pitifully, almost insignificant on a cosmic energy scale budget, but obviously, it's very important to us as a vital sign for our own personal health and lifespan.

7.64 So the universe seems to be flat. Astronomers have measured that there's unavoidable evidence for the accelerating expansion of the universe. And so what can happen in such a universe? We're acting like we have a repulsive form of gravity. It's not just that the universe is expanding. The expansion is accelerating. It's a very strange form of energy. It has what's called negative pressure. An ordinary balloon, if you try to compress, it, resists. A balloon filled with dark energy would collapse even faster if you applied energy to it. If you want to make a balloon expand, you have to let it, it will do work on the surrounding pressure in the room, and a dark energy balloon will be just the opposite. It's as if the universe is filled with a form of elastic or rubber, that wants to be contracted, that doesn't resist contraction. It actually wants to be contracted. And to make it expand, you have to supply energy to it, unlike a balloon, which will be expanding against the environment around it.

7.65 So Einstein was really premature in his ruling out of dark energy. We don't know, as I say, if this energy will someday convert into another form that will lead it to behave more like matter, in which case, the pie chart would get reorganized, and it could be that the universe could collapse again. If you convert dark energy, which is a constant to matter, then the universe will stop accelerating. And so astronomers are trying to determine if it's constant or if it's not.

7.66 And there's different names for these types of constant energy. One is called quintessence, for the fifth form of matter. And astronomers have been looking at this because if there is a solution where dark energy is varying, it can actually cause two different possibilities can occur. The dark energy can get more and more energetic. Somehow, it could convert to even more anti-gravitational behavior, causing the universe's acceleration to accelerate. That's a possibility. Or it could decelerate and convert to matter or decay.

7.67 Either way, astronomers are looking for variation very, very closely. So there's a couple of different possibilities, and I have to say, none of them are really that appealing, which leads to a very, maybe, perhaps terrifying question. Since the universe can have these different possibilities, either a crunch where all the matter gets compacted and eventually perhaps reaches even a singularity, the opposite of the Big Bang, a Big

Crunch—that would not be a very hospitable place to live. Or the universe can keep expanding at an accelerating rate, destabilize dark energy, running away forever in what's called a Big Rip, where all of space and time eventually accelerate apart, and even atoms will rip apart from one another, let alone galaxies and clusters. That'll happen long before.

7.68 So the question is, why do we live in the one type of universe right now? What made us so fortunate? Why do we live in a universe that has just the right amount of dark energy, of ordinary energy and matter and dark matter to permit the existence of life? Isn't that highly improbable? And people say, of course not. If we lived in a universe that prevented life, we wouldn't live in such a universe, because that's a paradoxical contradiction. So, here's a universe. These are just slides that show, as we transition to other wavelengths that we haven't yet encountered in this class, things you can't see. What will happen in a universe as it expands forever with a cosmological constant? That's dark energy that doesn't change.

7.69 And the universe could accelerate forever, perhaps for 100 billion years, 10 times the age of a current universe. And if that happened eventually, astronomers in such a time, they won't know that they live in a universe that has other galaxies. The galaxies will be so far apart that their light will be completely redshifted out of observable, even being potentially observable. In which case, they'll look at their galaxy, and they'll say, this is the whole universe, just like astronomers did 100 years ago. Now, that'll happen in billions and perhaps hundreds of billions of years from now. And that is a possibility.

7.70 And so astronomers work very carefully to see what is the nature of dark energy. Now that we believe dark energy exists, just as we search for dark matter, let's look at the properties of dark energy. Dark energy doesn't cluster into clumps of matter like galaxies and stars, can trace the properties of clumps of dark matter. It doesn't seem to be like that at all. It seems to be constantly distributed throughout all of space, maybe intrinsic to space itself, what's called vacuum energy density.

7.71 So astronomers using what's called the chandra, as I told you, the limit at which a star, beyond which a star explodes in a type one-A supernova, is called the Chandrasekhar limit. Chandrasekhar was a phenomenal Indian-American astronomer at the University of Chicago. And they have a satellite named after his nickname, which was Chandra. And this is an X-ray telescope. So now I'm showing you a completely different wavelength range. We did microwaves. We've studied a lot of optical. The last type of technology we'll encounter in this class are X-rays.

7.72 X-rays are extremely energetic photons. You've encountered them, you've been exposed to them, presumably, if you've been to the dentist. They cannot be focused the way light can be focused by a refracting telescope or a reflecting telescope, like we've seen before. Why do you think that is? What do you think happens if you had an ordinary mirror shaped like a parabola and a Newtonian telescope, or shaped like a,

made of lenses, with an X-ray, knowing what you know about X-rays. What would an X-ray do? That's enough energy.

7.73 Not nearly enough energy to destroy it, but the light will go right through it. In fact, it will go right through it unless you have the light come in and make a glancing blow. If you shot X-rays at this table, if this table were made of metal, depending on how thick the table was, it would either go right through it or get absorbed in the electrons underneath it. But if you shot an X-ray beam like this, it would just barely bounce off of it, scattering what's called grazing incidence.

7.74 So astronomers synthesize a bowl-like telescope out of very thin sheets, rolled-up conical sheets, brilliantly designed. These are actually nested hyperboloids, they're called, on the inner part, and nested paraboloids. These are little shells, so if you're looking at it, it kind of looks like an onion. Or the light would come in, the X-ray light, and just barely graze off one of the surfaces and then reflect and get detected on a certain, on a detector down there. And X-ray detectors are a sophisticated technology as well. And this mirror is only about 1 meter across. This isn't even close to the biggest optical telescope 100 years ago, and yet it's the most powerful X-ray telescope ever made.

7.75 And just to show you, remind you where it fits in with visible light, we see things like this. Optics, and then X-rays tell us about the extremely energetic universe. The mechanism to produce X-rays are highly, highly energetic phenomena, typically associated with things like black holes, accretion discs, exploding stars, and other very energetic phenomena. So another animation from the Chandra team shows what would happen for a gravitationally bound cluster. It's a cluster of galaxies like the Virgo cluster that we're being pulled towards, and we're kind of zooming through it. And these galaxies have as much mass as this total starlight is equal to the amount of gas, just pure hydrogen and plasma, in these clusters. It's an enormous amount of energy not associated with the visible structures.

7.76 Where does that energy come from? These are some mysteries that we're still studying. When there's acceleration of electrons in a magnetic field or electric fields, they emit very high energy X-rays, and this is what Chandra is looking for. And it can see the properties of these different objects depending on their distance, to measure the size of these objects versus distance, and measure how far away these objects are, giving us an important cross-check on the measurements done by the type one-A optical telescope supernovas. And the size of these clusters versus a distance is a proxy for the redshift dependence of the universe, the expansion rate, as a function of distance.

7.77 So it also confirms that the universe is accelerating. Totally different technology, different groups, different teams, different methodology, all coming up with the same result, as does the microwave astronomy that I do, that the universe is accelerating, that it has a critical energy density that makes all triangles add up to 180 degrees. So

this is looking at optical and also X-ray telescopes. This is called a composite image. This shows a different light inside of these clusters of galaxies, perhaps hundreds or thousands of galaxies that make them up. It also gives evidence for dark matter, dark matter being the unseen force that seems to keep these things gravitationally attracted to one another.

7.78 Now, another exotic possible future for the universe is that, depicted here, that dark energy could increase with time, as I say. There could be a slight change where this is not a constant, but actually, at very, very long times from now, will start to increase. Now, why that would be requires a lot of, you know, perhaps creativity, theoretical creativity, but people should look for it. Similarly, they should look for what happens if the dark energy starts to slow down. In other words, it's not constant. And so, in such a case, the universe will expand. And every atom, star, and galaxy would eventually get ripped apart. And long before that, we would cease to be able to see other galaxies because they'd be redshifted so far out that all of their light is completely invisible energetically. And no telescope could detect it.

7.79 So, again, keep paying your taxes. We don't think that will happen. Here's the opposite. A Big Crunch. If the matter were to decay or the dark energy were to decay, eventually the universe could collapse on itself, perhaps igniting a new Big Bang. This is the family history, where it may be that our universe emerged from another universe, a parent universe, deep, deep in the past. So, as I say, this is sort of like the different possibilities. None of these are accurately known as yet, because we have to measure many more galaxies to determine the properties in many different wavelengths to see if we can detect a slight change in the amount of dark energy. Is it evolving with cosmic time, or is it a true constant?

7.80 So this is the last sort of animation, I think, for now. The universe is expanding and then eventually starts to accelerate. This is the type of universe that is our best guess for what we observe, based on a universe that has a cosmological constant, was expanding for a long time, due to the initial expansion of space-time, due to the amount of matter in the universe. But if there were only matter, it would eventually collapse on itself, because matter is only attractive. Instead, in our universe, we see the universe gets to a maximum, gets to some size, and then the acceleration takes place and galaxies move ever faster apart from one another.

7.81 One last time, Big Bang occurs. Matter expands, slows down, piles up these galaxies. And then dark energy takes over and starts to accelerate in the expansion apart. Obviously, these are very sped up, very hypothesized. But gives some inkling of what the future may look like for our universe. So, as we talked about, there's lots of evidence from optical telescopes, microwave telescopes, X-ray telescopes, all sorts of evidence that seems to indicate a universe that has matter and energy that are causing the separation between any two points in space to increase—and not only increase, but to accelerate.

- 7.82** Now, so how do we get the structures that we see? How do we get to the fluctuations in the cosmic microwave background? If the universe started off perfectly uniform, perfectly isotropic, perfectly homogeneous, that's a great mystery. And in fact, the spots and the patterns that we see in the universe seem to indicate that there may have been a type of breakage of that symmetry, from perfect cosmological proportions of complete isotropy and homogeneity. And that subject is the subject of our last and final lecture, which will have to do not with the far distant future of the cosmos, but its far, far distant past, and maybe, just maybe, the origin of time itself.
- 7.83** This very image that I started the course off with, that came from Wikipedia, which really comes from NASA, indicates a universe that's evolving, that we've covered in this class in the preceding 7 lectures. We've accomplished a great deal, from the origin of matter in the universe, Big Bang nucleosynthesis, to the expansion of galaxies, to the formation of the cosmic microwave background, the oldest photon light in the universe, and later, the accelerated expansion of the universe. We have one milepost yet to encounter, which is really this point far, far on the left, indicated by this baffling terminology, quantum fluctuations. We'll see what that means in the very last lecture, which is not just the fate of the universe, but really the origin of the universe and the startling possibility that we may not be the only universe that exists. So stay tuned for the final lecture. Coming up.

Lecture 8: Cosmic Comprehension

At a glance — Cosmic comprehension, the conclusion.

- 8.1** Welcome to the exciting conclusion of our lecture series on cosmology, which ironically ends at the very beginning of the universe, with some ideas, some concepts, some mind-blowing philosophical challenges to our understanding of the nature of reality. That's the topic of the 8th and final lecture today. We've come an extremely long way here today. We've done so, grappling with evidence, data, ideas, models, and some mathematics. That was unavoidable, but hopefully enjoyable as well. And our subject matter will conclude with how the universe possibly began.
- 8.2** But before we go there, we'll take a philosophical detour, which will actually force us to confront our ideas of what is possible to be known about an event none of us witnessed, none of us have access to. How can we have indirect evidence for something as important as the beginning of the universe? And that will take us on a path that is familiar to those of us who have studied the history of science and the philosophy of science, ever since the days of the very first scientists to really practice and understand and attempt to confront with data their ideas of the universe.
- 8.3** Starting with Aristotle, all the way up to Galileo, Kepler, Newton, and Einstein, we've witnessed what are called paradigm shifts, new ways of understanding our position in the universe that totally overthrew previous understanding. In the course so far, we've already encountered different paradigms that regarded a revolution from a steady-state, eternal universe, to a universe that had a beginning, a Big Bang, to a universe that might have a future endpoint in a Big Rip or a Big Crunch. And we turn our attention now to a paradigm that can help explain some of the gaps, the flaws, the missing pieces of the Big Bang theory.
- 8.4** And this happens very often in science. An idea, hypothesis, is promoted. That hypothesis is found to explain some of the evidence. Later, investigators come upon that hypothesis and find it has flaws. Those flaws are filled in by a new successor theory model that's based upon evidence. And guess what? That model will also have unavoidable flaws, because no scientific result is ever final. They're always provisional, in that new evidence could always come in. And today we'll look at ways that scientists have come upon the stunning successes in the Big Bang, but never being satisfied, found flaws in it as well, and attempted to patch that series of flaws and gaps with a new model called inflation, which in turn—guess what—has its own challenges, flaws, and gaps that still need to be explained.
- 8.5** So look at how far we've come since the very first day of class, when I showed this image, this circular, cyclical model where the universe began, creating energy into matter. That matter then became in the form of life. That matter that became life eventually developed consciousness—brains and minds that could comprehend the

origin of the universe and everything that came afterwards. The question we turn to now is, can we really understand, with the only supercomputer that we carry with us on our shoulders, that actual moment of beginning and how something can come from nothing?

- 8.6** So before we get to the topic of the origin of the universe, it's important to take stock of how far we've come and give ourselves some credit for this vast, incredible journey that we've taken together so far. We've applied the laws of cosmology, the laws of physics, measurements based on geometry, on the properties of matter and energy, to determine that we live in a universe which has precisely measured properties—its age, its mass, its geometry, and possible fates of the universe. It's a stunning accomplishment.
- 8.7** All of this has been accomplished in the last 100 years with a lot of twists and turns along the way. It hasn't been smooth sailing, even though that's always the way we present it. There are many mysteries yet to solve, and hopefully those of you out there may play a part in this solution to some of the remaining flaws, realizing that you, too, may not be the final word. We also have to confront the underlying principles that lead us to believe that we should have faith and confidence in the model that we've constructed, the Big Bang model. We still don't know, for example, if the universe as a whole can be a part of the traditional scientific method. After all, the scientific method often relies upon controls and variables, and we don't have access to such things with a singular universe.
- 8.8** And the question that we'll ask today should always be grappled with this famous physicist, winner of the Nobel Prize, Lev Landau, who famously said, "Cosmologists are often in error, but we're never in doubt." So take whatever I say as provisional. We are today going to examine the case that the Big Bang may not have been a singular event, not only in the sense it may not have been a singularity, a point of infinite density, infinite temperature, and infinite pressure, but it may not have occurred only once. In fact, it may have occurred multiple times. It may have occurred even an infinite number of times.
- 8.9** And startlingly, it may be happening currently, surrounding us, in a vast cosmic network, a landscape known as the multiverse, which itself fills in and follows from the great Copernican tradition, as we'll see. We have to ask ourselves, in the grand landscape of science, underpinned by mathematics, which itself is underpinned by philosophy and logic, we always have to ask the ways at which we go about gathering evidence. Are they self-consistent? Are they explanatory? Do they add more than they subtract? Do they raise more questions than they answer? This is part of what is known as the philosophy of science. The tools and methods to obtain information.
- 8.10** Do they limit the types of information and the conclusions we can draw from those data? What is knowable based upon the structure of reality, which we call ontology? The methods that we go about that, to seek out new knowledge, to acquire them in an

epistemological search. And when we look at competitors to our scientific models, there are many types. There are models that are scientific, but are not explanatory. There are methods that are explanatory, but not scientific. We can think back to pagan gods that had control over not just human fate, but also the laws of nature. That had explanatory power, but it was not self-consistent and was not using reliable scientific terminology.

- 8.11** We always have to ask, what is possible to be known? What is real? How can you distinguish causes from effects? Should there be a causal structure to our investigation, that cause always precedes effect? There are, believe it or not, models in which that is not true. And in fact, the principle of relativity first led to notions that causality and reality may not be so interconnected, because one person's event that occurs simultaneous to another person's event is guaranteed not to occur simultaneous to another observer in relative motion. Therefore, to them, effect could precede cause.
- 8.12** And those types of philosophical conundrums are part of cosmology by virtue of the fact they're part of our understanding of the structure of space and time. We should also be cognizant of the fact that we want to have as many disparate sources of confirmation or of exclusion, that we should not trust a single methodology, a single individual. There's no point and no place for authority in science.
- 8.13** And there's always a backdrop, unavoidable by virtue of the fact that science is practiced by scientists who are human beings, despite the stereotypes to the contrary, that sociology can sometimes supersede true scientific facts. Famously, back in the mid-teens, 19-teens, 1915 and 16, after Einstein's theory of relativity, both the special and the general theory of relativity, Einstein was in Germany, and the society of its time was very anti-Jewish. They were horribly anti-Semitic, and in fact, they were refused to award him the Nobel Prize, primarily because of his being Jewish.
- 8.14** And overlaying, on top of that, the recognition for his scientific triumphs, which we recognize as perhaps the greatest in human history, was a backdrop of sociology, of politics. And in fact, 100 German scientists, the top German scientists of the age, wrote a book. It was effectively called why Einstein is wrong. And Einstein very perceptively said, "If I was wrong, it would only require one person." So, echoing Einstein and Galileo, their sense that you can sometimes stand up against the crowd, really means to be suspicious sometimes of consensus, of authority. Because somebody won a Nobel Prize doesn't mean that they can't be wrong. And that's part of the politics, sociological nature of science done by scientists.
- 8.15** So we've come a long way. We've described the creation of nuclei, the creation of the cosmic microwave background, the observation of galaxies expanding, and then the presence of elements and so forth in our solar system, on Earth. And now the question is, can we peer back from our vantage point to an earlier time, before the formation of the cosmic microwave background, 380,000 years after the Big Bang, before the epoch of the formation of nuclei, a few minutes after the Big Bang? Can we go back to

fractions of a second after the Big Bang? Can we go back to time equals zero? Can we go back to before time equals zero? And what does that even mean?

- 8.16** So, putting in context cosmology as a science, which again, was almost laughable when I began my career 30 years ago, that cosmology could be considered a science when we disagreed about the age of the universe by a factor of 2, to now, when we understand the age, the composition, the curvature, the presence of dark energy, dark matter, to sub 1%, one of the most precise sciences possible, just in 30 years, one billion seconds, is phenomenal. We still have to ask, how does cosmology fare when we examine it as a science, with regard to the pursuit of knowledge, the tools to acquire knowledge, what is knowable, and what ancillary or supplemental hypotheses we need to add in order to have a model that's self-consistently holding together?
- 8.17** So I won't go through all of these, these individual topics, but I will talk about the most important aspect from my perspective, is that multidisciplinary reasoning leads us from one branch of physics to understand another branch of physics, and to take those disparate branches of science to understand cosmology, and have it hang together, is not proof of the Big Bang, but it's evidence in support of it—and against or disproof of alternatives. Now, the Big Bang model, again, doesn't answer the question of what banged. That's a very important topic, very important question.
- 8.18** What was the nature of the source of the energy that became the first nuclei, the first light, the first galaxies? Where did that come from? And as usual, we'll say we don't know for sure. But there are tools available to us for the first time in human history to approach that question quantitatively, not just qualitatively. So the Big Bang theory came about almost 100 years ago, and for a long time, really posited there's nothing to see, there's nothing to possibly observe before the Big Bang.
- 8.19** And therefore it didn't make sense to ask the question, famously said by Stephen Hawking. He said, "If you ask what is, came before the Big Bang, that's like asking, what's north of the North Pole?" We all know it's above the North Pole at Santa Claus. But in the context of cosmology, the question is a valid one, because if time came into existence, and time is the change between one state and another state of a system, then how can you have a change in something before the quantity of time even existed? It's a mind-blowing philosophical paradox, and one that's not fully answerable, but there are ways to approach it scientifically from a quantitative perspective.
- 8.20** Now we'll talk a little bit about how you can give a grade to a scientific theory. You've heard of the theory of evolution. You've heard of the theory of gravity. You've heard of things like Pythagorean theorem. Are they the same types of things? Should we have more credence in the theory of plate tectonics than we do in the theory of evolution, or evolution and cosmology in the Big Bang? To appraise those types of questions, which are perfectly valid within the context of the philosophy of science, we should, in many scientists' opinion, assign scores based on a sort of rubric, a framework, a way to

determine the grade that one would give to, say, if you believe or have lack of confidence in the theory of cosmology.

- 8.21** So a scientist, philosopher of science, Michael Keas, has put together what he calls the theoretical virtues, the properties of a theory of cosmology, of geology, what have you—that should be adhered to in order for a person who may or may not be an expert to have confidence in that model's validity. And we can use this scorecard—it's not the only scorecard. There are others, and we'll encounter some of those, most famously, a scorecard put forth by Karl Popper that a scientific theory is only scientific if it can be proven wrong. We discussed that in earlier lecture. It's not that you can prove it right. It's that it's available to be proven wrong. That's called falsifiability, and that is an element.
- 8.22** So the theoretical virtues, we'll go through some of these in more depth than others. That your theory, number one, should account for the data very well. That if you have a theory that the universe is hot and dense, as we'll see, does that fit the data that we see? Next, does it adequately explain all the causes that produce the effects that are observed? Now, no theory can explain every type of evidence that's observed, but some can include it, and some can avoid including it by virtue of the fact that, for example, we couldn't detect dark matter necessarily at the very beginning of the inflationary epoch that we're going to talk about because it wasn't relevant. Or we couldn't detect dark energy until relatively recently because its effect to cause acceleration are not manifest until just a few billion years ago.
- 8.23** I know it sounds funny to say just a few billion years ago, but it's true that dark energy was completely invisible to cosmologists who may have lived 8 billion years ago, and there's no reason to expect that they couldn't. We have no evidence for that to exist. But nevertheless, if they had existed, they would say there is no dark energy. There won't be a Big Rip in the universe's future, and the universe is dominated by matter, and that's perfectly valid. And then the next question you should ask is, well, maybe there's some other form of energy that we cosmologists today are so smug about knowing not about, and yet it does exist. So that's a theory of causal adequacy. A virtue of causal adequacy.
- 8.24** Explanatory depth is that it applies to very wide variety of scenarios. The Big Bang model encompasses the early universe's formation of elements, of isotopes. It also explains galaxies rushing apart billions of years later. That's a sense of explanatory depth across a wide variety of phenomena. Consistency is that your theory does not contradict itself, which is very important. So in the context of a universe that's static and eternal, that is a valid model. But then it was internally inconsistent with the fact that we see things that are changing and evolving with time. So it could explain certain things, like why we don't see the presence of new galaxies spontaneously coming into existence. But it couldn't explain Olbers' paradox. So that was internally inconsistent. So that's a lack of a virtue, if you like.

- 8.25** Internal coherence is that you fit things together naturally and neatly. You don't have to use completely different arguments, say, arguments of Isaac Newton's theory of gravity. And then all of a sudden, to explain the orbit of another planet, you have to then apply—pagan gods have to come into play. OK, those are scientific theories. That doesn't mean that they're necessarily valid ones.
- 8.26** Universal coherence in cosmology is very important. We're saying the laws of nature apply here on Earth, in the laboratory that we have here, all the way out into the heavens above, that they're not just limited to something on Earth. That violates the Copernican principle. That would say there's something special about Earth. It may be a bad special, but it could be that it's a very particular place. And just because you live at this very specific time are you able to construct your model. That's not necessarily a virtue.
- 8.27** This one I disagree with Dr. Keas about. I don't believe that beauty should count as a theoretical virtue. I think this bleeds into the kind of tendency for scientists to appreciate things that are beautiful for the sake of their own beauty. But I have to say that I disagree with this. I don't think that beauty in itself can be used as a proxy for the validity of a scientific model or its virtues. There are many things that are beautiful externally that aren't as such internally. And there's no reason to suppose beauty, which is something that is esthetic, literally, to the human mind, although we don't like to admit it, that that should somehow influence something scientifically.
- 8.28** And I like to point out that, actually, it's the deviations from perfect symmetry. There have been wonderful kind of, maybe even memes or demonstrations about this. But if you take, like, the world's most handsome man, not me, I'm going to exclude myself. But if you take Brad Pitt, OK, he's George Clooney, whoever. And you look at his face, and they've done this. They take his face, they take half of his face, and then they reflect that other half on the other side, OK? Just his face. He looks grotesque, hideous, horrible. The human mind hates perfect symmetry. There's imperfect symmetry between the two halves of our own face, and that imperfection is viewed as beauty. Ironically, imperfection is beautiful. So I disagree, as I say here, with Dr. Keas. But that's OK. We can have disagreements.
- 8.29** Simplicity. That you can explain a lot with very little inputs, that you can explain how things interact with a minimal number of assumptions is very, very important to a good theory. Now, these theories may have vast numbers of input parameters, the masses of all these particles, the lifetimes of all these different subspecies of particles, et cetera, et cetera. That doesn't mean it's not simple, because if you took one away, then things would not hold together. So you should make it as simple as possible, but not any simpler.
- 8.30** Lastly, the final one is unification. It explains more facts than another rival theory. So both the steady-state model did have some of these virtues. And since it was beautiful in a certain sense, that it's unending, it exists forever, it's permanent, eternal, you

know, forever young, it's really beautiful, but it's not necessarily unified because that same property made it impossible to account for the evolution of expansion of the universe. So it has some virtues. The more virtues the better, at least in Dr. Keas' estimation.

- 8.31** So let's compare these in the Big Bang model. I'm not going to go through all of these, each one in sort of a scorecard, and look for evidential accuracy. Does the Big Bang fit a wide variety of evidence? Yes. Cosmic background radiation, the formation of elements, the expansion of galaxies. We'll see that it has some inadequacies in that it can't explain its own origin. And that's OK. It explains a lot, since the origin of the universe. Now, asking where did the Big Bang come from is another request. It doesn't mean that the universe has to provide us with that answer internally consistent for us to believe in the Big Bang theory.
- 8.32** In fact, that may not even be part of the Big Bang theory. It may be an addendum that you add to it, which then affects the simplicity of the model, that you can't understand it. And then analogy might be something like the understanding of evolution. Evolution based upon heredity, natural selection, general mutations, and stuff like that, survival of the fittest—that is not necessary for us to understand, say, the biology of the lungs of a salamander. You can still learn a lot about the biology of the lungs of salamander and not know anything about DNA.
- 8.33** But once you have the DNA hypothesis, you can say, well, how does this animal relate to another animal? Is there a unified explanation for all their different causes of their genetic manifestations and of environmental or epigenetic effects? That's all part of biology. You don't have to understand every single one to understand, say, the origin of the species of salamander, and so on. So the unification, the Big Bang theory unifies different aspects, from physics of particles and nuclei, the smallest things in the universe, quarks, all the way up to the largest gravitational structures in the universe, clusters of galaxies—completely coherent and self-consistent and unified within one global theory, the Big Bang theory.
- 8.34** Now, did the CMB, did the formation of nuclei, and do the galaxies themselves uniquely specify a single Big Bang? No, they do not. It is possible there could have been multiple Big Bangs in a variety of ways. We could be part of an unending cycle of universes that expand and collapse, that have different values for the constants of dark energy, for example. In the previous universe, there wasn't enough dark energy to keep it expanding forever, so it collapsed. That collapse created a new Big Bang. That may have happened once. It may have happened twice. It may have happened an infinite number of times.
- 8.35** So the discovery of the Big Bang does not preclude the fact there may have been an unyielding, unending series of Big Bangs before our current universe became observable. And that's completely compatible with the observations that we've discussed of the cosmic microwave background, of the formation of nuclei, and of the

expansion of galaxies. In other words, in a universe that features multiple Big Bangs, maybe an infinite number of Big Bangs, cycling through time in the same volume of space, that could be compatible with all the observations we've talked about in the previous seven lectures.

- 8.36** And there could also be a universe that's being spawned outside of the observable universe that we inhabit. In other words, the universe next door, far enough away that light has not had time to reach us, just as light over the horizon has not had time to reach from a sailboat to the shore. And you must use other tools to access that information. One of those models is called the conformal cyclic cosmological model.
- 8.37** This model features expansion and dilution, but never features a Big Crunch. The universe expands. The photons and the energy inside that universe eventually create enough energy to create another universe, the universe sort of getting larger and larger with time. And this model of Sir Roger Penrose, who won the Nobel Prize in 2020, has certain attractive features to it, but in no case does it have just one creation, even. It has multiple origins and multiple, if you like, expansions that lead to that illusion of the existence of the current universe. That's called a cycle, a cosmology, or cycles of these eons, as they're called.
- 8.38** There could also be a single crunch and bang illustrated here, kind of the Big Bang in reverse. That could happen once, twice, infinite number of times. There's a bouncing model that features not a collapse to a point, as this Big Crunch would feature, down to a singularity in space and time, the inverse of the Big Bang, but actually just classically oscillates into and out of a very small amount of space, but not a singularity of infinite temperature and density. That's called a bouncing cosmology model.
- 8.39** And then there's a third type of cosmology genesis event called inflation. Inflation predicts in our region of the universe, there was just one Big Bang, but there's a vast cosmic landscape called the multiverse, in which creation and annihilation of universes are happening all the time. In other words, there's not just one Big Bang in the multiverse. We can only see the effects in our observable universe for now. But there's a possibility we could come into contact with another universe in the very near future that would give us evidence that we are not alone universally.
- 8.40** So inflation is the underpinning model. It's a particle physics theory created in 1979 by Alan Guth that suggests that the universe went a sudden phase transition. That phase transition was the result of the decoupling of the four forces of nature we spoke about in lecture three, that the inflation event took place, generated in tremendous expansion. That expansion solves at least two problems in the Big Bang model that we've discussed so far, and we'll discuss those in some detail.
- 8.41** The way that this comes about is a quantum field theory. That quantum field theory is relying on the fact that all of space is filled with what's called a quantum field. And all quantum fields are unstable. They all fluctuate all the time. We have names for those

fluctuations. We call them photons. We call them particles. But, in fact, there's nothing that prevents an entire universe from spontaneously fluctuating into existence with enough energy—which is an arbitrary quantity that we just described as relative to things that we can comprehend—but there's nothing that prevents an arbitrary fluctuation with enough energy to nucleate all the energy that was needed to form the first nuclei in the universe, to form the first dark matter in the universe, to form the first luminous matter in the universe.

- 8.42** So the theory of inflation is a cosmogenesis story, a story that can account for many of the features that we haven't discussed but may have been implicit in your mind or perhaps puzzling to you—as in, how did the universe go from being perfectly smooth, perfectly homogeneous, and even as smooth as it was at the surface of the cosmic microwave background radiation—how did that go from perfectly smooth to form very, very non-smooth, inhomogeneous, anisotropic patterns, like the clusters of galaxies, stars, etc.?
- 8.43** Inflation accounts for that. It relies on an abstract notion of what's called a quantum field. All quantum fields have unavoidable amounts of energy fluctuations in them. We are familiar with them in the context of light, which is a quantum fluctuation in the photon field. Another type of field, called a scalar field, is the inflaton. And as the inflaton evolved in its early, early history, it had jitters, it had fluctuations, it had uncertainties. Those uncertainties contributed to the universe's energy. A little bit more over here, a little bit less over there. Those perturbations in the energy changed the rate at which the universe expanded. More energy, more expansion. Less energy, less expansion.
- 8.44** If you have more expansion, that region of the universe will be more diffuse, more dilute. At regions of the universe that expand less will have more density because they will be more compactified, higher density. That could then explain those fluctuations in the curvature and the energy density of space-time, could explain how the universe had these tiny ripples in its amount of photon energy, which themselves trace to the origin of the distribution of matter, dark and luminous, in the early universe.
- 8.45** So inflation is supported by several different pieces of evidence. We'll go into some detail about them versus others, but we've all encountered them in one variety or another in the previous lectures. We've talked about the fact that the universe is close to the critical density, perhaps exactly equal to the critical density. That means the universe is almost perfectly flat. That is a mystery. That is not part of the Big Bang theory. There's no reason in the Big Bang theory that the universe should be completely flat. In fact, astronomers throughout history have thought it could be flat, it could be open, it could be closed. We showed beyond a reasonable doubt the universe is flat with zero curvature.
- 8.46** How did it get that way? The Big Bang doesn't address that. It just, we take it as a fact. There's another feature of the universe which we encountered. We talked about the

cosmic microwave background, that it has these tiny fluctuations. But I didn't point out until now how remarkable those tiny fluctuations are, because it turns out that the regions of the sky that we look at in the cosmic microwave background, not only do they have this tiny 1 part in 100,000 level of variation, so tens of microkelvin compared to 3 Kelvin, but that level of fluctuation is exactly the same, on average, over the entire universe, which is quite puzzling, because it shouldn't be that way.

- 8.47** It should be big variations over here, smaller over there. But in fact, the variations are identical, no matter where you look on the sky. The temperature of the universe is the same over there on the night sky as it is 180 degrees away over there, even though those parts of the universe are separated by twice the age of the universe times the speed of light. They could have never told each other what temperature to be at, and yet they agree to within 1 part in 100,000. That's a huge mystery, completely unsolved in the Big Bang model, without the addition of the inflationary universe addendum.
- 8.48** So inflation explains the flatness of the universe by the mere name inflation. That space-time increased so rapidly, and its volume increased so much, that just as if you're near the surface of the Earth, it looks perfectly flat, even though it's actually curved. In the early universe, the universe blew up so fast that it drove the curvature of the universe to be identically equal to zero, completely flat, completely Euclidean, such that lines never meet if they're parallel. So the flatness of the universe is postulated to be a byproduct of the expansion at near, or perhaps the speed of light, but it only lasted for a microscopic period of time, perhaps only a trillionth of a trillionth of a trillionth of a second.
- 8.49** So the evidence that we've collected so far in the previous seven lectures, we talked about the abundance of elements. We've talked about how much helium and hydrogen there are in their isotopes. We talked about the later epoch of the cosmic microwave background, the even later epoch of galaxies. And the question is, how far down could we go to understand the actual formation of the curvature fluctuations in the universe's structure that led to the imprint of all these other factors coming together?
- 8.50** Now, the horizon problem is the following, as I said. This is a map of the temperature of the CMB if you had microwave eyes. This is the whole sky seen, if you like, by God, or projected like a map of a sphere onto a flat surface. These fluctuations mean, where it's red, that the universe is hotter than the average 2.7 Kelvin temperature by a mere 10,000,000ths of a Kelvin. In contradistinction, this region is equal—a blue region is cooler by an amount of average 10 microkelvin. So there's 10 microkelvin fluctuations against the average temperature of 3 Kelvin. So that's a part in 100,000.
- 8.51** And yet, if you look at this region, remember this image comes from when the universe was 380,000 years old. The maximum size scale that the universe could have communicated sent a light signal traveling at the speed of light to another part of the universe to say, "This is my temperature. Make sure you set your thermostat to the exact same temperature." The maximum volume, the maximum radius that light could

have traveled, is the speed of light times 380,000 years. And yet what this image is showing is that not only do two regions that are separated by 380,000 light years have the same temperature, but a region that's billions of light years away from it has the exact same temperature that couldn't possibly have been in contact with another region on the opposite side of the universe.

- 8.52** It's as if these two regions were in collusion, conspiring to have the same value. But scientists don't like conspiracy theories. We want to have an explanation, and it's a better explanation to say that there's some underlying time period when the universe was much, much smaller, when they were in thermal equilibrium, to share the same temperature, that later expanded and inflated so much that they appear to not be possibly in causal contact. And in fact, that's the basis of the inflationary theory.
- 8.53** So if you take the surface of the night sky in microwaves, you can divide it into 40,000 circles. Each one has a radius of about one degree across. That one-degree patch can be in causal contact. That's the size of, effectively, the size of the universe at the time when the CMB was produced. But we see every single patch of the sky, a region over there, identical to a region over there, to within one part in 100,000. It's a very, very big coincidence, given that all 40,000 patches have almost exactly the same temperature. They must have been in contact with one another at an extremely early phase in the universe's history, and then blown up and expanded so far that, today, we can't conceive of them being at the same temperature if they were in thermal equilibrium. But inflation solves that problem.
- 8.54** An analogy is like, if you tell 40,000 people to come to your dinner party and bring a different dish, and they all brought the exact same dish, even though there's millions of different possibilities, your only explanation for that would be—and besides sheer random luck, which is incredibly improbable—would be to say that they collaborated. "Let's all bring potato salad or whatever." And in fact, the only way to do that, other than to say it's a random fluke, is to establish that they were in communication with each other before we found out about it. And that's effectively what inflation does.
- 8.55** Inflation explains the flatness through this rapid expansion of space-time for a very short period of time, and it explains this horizon problem. Why is the temperature of the CMB equal on opposite sides of the universe? Those were two huge mysteries in the Big Bang. So those are solved by the addition of the inflationary idea. So now we have to say that the Big Bang model explained a lot, but it didn't explain these two different features. Why is the universe flat? Why does the universe have the same temperature in all directions?
- 8.56** Now there's a problem, as always happens. The Big Bang model explained and got rid of the flaws of the steady-state model, kept some of its features, but not many. Inflation explains the flaws in the Big Bang model, but has some flaws of its own. Perhaps the most important one, philosophically, to my colleagues and I, is the fact that it predicts a multiverse. The only way that Petri dish analogy really works is if our

universe is just one of the crystals, and there's many other crystals in this multiverse filled with this inflationary field. So the inflationary quantum field fills all of the multiverse, and our universe is one of those crystals, one of the fluctuation points where the density was just right to produce a universe, a crystal universe.

- 8.57** So that's a problem, because that's adding on something else in this case, which may not be falsifiable. You may not be able to prove the multiverse wrong, even if it is wrong. But the multiverse follows a grand tradition that we started talking about in the very first course lecture, in that we've known for centuries that there's other places that we can't see by eye, but they still exist, just because you can't see them. There's places beyond the horizon, on land, on sea. The Earth is just a planet in the Copernican context. The sun is just one star amongst 100 billion in our galaxy. Our galaxy is but one of a hundred billion or more in our corner of the observable universe.
- 8.58** So the logical progression might have you believe our universe might be one of a potentially infinite number of other universes. So it actually fits the virtue of the Copernican principle, and that there's nothing special about our universe. That might be a virtue, in that it is unified with all these other disparate forms of evidence that point to our existence not being unique or special. You don't want us to be at a unique vantage point in time or space. And so, too, with the multiverse, you don't need that.
- 8.59** Another example of this is, imagine these various bacteria cultures. On the left is shown a petri dish, and there are different colonies of bacteria. They have different properties. Some secrete toxins that kill off other bacteria cultures, and they actually end up with a counter-effect that the bacteria will secrete an antitoxin to defeat the toxin of its neighbor. And you end up with these separated bacteria culture universes, if you will. Now, could a bacterium, a bright bacterium, living in one of these colonies, deduce the existence of other colonies that they can't see, they can't sense, they can't get to, because the toxins that would kill them and prevent anyone from returning with information about it—could they deduce it?
- 8.60** Could they say, "My mere existence and the existence of this gel called agar means that it could be. There could be other cultures of bacteria that we cannot see. They're beyond our chemical horizon." That's an analogy that we could think about. So in this way, you might gain evidence for the multiverse, the petri-verse here, from the existence of your own personal universe with properties that make it suitable for you to exist. It's an example of what's called an anthropic principle, reasoning based on our own existence. We could not live in a universe that doesn't allow us to live. It sounds like a tautology. In a sense, it is.
- 8.61** But in the context of cosmology, it allows the explanatory explanation, if you will, of why our universe has the properties that allow us to observe it and make predictions about its past and its future. And it was started from the founding fathers of inflation, two men that are the theoretical pioneers. And they both agree that you need to have the multiverse in order to have inflation. And many cosmologists believe you need to

have inflation to explain the flaws in the Big Bang. So if you want to keep the successes of the Big Bang, you're driven to explain the flaws in it, and the flaws of it are explained by inflation.

- 8.62** And the one new flaw, if you like, that inflation brings up, is a possibly untestable, possibly unfalsifiable, multiverse. So it's a trade-off. Do you want to keep the Big Bang theory? If so, are you comfortable accepting the theoretical flaws of the multiverse, namely its lack of falsifiability? Now, it turns out you can actually detect evidence supporting inflation, and it's called inflationary B modes. It's an aspect of the polarization of the cosmic microwave background.
- 8.63** So the question about detecting whether or not inflation took place brings us back to this philosopher of science named Karl Popper. He said, "Scientific theories can never be proven." We've said that many times. "But they can be falsified, proven wrong. They can be falsified and scrutinized by decisive experiments." In other words, experiments that can prove the theory wrong. And it turns out that the alternatives to inflation that we've encountered—the bouncing model, the conformal cyclic model—those can be falsified. You can prove those wrong, but you can't prove inflation wrong. Inflation is unfalsifiable, which is a problem, because inflation may have occurred, but we can't detect it, or it may not have occurred, in which case we also couldn't detect it. So it's not possible to falsify inflation. And inflation comes hand-in-hand with the multiverse.
- 8.64** Fortunately, you can falsify rivals to inflation. So if those are falsified, it lends more support to inflation. Inflation survives, if you like. And the way that you prove those models wrong is with the type of science that I and my colleagues do, which is called the polarization of the cosmic microwave background. We're not going to describe what polarization is. Suffice it to say, it's one property of light that leads to it having a preferred orientation in space, that light waves are waves. They oscillate like a rope swung between two people. The plane of oscillation is called a plane of polarization. You have polarized sunglasses, probably many of you, that blocks out one type of light that leads from glare off of a liquid or off of the road. And that's why those are useful.
- 8.65** But if inflation took place, there should be a characteristic signature, an imprint on the microwave background's polarization—not its temperature, but its polarization. And that will be a decisive test that would prove wrong the models of cyclical or bouncing cosmological origins for the universe. It would prove that if there were multiple Big Bangs, that it didn't come from a previous collapsing universe or a cyclical universe or a Big Crunch. These are all rivals that can explain the origin of our current universe. Remember, that's what we're trying to do. How did our current universe come to be?
- 8.66** One model is it's been an eternal cycle of cosmological origins, stretching and so forth. That's called the conformal cyclic model, or that it's a bouncing, non-quantum, singularity-type event which features multiple Big Bangs and collapses or bounces. We can prove those wrong if we see this type of signature. And in fact, we did see this type of signature with a telescope that I co-invented called Bicep, and its successor, called

Bicep 2, which is a simple refracting telescope made of lenses with about 1.5 feet in diameter. And that refracting telescope was different from what Galileo uses over here, in that you had to cool it down to one quarter of 1 degree above absolute zero Kelvin. In order for these sensors to detect microwaves, they must be cooled to superconducting levels. And that required us to cool this down to near absolute zero. Otherwise, it just uses lenses like a camera or a telescope that you could get or have probably used before.

- 8.67** It was about 10ft tall, weighed about 10,000 pounds, and instead of taking it to San Diego, which would have been convenient for me, we had to bring it to the South Pole, Antarctica, because the South Pole is actually one of the driest and highest points on Earth. Dryness is important because water absorbs microwaves. Microwaves from the Big Bang can get absorbed in the Earth's atmosphere. But over the South Pole, the atmosphere is so dry and so cold, there's very little water impeding the photons to our telescopes.
- 8.68** The South Pole is the last place on Earth that was really reached by human beings. And it was only about 100 years ago. There were two teams that went there trying to be the first. They both reached the South Pole. The Norwegian team, led by Roald Amundsen, got there three weeks before the British team, led by Robert Falcon Scott. And all of these men died. And the reason that they were late by three weeks, and that made a huge difference in the Antarctic cruel winter, is that they refused to use this novel technology called sled dogs. The Norwegians would use the sled dogs skiing up from the coast of Antarctica. Antarctica is a continent, unlike the North Pole. The South Pole was above a continental shelf.
- 8.69** They landed in the Ross Sea and skied up 9,000 ft in elevation over 700 nautical miles in just a few weeks, 100 years ago, with no technology. But the Norwegians would use the dogs to climb up to the top of the 9,000-foot polar plateau. And then they ate the dogs. Sorry to tell you, PETA wasn't around back then. And not only did it help them carry the food and their tools and equipment that they needed, but the British refused to eat dogs. You know, I won't make a joke about British cooking, but they were reliant themselves on skiing, which meant they had to carry more food, more fuel, and it took them much longer. And that 3-week difference made the difference between life and death. They all died.
- 8.70** When they got there, they saw a Norwegian flag. This is the British team seeing the Norwegian flag, which kind of would have been like Neil Armstrong stepping out of the Eagle onto a Russian flag. That would have been so disappointing, so depressing. And so he said, "Great God, this is an awful place." And it's really quite boring and dangerous, even to this very day.
- 8.71** So when you get to the South Pole, you are actually nowadays in an awesome place. Not an awful place. It's actually a station that looks like an outpost from Star Wars. And this is the axis on which the Earth is spinning. This marker indicates the South Pole, the

geographic axis on which the Earth is spinning. The one on the North Pole, you can't plant because there's no continent there. There's ice there that sometimes melts. The station is paid for by the US taxpayers, and any country that does scientific research represented by those flags in the right corner is welcome to go to Antarctica. But you can't do any business or mining or military operations. It's purely for scientific research. You check in on Instagram. Prove to your friends you went there.

8.72 That's the telescope that I showed you before, a refracting telescope cooled to almost absolute zero. And I like to say it's in the second most important building. It's called the Dark Sector Laboratory. And that building there is the outhouse. That's the most important building. So we observed for three years, and we detected that signal of polarization that was claimed to be the decisive test to rule out the alternative models to inflation—the cyclical model and the bouncing models. We did that exactly about 10 years ago. It's depicted artistically up there, but really, this is our data, and this is an image from the Washington Post.

8.73 They depict the pattern that you would see if inflation took place. It would produce waves of gravity. Those waves of gravity would make a particular type of polarization called B mode polarization. That's what we designed Bicep to do, and that's what we detected, or so we thought. So this made headlines around the world. This is now the real data, not just a depiction in a newspaper, but we got plenty of coverage in the newspapers. New York Times, my hometown newspaper. This is the Economist. You know that you're successful when it's on CNN. We even made it to The Onion, the foremost magazine of all scientific reason. And they made a joke headline based on our discovery that the B mode polarization was comparable to the love between musicians and their, and their partners. And they may have been around forever, and that's what they were debating, whether or not you could give someone your heart forever.

8.74 But like good scientists, we were always worried that what we saw was just what we hoped to confirm that we saw, and we did the best that we could do. But it turns out we were flummoxed. We were duped into seeing a pattern of polarization that we thought came from inflation, but actually came from microscopic versions of the dust that I gave you and I give away on my website, BrianKeating.com. This meteorite is a giant chunk of interstellar space dust. And interstellar space dust has the following properties. It's highly magnetic because it came about from the explosion of a supernova, a type-2 supernova, that produced, in its dying moments, iron and nickel and cobalt. And when iron and nickel and cobalt are fused together and made from lighter elements, they don't give off enough heat to keep the massive gravitational bulge of the star apart. And it collapses and explodes in the supernova, and it sprays out into the interstellar medium the chunks that it last made, namely these meteorites.

8.75 Now, our galaxy also has, in addition to stars and supernova, a magnetic field. And that magnetic field can be attracted and orient these grains of meteorites and the grains of dust. So, in fact, what we saw was not the explosion, the aftermath of gravitational

waves in the Big Bang's early moments, but, in fact, was the presence of little, tiny dust grains aligned by their magnetic field of the galaxy, mimicking the signal that we thought we saw. Here's a depiction from my book that shows how these particles can get aligned, produce the type of polarization that we claim we saw or we did see. We actually saw it. It didn't come from the Big Bang. It did come from the cosmos, but not from the origin of the universe. It came from parts of the origin of our solar system, which is interesting, but tells us nothing about the Big Bang.

- 8.76** So these signals are still there to be discovered. So I like to say, from San Diego, we thought we were looking back into time and space in this pristine cosmos, but really, it's more like Los Angeles. It's very smoggy, dusty, dirty, filled with things that get in your way when you're trying to look at the stars, trying to be a paparazzi. You get flummoxed and confused sometimes, and it's your job to go back and do a better job next time as a good scientist. So, I talk about this in my book, my first book, how our drive to discover it caused us to overlook evidence to the contrary. Confirmation bias was at play, and that's very dangerous, and we have to be aware of it.
- 8.77** So the only way to defeat this cosmic dust is to do another experiment, and that experiment would have a dedicated capacity to detect dust. So imagine if what we saw, the total signal that we saw, was comprised of the following factors. We detected a signal S . That signal S is made up of the true cosmic signal, where this is produced by inflationary gravitational waves, which would mean that inflation took place, and this would be a byproduct of the multiverse. So the multiverse leads to inflation, which produces gravitational waves, which leads to this B mode polarization signal.
- 8.78** But we know now that we also observe dust. So now what we have to do, as good scientists, is make an experiment that only looks for the dust. A dust buster, a dust detector. You have an experiment that only measures dust, call that signal 2. And now you'll take signal 1, you'll subtract signal 2, or the dust signal. And what you're left with is the pure cosmic signal, or so you hope. Of course, there'll be unavoidable noise from the instrument, from the surroundings, from the observatory and more, and we can account for those as well.
- 8.79** So the new experiment that we're doing is called the Simons Observatory. It's a huge observatory funded by various foundations and our member partners. There's over 300 of us. That's a picture of us in San Diego last summer. And we're trying to build a detector, not at the South Pole—a series of telescope cameras in the Atacama Desert of northern Chile. And in so doing, we can measure dust and also the cosmic signals that can reveal the presence of these B modes, which could refute these alternative hypotheses and give us more credulity in inflation and perhaps in the multiverse itself. It uses refracting telescopes and reflecting telescopes. One is enormous. It's 6-meter diameter mirrors, enormous telescope. The other three are small refracting telescopes.
- 8.80** It operates at multiple frequencies, so we can measure the imprint of dust and separate it from the pure cosmic signal. Remember, a black body has a characteristic

signal that we talked about in lecture five. And the dust signal has a very different black body signature that we can also measure, as long as we have multiple frequency channels. The big telescope looks at very small features on the sky. Small telescopes, counter intuitively, look at big aspects on the sky. We're about to get our first data from these instruments, after 7 years of building, designing, enduring Covid, and building materials up at 17,500 ft above sea level. It's not easy to do this.

8.81 And there are some of my students and colleagues at UCSD building the telescope, fielding it. That kind of gives you a scale of how big it is. And again, coming back to Popper, coming back to doing a decisive experiment that can determine whether or not a scientific theory should be given more credulity or less. And in the case of B mode, as a decisive experiment, like the Simons Observatory, we hope, as I've been asked before here, what do I hope to see? I hope to rule out as much as possible and be left with as much and as close to the actual truth as possible. And never forget, the stakes of this involve, if we detect this, it will be circumstantial evidence that inflation took place, and if inflation took place, that the multiverse exists.

8.82 And many people feel like this is the famous elephant in the room. And we haven't talked about theology or religion, but the question of what the multiverse does, it allows different universes to have different physical properties. And my friend and colleague Paul Davies at Arizona State University has said that, even though the multiverse sounds very scientific, it's actually sort of a theological explanation just the same. He said, "Multiverse explanations are reminiscent of theological discussions involving a possible infinity of unseen universes to explain the features of the universe that we do see. And this is just as made up or ad hoc as invoking a creator. And so while the multiverse theory seems quite scientific, it does require something that is anathema to science, which is to have faith."

8.83 I always say, I don't believe in gravity. I don't have to. I have evidence for gravity. I don't believe in evolution. I have evidence for evolution. I might say I believe in God, or I don't believe in God. That's a question of faith. So you'd like to separate faith-type questions that have no proof or aren't amenable to proof, necessarily. You can have personal relationships with God, I suppose, but it's not rising to a level you would publish in a scientific journal. And that's OK. There are different fields of expertise, but nevertheless, we should be careful if we confuse the two, as being separate and explaining or getting rid of the need for a creator.

8.84 And I just want to close with our discussion, which has been fascinating and fun for me—I hope for you as well—that it's amazing we've been able to do this, to study the properties of the cosmos using primarily the human mind and going on journeys through things that are completely incomprehensible, really. And it's a wonder that we can comprehend them at all. Time scales of billions of years, when we only live and are conscious for decades. Size scales billions of times bigger than our own. Time scales that are billions of times shorter than anything we could imagine. Size scales that are

infinitesimal, almost. And we've been able to comprehend it and apply quantitative laws and make real scientific predictions.

8.85 So we've encountered these vast questions. There are many, many more to come. And the last kind of portion on that circle of fascinating questions that I like to think about and I mentioned at the very beginning of the course, it's fascinating that inanimate matter created life, and that life emerged into conscious matter that can contemplate the origin of matter, and yet we don't know. Is consciousness a byproduct, as some say, intrinsic to every particle in the universe? Is there something about consciousness that can be explained in life and its evolution? Is consciousness merely an evolutionary byproduct helping us to live and exist and rationalize things in our brain? And the last question, which I think is very important, is one of freedom. Freedom, free will, causality, cause and effect. This is where a next phase might take over once we understand better how our cosmos came to be the way it is.

8.86 And I want to conclude with just two quotes, one by my favorite scientist of all time, perhaps along with Galileo. And he gives a warning, does Michael Faraday, who knew almost no math, and he made contributions as vast as the first person to describe what a field is, electric field, magnetic field. He invented the motor, the generator. He did incredible things without knowing any math, really, whatsoever. He said, when he would give a speech, as I had the honor to do at the Royal Institution, he said, "I think it's likely that I've made many mistakes in the preceding pages or PowerPoint slides, for even to myself, my ideas in this point appear only as a shadow of speculation or as one of those impressions on the mind, like a dream, which are allowable for a time as guides to thought and research. He who labors in experimental inquiries knows how numerous these mistakes are and how often their apparent fitness and beauty vanish before the progress and development of natural truth."

8.87 So we should have a little bit of humility when we confront these questions. But we should also have a little cosmic ego, too. Because if you're just saying, "I'm not adequate, I'm not worthy," you're never going to make any accomplishment whatsoever. And I'll close with a quote from the great Winston Churchill. He said, "Now is not the end. It is not even the beginning of the end, but it is, perhaps, the end of the beginning." Thank you very much.